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Benjamin Spada

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THÈSE PRÉSENTÉE

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L'UNIVERSITÉ DE BORDEAUX

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OPTION : PHILOSOPHIE DES SCIENCES

Par Benjamin SPADA

Comprendre l'évolution du cancer : le rôle de l'oncologie comparée

Understanding the evolution of cancer: the role of comparative oncology

Sous la direction de : Thomas PRADEU

Soutenue le 28/10/2025

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Comprendre l'évolution du cancer : le rôle de l'oncologie comparée

Résumé :

Cette thèse interroge ce que l'étude de l'évolution du cancer à travers le vivant peut apporter à la recherche en oncologie. Alors que la majorité des travaux philosophiques se sont concentrés sur l'évolution clonale des tumeurs à l'intérieur d'un individu, ce travail déplace la focale vers le cancer à l'échelle de l'arbre du vivant. L'objectif est de montrer que la biologie de l'évolution constitue un cadre puissant, non seulement pour replacer le cancer dans son contexte historique, mais aussi pour le reconceptualiser comme phénomène biologique.

Le premier chapitre analyse les approches de l'évolution du cancer à travers le vivant en particulier l'idée influente du cancer comme « *cheating* » au sein de la multicellularité. Si cette approche met en lumière l'universalité du cancer, elle reste trop vague pour servir de base conceptuelle solide. Le deuxième chapitre retrace l'histoire de l'oncologie comparée, des premières descriptions naturalistes à son tournant évolutif récent, et montre comment ce champ constitue la meilleure manière d'étudier le cancer en tant que phénomène inter-espèce.

Le troisième chapitre constitue la contribution principale de cette thèse : une définition universelle du cancer fondée sur l'évolution. En mettant l'accent sur les processus plutôt que sur les résultats, cette définition dépasse le prisme anthropocentrique habituel de l'oncologie et permet des comparaisons interspécifiques plus rigoureuses. Enfin, le quatrième chapitre s'appuie sur les analyses critiques précédentes pour proposer des pistes de développement pour l'oncologie comparée : préciser les définitions, mieux articuler méthodes et objectifs, et clarifier les ambitions théoriques du domaine. La thèse défend l'idée que l'étude de l'évolution du cancer à travers le vivant n'apporte pas une complexité supplémentaire mais une clarification et unification conceptuelle. Elle met en avant une contribution académique : définir ce qu'est le cancer, cartographier le champ de l'oncologie comparée de manière exhaustive et montrer comment la philosophie peut aider les sciences du cancer à mieux formuler leurs questions. Situer le cancer dans son histoire évolutive révèle qu'il n'est pas seulement une maladie humaine mais bien un processus biologique fondamental, que la philosophie est particulièrement bien placée pour éclairer.

Mots clés: **cancer, évolution, philosophie de la biologie, oncologie comparée**

Understanding the evolution of cancer: the role of comparative oncology

Abstract:

This dissertation explores what the study of the organismal evolution of cancer can bring to cancer research. While most philosophical work has focused on tumour evolution within individuals, this thesis shifts the perspective to cancer across the tree of life. It shows that evolutionary biology provides a powerful framework not only to situate cancer historically but also to reconceptualize it as a biological phenomenon.

Chapter 1 examines attempts to explore the different approaches of organismal evolution of cancer, especially the influential view of cancer as "cheating" in multicellularity. While this perspective highlights cancer's ubiquity, it lacks conceptual precision and leads to an overly broad definition. Chapter 2 traces the history of comparative oncology, from early descriptive studies to its recent "evolutionarisation," showing how the field is the best approach to organismal evolution of cancer.

Chapter 3 presents the core contribution of this dissertation: a universal, evolution-based definition of cancer. By focusing on processes rather than outcomes, this definition avoids the anthropocentric bias commonly found in general oncology and allows for more robust cross-species comparisons. Chapter 4 then builds on earlier critiques to propose avenues for comparative oncology, including sharpening definitions, aligning methods with goals, and clarifying the field's theoretical ambitions.

The central claim defended here is that studying the organismal evolution of cancer enriches research not by adding complexity but by providing conceptual clarity and unity. The contribution of this work is both academic and conceptual: clarifying what cancer is, mapping the field of comparative oncology, and suggesting how philosophy can help science ask to better formulate their questions. Ultimately, situating cancer in its evolutionary history shows it to be not only a human disease but also a fundamental biological process, one that philosophy is uniquely well equipped to clarify.

Keywords: cancer, evolution, philosophy of biology, comparative oncology

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"The philosophers have only interpreted the world, in various ways; the point, however, is to change it."

— Karl Marx, *Theses on Feuerbach* (1845)

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I am deeply grateful to every scientist who took the time to interact with me, whether through discussion or collaboration on a paper. Even as a philosopher among scientists, I have never once felt like an imposter.

On a more personal note, I wish to express my deepest gratitude to everyone who has supported me over these four years — my friends from Bordeaux, Lyon, and from across France and around the world. To Clara, of course, whose light has been a constant and precious presence in my life. My roommates, Andrea and Ekin, who made me feel at home and offered more love and support than anyone could hope for; Mrs. Coadic, who gave me the keys to endure even the toughest times; and my trusted companions in struggle, in whom I place all my faith for the years to come. A special thanks goes to Saturne, who generously offered to share some of her otherworldly and celestial art to grace this manuscript. Thank you for your dedication and for your humanity.

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List of abbreviations

ADN	Deoxyribonucleic acid
CTLA-4	Cytotoxic T-lymphocyte antigen-4
dMyc	Drosophila Myc
EGFR	Epidermal growth factor receptor
ERCC1	Excision Repair Cross-Complementation group 1
HLA	Human leukocyte antigen
IGFBP	Insulin-like growth factor-binding protein
ISIS	International Species Information System
JNK	c-Jun n-terminal kinase
l(2)gl	Lethal (2) giant larvae
MHC	Major histocompatibility complex
PCNA	Proliferating cell nuclear antigen
PD L1	Programmed Death Ligand 1
PD1	Programmed cell Death protein 1
Pi3K	Phosphoinositide 3-kinase
PinS	Philosophy in science
TOR	Target of rapamycin
TP53	Tumor protein 53
Wnt	Wingless-related integration site

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Introduction

Over decades of intense research, our understanding of cancer has remained largely framed by what we observe in humans. Yet cancer is not a uniquely human disease: it occurs in many non-human organisms and even, as some claim, across the tree of life, from mammals to fish, and even in some invertebrates. This fact, though largely overlooked, carries substantial consequences. If cancer emerges repeatedly in such diverse organisms, it could reflect something fundamental about the nature of multicellular life—about cooperation, conflict, and the evolutionary trade-offs that shape complex organisms. Yet how does one study this evolutionary dimension of cancer? Building an exhaustive evolutionary history of cancer —for instance by relying on fossil evidence— would be tempting. However, the data needed to determine precisely when cancer first emerged and in which species it exists are not available. What remains possible, therefore, is the reconstruction of a partial history of cancer based on the evidence currently accessible. This is one of the main reasons that led me to focus on the scientific field of comparative oncology: as I will argue, it offers the only viable approach to investigate cancer's evolutionary trajectory. Another reason is that the field is already sufficiently established to provide philosophers like us with appropriate interlocutors — a crucial condition for any philosopher who wishes to contribute to the scientific endeavour.

The most significant contribution of exploring the foundations of comparative oncology lies in the conceptual unification it entails. Such unification is crucial for comparative oncology in two respects. First, studying cancer across species requires a definition broad enough to enable cross-species comparison. Clinicians, by contrast, do not need a universal definition: they only require a characterization precise enough to identify and treat the specific type of cancer in front of them. Comparative oncologists, for their part, must rely on a sufficiently unified concept to determine what should count as cancer and what should not across many diverse species. Second, the examination of cancer across the tree of life also results in the production of novel definitions or characterizations of cancer. Tracing the occurrence of cancer across the tree of life inevitably pushes researchers to revisit and refine their initial definitions of cancer. In this process, comparatists encounter diverse phenomena that may claim the label of cancer. It will irremediably provide a new perspective on cancer — making it less anthropocentric and more applicable to a larger part of the living world. This need for conceptual unification alone justifies the involvement of philosophers, who are particularly well equipped to assess and/or carry out themselves this kind of work. As a matter of fact, the main proposition of this PhD dissertation will be a universal definition of cancer – which, we argue, will be helpful for comparative works on cancer. Another

part of this dissertation will explore the central ambiguity which the field faces. Is the primary aim of comparative oncology to advance therapeutic knowledge through interspecies comparisons, or instead to uncover the evolutionary foundations of cancer? Both goals are valuable, but without careful conceptual work the tension between them risks going unexamined.

In addition, while characterizing cancer as a complex and heterogeneous disease is accurate, these terms are often repeated without sufficient clarification, and risk becoming little more than placeholders. Philosophers, especially leading philosopher of cancer Anya Plutynski, have provided valuable analyses of why cancer should indeed be understood as complex and heterogeneous (Plutynski, 2018, 2021). Cancer, in its many dimensions, resists easy universal definitions. As Plutynski has argued, one important source of this difficulty lies in cancer heterogeneity: even when one considers humans only, cancer is complex because it manifests differently across multiple levels and contexts. This multidimensional heterogeneity hinders any attempt to establish a universal set of defining criteria. Put simply, each time one claims to have captured the true nature of cancer, counterexamples emerge to undermine that definition. This difficulty gives rise to a further kind of complexity—not inherent to cancer itself, but to our conceptual efforts to define it, a form of conceptual complexity. In this dissertation, I suggest that rather than adding more complexity and heterogeneity around an already elusive object (as might be argued by some), adopting an evolutionary perspective on cancer offers a way to dissolve this conceptual complexity. In fact, cancer takes different forms across species because it is a biological phenomenon with an evolutionary history. By stepping back and adopting a broader focus, we should be able to better identify the commonalities between all instances of cancer. However, this approach carries the risk of overlooking one of cancer's defining features: its heterogeneity. A central challenge of this dissertation, therefore, will be to propose a conceptual unification that both provides a characterization applicable to all cancer cases across the tree of life and accounts nonetheless for the multiplicity of ways in which cancer manifests itself at different levels in different living forms. Following this line of inquiry, the present work will demonstrate that one fruitful way to dissolve the aforementioned conceptual complexity is to situate cancer within its historical context and to examine its evolutionary trajectory. What is needed, then, is a critical reflection on the foundations of comparative oncology—an effort to make its assumptions explicit, sharpen its questions, and reveal what is at stake in its practice. This dissertation takes up that challenge: to examine the foundations of comparative oncology, bring conceptual clarification and unification, which requires the sustained attention of philosophers of science.

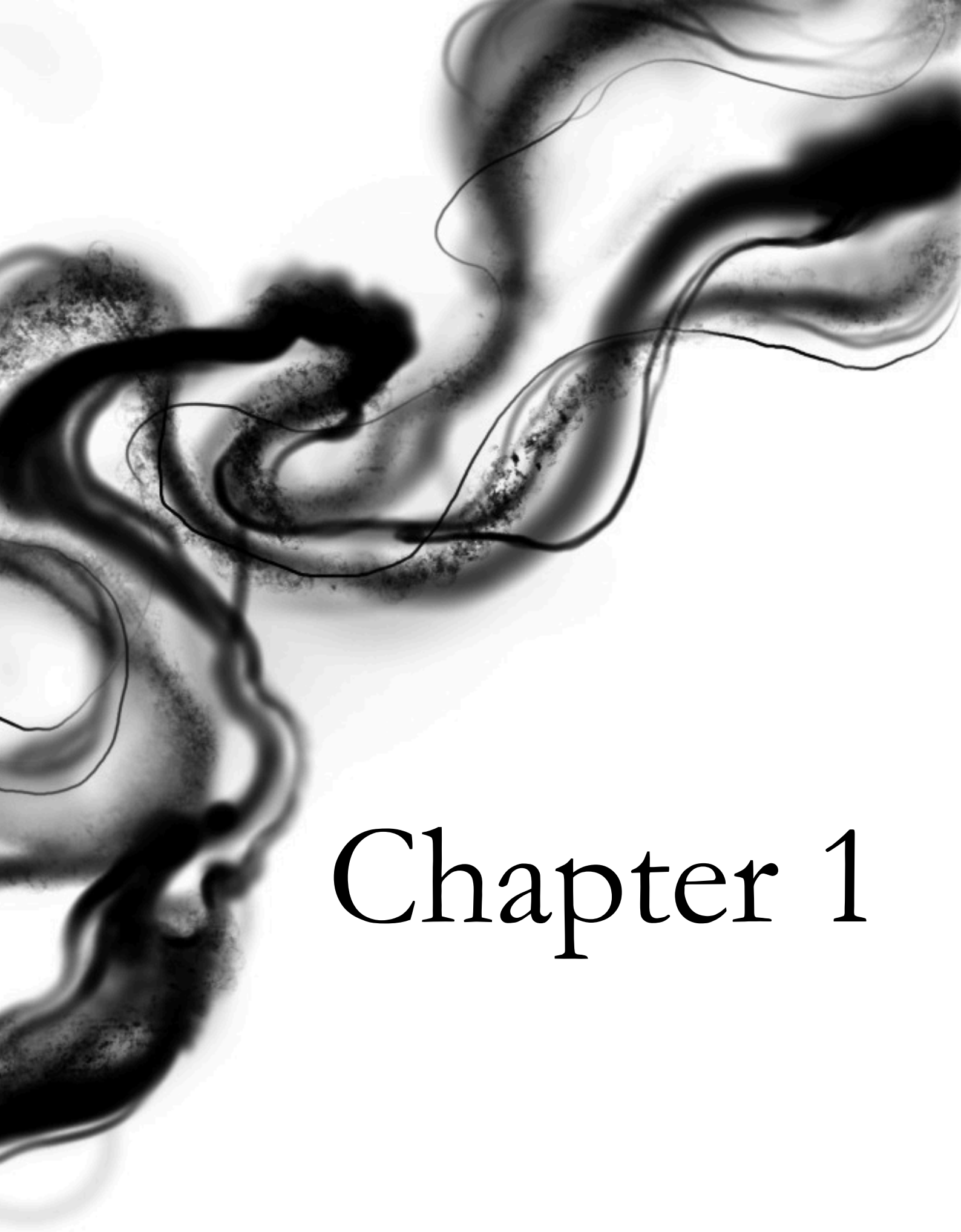
Our research group in Bordeaux strongly believes that philosophical expertise can play a valuable role in such an inquiry, especially given philosophy of biology's long-standing engagement with evolutionary questions. With that in mind, I set out to study cancer through an evolutionary lens. Crucially, in no more than five years (2020-2025), our research group has built connections with some of the best specialists of this field in the world, especially through our collaborations with Mathieu Giraudeau and Orsolya Vincze at the LIENSs lab in La Rochelle, France and with Carlo Maley at the Arizona State University, USA.

A crucial point to clarify before delving into the core of this project is the nature of our approach. The present work falls within a specific subtype of philosophy of science that my group refers to as *philosophy in science* (PinS). Traditionally, philosophy of science has treated science as its object of inquiry. Philosophers of science in this classic sense "contribute to our knowledge of the methods, concepts, objects, and problems of science, and/or address philosophical problems using lessons taken from science" (Pradeu et al., 2024, p.1). Sometimes, productions in philosophy of science can be helpful to scientists but it is not the primary goal of the field. On the contrary, PinS aims at participating in the resolution of scientific problems. In this case, philosophers "use philosophical tools to produce scientific knowledge rather than knowledge about science" (*Ibid*). As Pradeu et al. phrase it, philosophers in science "permeate through science". This approach underpins the entirety of the present work. What I seek to do is to identify scientific problems about the evolution of cancer and, in close collaboration with scientists, use the range of philosophical tools at my disposal to help, as much as I can, in the resolution of those problems. In that regard I worked with various scientists, most of them playing a central role in the field of comparative oncology. In addition to the scientists mentioned above, I have had interactions with David Bilder (UC Berkeley, USA), Vera Gorbunova and Andrei Seluanov (University of Rochester, USA), Joshua Schiffman (University of Utah, USA), James De Gregori (University of Colorado, USA), Elizabeth Murchison (University of Cambridge, UK), Thomas Bosch (University Kiel Germany), Hannah Kokko (University of Mainz, Germany), most of whom have been involved in a co-authored interdisciplinary paper that we have recently published in *Nature Reviews Cancer* (Vincze et al., 2025). The article, which the reader will find in appendix, is a product of this international collaboration.

This thesis constitutes the first philosophical attempt to evaluate the scientific endeavour of comparative oncology to explain cancer. It is part of a wider group reflection about the nature and origins of cancer and the benefits of evolutionary approaches for studying cancer. This group is constituted of Bertrand Daignan-Fornier (IBGC, Université de Bordeaux, France), Lea

Mascherrek (Université de LaRochelle & Université de Bordeaux), Orsolya Vincze (Université de LaRochelle), Guglielmo Militello (Université de Bordeaux), Thomas Pradeu (Université de Bordeaux) , and works on several related topics, including the connection between cancer and multicellularity (Pradeu et al., 2023), the critique of the atavistic theory of cancer (Daignan-Fornier & Pradeu, 2024), and an exploration of why cancer exists in certain invertebrates but not others (work conducted by Lea Mascherrek in her PhD dissertation, started in 2024). My role in this collective project was to thoroughly explore comparative oncology and evaluate its ability to address cancer with an evolutionary perspective.

This dissertation is divided into four chapters. Chapter 1 examines the utility of evolutionary accounts of cancer. It outlines the types of questions such accounts are expected to address and evaluates the extent to which various approaches succeed in doing so. Chapter 2 provides a historical and conceptual review of the scientific field of comparative oncology. In this chapter, I trace the development of the field, arguing that its origins lie much earlier than commonly assumed by its present-day practitioners. I show how, due to two major theoretical shifts, the field's primary objective transitioned from studying cancer across the tree of life to developing therapeutic solutions through comparative research. Chapter 3 presents the outcome of a collaboration between philosophers and scientists, in which we propose a universal definition of cancer grounded in the concept of affordances. It reflects the state advancement of this conceptual project. In this chapter, we argue that this definition holds value for both clinical and comparative approaches. Chapter 4 is a propositional chapter in which I put forward future directions for comparative oncology, drawing on my analysis and my critiques of the field.



Chapter 1

Chapter 1 : Organismal evolution of cancer

1.1 Seeing cancer through an evolutionary lens

In the last two decades, many cancer experts have expressed concerns about the current state of cancer research. Some claim that progress in cancer treatments has been, overall, quite limited, in particular because precision oncology has turned out to be not as promising as initially thought (Prasad, 2016). Some, like Robert Weinberg (Weinberg, 2014), argue that the field, after much enthusiasm around the promises of reductionist gene-level approaches, entered a form of perplexity in front of the complexity of the disease. Additionally, scholars such as Mina Bissell and her colleagues (Bissell et al., 1982, 2002) have highlighted the limitations of genetic reductionism, asserting that it cannot fully account for how cancer works. In front of those challenges, there has been a growing and pressing recognition that cancer research may require new conceptual and theoretical frameworks[1] (Jassim et al., 2023; Paduch, 2015; Pradeu et al., 2023; Strauss et al., 2021). Evolutionary biology, with its ability to provide a broader perspective on virtually any biological phenomenon, is a promising candidate for developing such frameworks. Life on Earth possesses an historical dimension (Gayon, 1992; Mayr, 1982); therefore, a comprehensive understanding of any biological phenomenon necessitates an examination of its evolutionary trajectory. Evolutionary biology provides precisely this historical narrative, reconstructing and explaining the origins and transformations of biological entities or processes. Evolution does not only chronicle isolated biological stories, but it also connects those stories to draw a larger picture. An illuminating comparison, which will be used at different moments in this dissertation, is with the study of cognition. Approaches to cognition anchored in evolutionary history have shown that behaviours and cognition are far more distributed and varied than initially thought. In addition, it led to the realization that cognition does not require a brain and not even a neural system (Allen, 2017). As far as cancer is concerned, we can similarly hope that adopting an evolutionary approach will allow us to shed a new light both on the diversity of biological realizations of cancers and on conserved features of cancer which play a role in human cancers or general features which help better understand cancer not only as a disease but also as a biological phenomenon. An evolutionary account of cancer should offer dual insights: it should elucidate the history of cancer across species at a microscopic level, while simultaneously clarifying its role at

[1] In the present dissertation, I distinguish between conceptual work, which clarifies definitions and categories, and theoretical work, which generates explanations and predictions, while recognizing that conceptual refinements can directly inform theoretical development.

the macroscopic level. A microevolutionary account of cancer raises the question of whether cancer could act as a selective pressure. If cancer is indeed a strong selective pressure, an evolutionary framework would allow us to enquire about the mechanisms that arose in response to that pressure. At the macroscopic level, it should tell us about the impact that cancer had on life history of most species which are susceptible to develop cancer. Therefore, it is not surprising that researchers have begun to explore cancer through an evolutionary lens for the past thirty years, hoping that this account will generate novel insights to advance the field and maybe address the practical challenges of curing cancer. Many of those researchers have used evolutionary biology to enlarge the scope of analysis and study cancer across the tree of life.

This dissertation aims to demonstrate that the study of the evolution of cancer is a way to reconceptualize cancer. By 'reconceptualizing', we do not mean merely offering an alternative definition of cancer within the same research paradigm but rather initiating a paradigm shift—one that modifies the working hypotheses of research programs in relation to their object of study. A well conducted reconceptualization should lead to new working hypotheses. Other frameworks have been proposed to reconceptualize cancer. For example, some researchers have proposed a reconceptualization of cancer through the lens of developmental biology. Some aspects of tumour development, they argue, resemble processes seen in developing organs, whereas others are closer to tissue remodelling. Based on those observations, they proposed to characterize cancer as a developmental disorder (Da Costa, 2001; Rubin, 1985). While their immediate impact was initially conceptual and theoretical, those ideas anticipated and helped shape several major shifts in cancer research from the 1990s onward including the rise of cancer stem cell theory, the study of epithelial-mesenchymal transition, and the importance of the tumour microenvironment. It also informed new therapeutic approaches, such as differentiation therapy and microenvironment modulation.

In parallel, there are cases of reconceptualization of other biological notions through an evolutionary framework. The case of consciousness is particularly interesting in that the reconceptualization of consciousness through an evolutionary account changed the way consciousness was conceived and studied. The evolutionary work about consciousness generated several working hypotheses and models about its adaptive origin (Arias Domínguez, 2024) and the historical component of evolution helped with the debates about the definition of consciousness and cognition (Allen, 2017; R. Brown et al., 2019; Dennett, 1995; Godfrey-Smith, 2016, 2020; LeDoux & Sorrentino, 2019). Even sceptics regarding the distribution of consciousness among non-human animals, such as Fleming and Michel, base their claims on evolutionary scenarios

(Fleming & Michel, 2025). Others like Adams and Aizawa (2001) oppose overly broad definitions (like the extended mind theory), arguing that true cognition must be constrained by clear, biologically grounded criteria. Their critique implicitly supports the need for an evolutionary framework: if cognitive processes are products of natural selection. Conversely, Baluška and Levin (2016) expand the concept of cognition to include organisms without nervous systems, like plants and unicellular organisms. This perspective implies that cognition could have evolved from basic cellular communication and adaptability, not just neural complexity, forcing us to rethink what constitutes "conscious-like" behaviour. Those contrasting positions illustrate how evolutionary accounts of consciousness or cognition are used to determine the inclusion of exclusion of the characterisation of those phenomena contributing to the shaping of scientific agenda.

These examples demonstrate that efforts to reconceptualize cancer have already been attempted, and that evolution-based reconceptualization of other scientific objects have already proven successful. One of the central claims of this chapter is that evolutionary accounts of cancer hold significant potential to contribute to this kind of conceptual shift.

In this context, it is crucial to distinguish between two applications of evolutionary thinking to cancer. The first is somatic evolution of cancer or *tumour evolution*, which examines the evolution and progression of tumours within an individual. *Tumour evolution* is mainly studied through the clonal evolution model (Nowell, 1976) Clonal evolution emphasizes the expansion and diversification of cell clones within a tumour through genetic mechanisms (in fact, though, any cell-level variations, be they genetic or epigenetic for instance, may contribute to the appearance of these different clones: see (Laplane, 2018, 2024; Laplane & Maley, 2024; Plutynski, 2021). The second is *organismal evolution*, which explores how, through evolution, cancer emerged in multicellular organisms and how it has subsequently influenced the evolutionary trajectories of certain taxa. In essence, somatic evolution focuses on the cellular level, while *organismal evolution* examines the organismal, populational, and species levels. This dissertation will focus primarily on the *organismal evolution* of cancer, a topic that remains largely unexplored by philosophers of biology, as opposed to the topic of tumour evolution, to which a significant number of studies have been devoted (Laplane, 2018, 2024; Laplane & Maley, 2024; Lean & Plutynski, 2016; Plutynski, 2018).

If cancer is to be studied through an evolutionary lens, then it leads to the question “Where can we find cancer across the tree of life?”. Cancer has been known and studied since antiquity (Breasted, 1930), and cases of cancer in non-human organisms have been found in extinct species

dating back up to 240 million years ago (Vincze et al., 2025). One might therefore expect this question to have a straightforward answer. Early comparative studies of cancer emerged in the early 20th century (see chapter 2), and while the idea that cancer might exist throughout most of the tree of life was already suggested at that time (Smith, 1916a, 1916b, 1916c) no systematic empirical exploration of cancer distribution at this scale had been attempted. The endeavour to map cancer distribution comprehensively across the tree of life (in order to address the topic of organismal evolution of cancer) is, in fact, relatively recent, with significant efforts only appearing from 2015 onward. Indeed, in 2015, a theme issue of the *Philosophical Transactions of the Royal Society B* titled "Cancer across life: Peto's paradox and the promise of comparative oncology" was published. Among the contributions, a paper by Athena Aktipis and colleagues on "Cancer across the tree of life: cooperation and cheating in multicellularity" (Aktipis et al., 2015) proved particularly useful and ground-breaking. This article pursued two key objectives. First, it endorsed and expanded a framework that had been suggested by other authors in earlier works (Cairns, 1975, 1981; Nunney, 1999, 2013; Tomlinson, 1997; Weinberg, 1999) but had remained relatively allusive, namely the concept of cancer as an instance of "cheating" within cooperative multicellular organisms, representing a breakdown of the normal rules of multicellularity. Cheating refers to the idea that some individuals in a group might not cooperate while still benefiting from the cooperation of other members of a group, potentially jeopardizing the unity of the group. Second, the paper reviewed cases of tumours found across the tree of life based on the authors' proposed definition of cancer as a breakdown of multicellularity. This article, which has had a very significant impact overall (552 citations according to Google Scholar as of August 2025), served as a critical entry point for us into the topic of the organismal evolution of cancer. Notably, it was characterized by its significant conceptual contribution, offering both a theoretical framework (social evolution theory) and a definition of cancer (cancer as cheating). Since conceptual frameworks and definitions lie at the heart of philosophical inquiry, this work was an indicator that philosophical engagement with the topic was not only possible but, it seems, very much needed to move the field forward. Here I propose to take a close look at the conceptual proposition made by Aktipis and colleagues and see how they address organismal evolution of cancer.

1.2 Multicellularity as a keystone for a new conceptual framework of cancer?

For the purpose of this PhD dissertation, it is essential to dive without further delays into the paper by Aktipis and colleagues, because it provides reflections about organismal evolution of cancer that had previously been scattered throughout the literature but had never been consolidated to this extent in any single work, which overall makes of this paper a landmark source of reflection for the present work. The authors propose examining cancer at the scale of the entire tree of life and stand out as the only authors to offer a distinctly conceptual approach to this topic. They conceive something to be a cancer when rogue cells become cheaters (i.e., stop cooperating with other cells in the body) and do not follow the rules of the "foundation of multicellularity". The concept of cheating was initially formulated by Robert Trivers (Trivers, 1971) who proposed a model to account for the natural selection of reciprocally altruistic behaviours. It falls within the tradition of unveiling how biological systems succeed in selecting for cooperating behaviours and repressing selfish behaviours within and between biological systems. In this regard, even though the concept was not formally defined at the time, the field of social evolution theory—led by William Hamilton—anticipated the notion of cheating (Hamilton, 1964). More broadly, the concept of cheating plays a key role in evolutionary biology's discussions on how new levels of selection emerge through mechanisms that enhance group fitness while reducing individual fitness (Michod, 1999; West et al., 2015). This topic, later formalized as the theory of major evolutionary transitions, is of direct relevance to cancer biologists seeking to conceptualize cancer as a challenge to multicellular organization. In fact, one of the main 'major transitions in evolution' is the shift from unicellularity to multicellularity. Since this transition is thought to be stabilized by mechanisms that prevent individual cells from reproducing selfishly at the expense of the collective, and since cancer can be seen as the breakdown of such mechanisms, it is unsurprising that some cancer biologists endorse this type of vocabulary borrowed from the body of literature cited above. As Samir Okasha writes: "The language of selfishness, cheating and cooperation has its original home in the study of animal social behaviour, but in principle applies whenever there is conflict between levels of selection" (Okasha, 2021 p.2). In fact, several people argued that cancer is a conflict between levels of selection (Aktipis & Maley, 2017; Burt & Trivers, 2006). Even though his work has a more general scope than the work of Aktipis and colleague, Steven Frank who also applied social evolution theory to cancer (Frank, 2007), commented on the cheating hypothesis. Frank explains how Nunney (Nunney, 1999) sees cancer as a result of short-term competition among cells within an organism, and suggests that larger, longer-lived organisms evolve stronger growth controls to balance this internal conflict, a principle that also applies to social dynamics in human groups (Frank, 2003).

The framework developed by Aktipis et al.(2015) is thought to match with the hallmarks of cancer framework (Hanahan & Weinberg, 2000), each hallmark corresponding to one of the following broken foundations:

i) Inhibiting cell proliferation, ii) regulation of cell death, iii) division of labour, iv) resource transport, v) creation and maintenance of the extracellular environment. (see Figure 1)

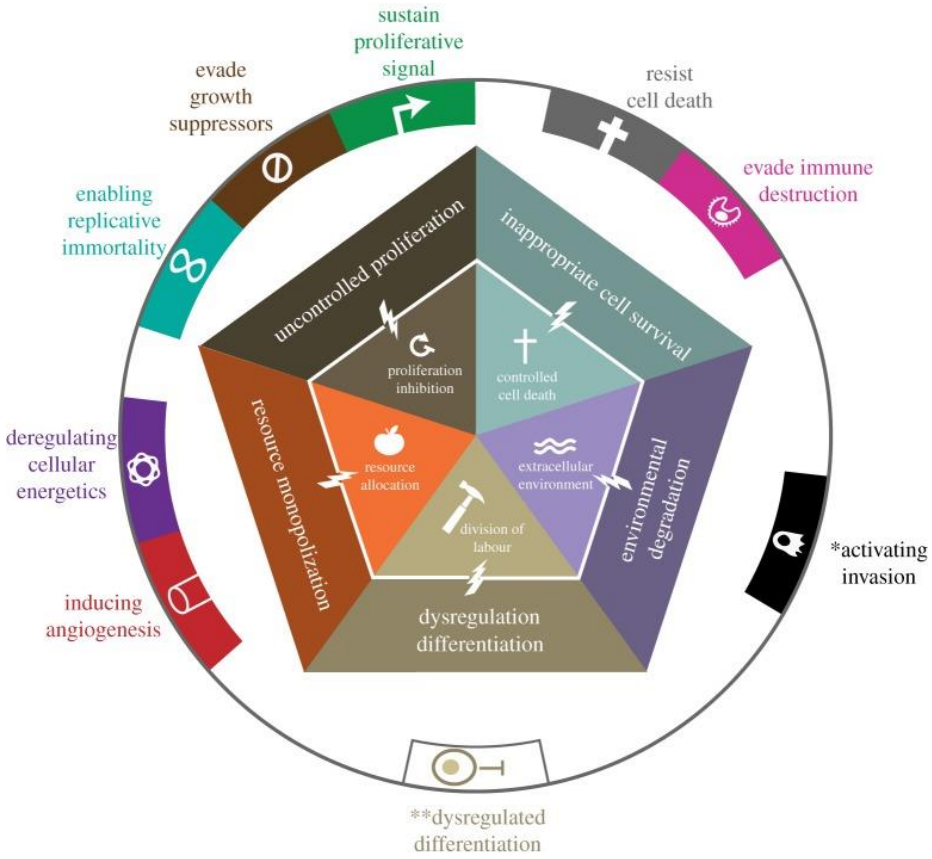


Figure 1. "Foundations of multicellularity and cancer hallmarks" from (Aktipis et al., 2015, p.13).

On the outer circle are the hallmarks of cancer proposed by Hanahan and Weinberg. On the internal side of the pentagon are the proposed foundations of multicellularity by Aktipis et al.; on the external side of the pentagon are the respective breakdowns for each foundation.

The scientific question to which Aktipis et al.'s article tries to answer is: in the tree of life, is cancer found wherever multicellularity appeared? In order to answer that question, Aktipis et al. first needed to provide detailed characterisations of cancer and multicellularity, which required a conceptual approach. According to the authors, multicellularity is characterized by the five foundations mentioned above and cancer as a manifestation of the breakdown of those foundations. This framework has been subsequently explored more thoroughly in Athena Aktipis' book: *The*

Cheating Cell[2] (Aktipis, 2020). Based on their characterisation of cancer, they also propose a phylogenetic tree mapping cancer distribution across taxa (see Figure 2).

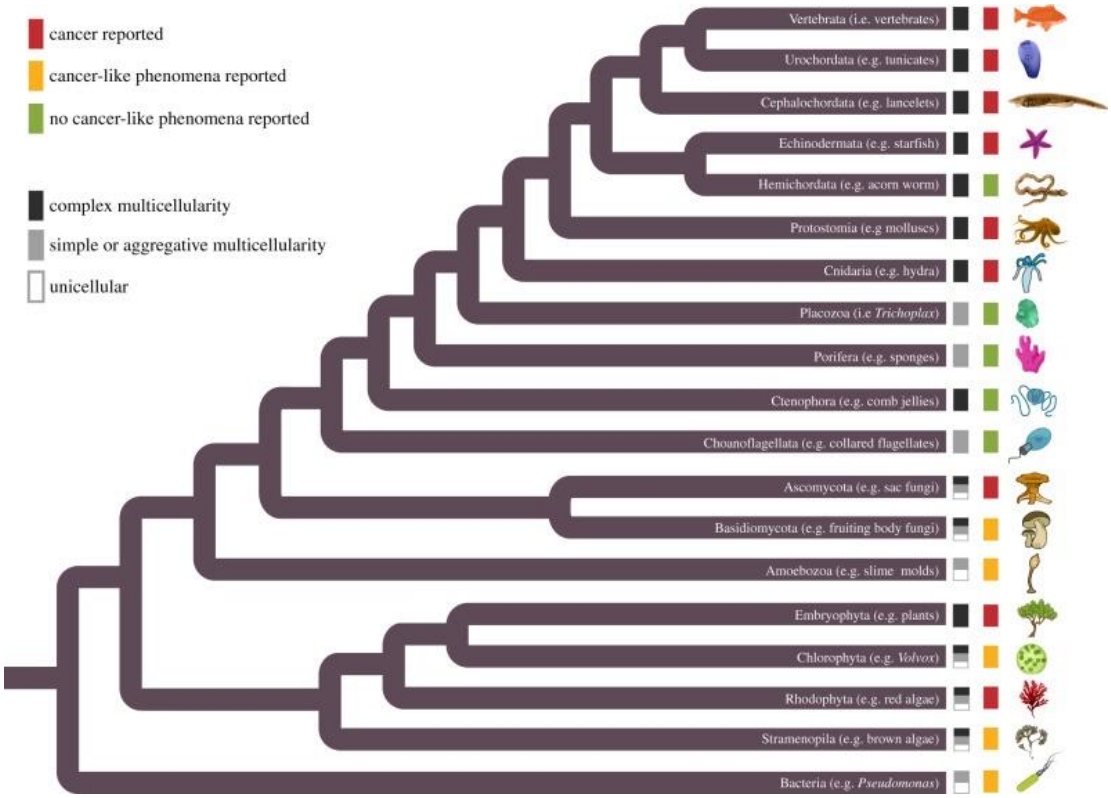


Figure 2. "Cancer across the tree of life" from (Aktipis et al., 2015, p.4).

Black, grey and white boxes indicate the types of multicellularity and green, yellow and red boxes indicate whether cancers were reported in those phyla. The notion of "cancer-like" phenomena remains conceptually vague.

As philosophers of science[3], this type of conceptual approach to scientific problems aligns very well with our objectives. Thus, the fact that scientists—who often downplay the significance of such work—engage in these efforts naturally sparked our interest. It prompted us to consider whether there is potential for meaningful philosophical contributions in this area. However, even though we were interested in the approach, disagreements pushed us to take over such conceptual questions. The central claim of this chapter is that since 2015 and until now, evolutionary approaches to cancer across the tree of life have largely been interpreted through the social

[2] As this work is intended for a broader readership rather than as a scientific publication, both its tone and pedagogical style prioritize accessibility over technical rigor.

[3] Part of the work I did during this PhD was done hand in hand with several other philosophers from my team. The use of "we" refers to this group.

evolution framework proposed by Aktipis et al. I also claim that the comparative approach that I put forward, and which is also shared by Aktipis et al. to some extent is the best approach to address questions about organismal evolution of cancer. Thus, in this chapter I will compare all evolutionary approaches of cancer and demonstrate that the comparative approach is more suited to account for organismal evolution of cancer. Given the limitations of the social evolution framework, I claim that comparative oncology should not depend on it. Here follows what I identified as potential flaws of their framework.

1.2.1 The characterisation of breakdown

First, it is unclear what the term "breakdown" refers to when Aktipis et al. write about "a breakdown of cooperation in each foundation of multicellularity" (p.2). It is not stated whether all of the foundations must be broken in order for a phenomenon to be labelled as "cancer" or if only a subpart of them suffice. If that were the case, it would be necessary to clarify whether some foundations are more important features of cancer than others. For example, one could argue that only three are enough, or maybe that only the inhibition of proliferation and controlled cell death truly matter. Aktipis et. al do not clarify this, despite the fact that it is a crucial point to be able to correctly identify cancer, which is one of the explicit purposes of the article. Additionally, Aktipis et. al. write "*cancer represents a breakdown of this multicellular cooperation*" (p.2). The concept of breakdown, though, can be understood in different ways:

1- Causal: cancer is a consequence of the breakdown of the rules of multicellularity.

2- Correlational: cancer occurs whenever the rules of multicellularity are broken.

3- Instantial: cancer is an instance of a breakdown of multicellularity. [4]

Only a causal understanding would really make the point formulated by the authors, and yet this causal understanding is not made explicit nor is it really demonstrated in their paper. The problem with this ambiguity is that it prevents to correctly identify how multicellularity relates to cancer, which was one of the initial and crucial goals of the article.

1.2.2 Is cancer cheating?

Another issue is that the characterisation of cancer as an instance of cellular cheating is insufficiently demonstrated by Aktipis et al. to be accepted as is. The issue is that framing cancer

[4] I am grateful to Ding Sin Qiao for suggesting this typology.

in this way results in a definition of cancer that is too wide. This is especially the case when it is not clear whether all foundations of multicellularity must be broken at the same time in order for something to be labelled as cheating and thus cancer. As explained earlier, the concept of cheating originates from the framework of social evolution in evolutionary biology, where it has a specific meaning. Yet, as showcased in (Pradeu et al., 2023), "*Many evolutionists [...] reject the idea that cancer cells are strictly speaking 'cheaters'*" (p.1676) mainly because cancer cells are evolutionary dead-ends. For instance, Gardner argues that cancer being an evolutionary dead-end, the advantageous phenotype of cancer cells cannot be transmitted to subsequent generations (Gardner, 2015a, 2015b). Shupak and Liu emphasize that "in true multilevel selection, the lower-level replicator must have the potential to contribute to the genotype or phenotype of the next generation (with respect to the higher-level replicator's generation time) (Shpak & Lu, 2016). On this topic, Okasha (2021) provides a comprehensive mapping of the debates surrounding whether cancer is a *bona fide* case of multilevel selection. He argues that the only possibility for someone to claim that cancer is a case of multilevel selection (and thus that cancer cells are selfish cheaters) is to demonstrate that the evolutionary dead-end argument is false. Okasha explains that thinking of interlevel conflict through the prism of multicellularity leads to a specific way to deny the evolutionary dead-end argument. He writes: "the justification for viewing cancer as a form of selective conflict between cell and organism lies in the fact that such a conflict occurred in the evolutionary past, and that the mechanisms that organisms evolved to control it are precisely those whose breakdown leads to cancer today." (p.17). This can be done, as Okasha shows, in two ways: Leo Buss' theory of germ line sequestration (Buss, 1987) and the atavistic hypothesis[5] (Davies & Lineweaver, 2011) both being highly debated ideas.

On another note, Ghoul et al. (2014) point out that the concept of cheating "has been inconsistently used across a diversity of biological systems". They conclude that even though they do not condemn the use of semantically loaded terminologies (such as "cheat" or "altruism"), the use of those terms should be restricted to cases where they make a real difference on building conceptual overviews of a biological phenomenon. This raises an important question: does labelling cancer as cheating actually make a difference? It could potentially matter in two ways: (1) in the context of debates within the literatures on cheating and major evolutionary transitions, and (2) in shaping our understanding of what cancer is. Regarding the first, it is not the goal of Aktipis et al. to engage directly into these debates; however, they do make a definitive claim about

[5] According to the atavistic theory of cancer, cancer cells revert to an ancestral unicellular lifestyle due to the breakdown of regulatory mechanisms that evolved to suppress such primitive phenotypes.

what constitutes cheating in the context of multicellularity. While the idea that cancer is a form of cheating may seem intuitive, this intuition requires to be put on trial. Aktipis et al. apply the concept of cheating to cancer without sufficient justification, despite longstanding debates over how cheating should be understood in the transition from unicellularity to multicellularity. The consequence is that any phenomenon appearing to violate the foundations of multicellularity is considered as cheating, therefore cancer. It becomes even more permissive when the authors introduce the notion of "cancer-like" phenomena. As a result, their claim does not make a difference for our conceptual understanding of cancer. Thus, labelling cancer as cheating does not advance our understanding of cancer either.

If one wants to redefine cancer through the lens of cheating, the first step must be to rigorously demonstrate how and why cancer qualifies as such. But Aktipis et. al. not only present cancer as an example of cheating in multicellularity but suggest it exemplifies what cheating in multicellularity is like. In response to a complex question that demands careful analysis, they instead presuppose their conclusion, begging the question of whether cancer truly constitutes cheating. The reason they engaged in question-begging is that it aligned with their broader agenda: to argue that, because rule-breaking behaviour—interpreted as cheating—occurs universally at the cellular level, cancer must therefore be ubiquitous.

What readers ultimately retained from the article was not so much the claim that cancer constitutes cheating, but rather the striking assertion that cancer is found everywhere in plants, fungi, algae, and beyond – without necessarily realizing that as a matter of fact this strong and bold conclusion only reflects the choice of their stipulative definition of cancer as a form of cheating.

To summarize, the very legitimacy of the concept of cheating as well as the idea that cancer is an instance of cheating have been highly debated in the literature and Aktipis et al.'s approach does not fully address those debates.

1.2.3 An all-encompassing framework

Finally, on a more general note, the broadness of the framework is fundamentally what undermines it the most. The framework attempts to integrate three distinct concepts: multicellularity, evolutionary cheating, and the hallmarks of cancer. This creates a catch-all structure in which virtually any form of cellular proliferative anomaly fits. Consequently, it is unsurprising that cancer appears in most multicellular groups. The incorporation of the hallmarks of cancer is similarly unsurprising, as the authors note: *"The current paradigm for understanding*

the characteristics of cancer is the 'hallmarks of cancer' approach" (p.13). Their strategy seems aimed at convincing their audience of the validity of their framework by demonstrating its alignment with the mainstream paradigm. However, I argue that this move is counterproductive, as it forces an artificial linkage between two frameworks conceived for entirely different purposes. As the authors themselves remind the reader, the hallmarks of cancer framework *"was developed from observations of common phenotypes across human cancers."* In contrast, Aktipis et al.'s framework extensively broadens its scope to encompass all multicellular groups. In other words, they want to articulate a framework that emerged from empirical observations of only one species to a framework that they forged to apply to all multicellular species. To this end, they incorporated the concept of evolutionary cheating, which remains conceptually vague, as anything that breaks the presumed foundations of multicellularity can be classified as cheating. This vagueness is further compounded by the introduction of the concept of "cancer-like phenomena," which blurs the boundaries of the authors' definition of cancer even further. Due to the broad nature of Aktipis et al. 's definition, it encompasses a wide variety of features, making it challenging to arbitrate over edge cases with clarity and precision.

Does this mean that the entire enterprise is in vain? I don't believe so. In fact, I see this work as paving the way for conceptually oriented scholars, including philosophers and theoretical biologists, to engage with these issues.

1.3 Why organismal evolution of cancer needs more conceptual work

As argued by (Pradeu et al., 2023) the literature on multicellularity helps us to better understand cancer, specifically why and how it occurs. Aktipis et al. rightly saw the benefits of nurturing cancer biology with this knowledge on multicellularity, but their proposition wasn't conceptually strong enough to create major changes in how cancer is conceived in general —for example by more traditional oncologists or clinicians. I do not claim that every conceptual breakthrough about cancer must be directed toward such audiences; contributions can be valuable in their own right and can significantly benefit non-traditional cancer research. However, the extent to which a conceptual proposition influences broader audiences, including that of traditional oncology, can serve as a useful metric for evaluating its strength and impact, especially after a few years (in the present case, 10 years for the paper by Aktipis et al.).

One critical issue of this work lies in their attempt to explain how cancer occurs, which heavily depends on the definitions they provide. The strength of any definition rests on its evidential basis; a definition must be well-supported to inspire trust and facilitate its application.

From their definition, Aktipis et al. can address the question: “Where can we find cancer in the tree of life?” This question serves as the foundation for other inquiries related to the organismal evolution of cancer. If we fail to take this initial step of identifying the species in which cancer occurs, all subsequent attempts to tackle these related questions risk failure or, at the very least, becoming significantly more challenging to navigate. More broadly, the validity of any scientific work is constrained by the quality of its underlying conceptual framework. As I demonstrated above, while the framing of cancer as a breakdown of multicellularity offers a promising perspective—particularly at a time when cancer research requires new conceptual and theoretical frameworks—it falls short of the robustness needed to withstand scrutiny or to be reliably applied across a wide range of contexts. The issues with their definition undermine the ability of their framework to serve as a dependable conceptual tool, limiting its potential impact on how cancer is understood and studied. For those reasons, and without taking away the merit of their work, it is important for rigorous conceptual work to be done on this topic in order to fulfil what was initially attempted by Aktipis et al. I argue that while the initiative was well-intentioned, the conceptual foundation was flawed due to its excessive broadness and ambition. By attempting to use the entirety of multicellular life as the basis for their framework, Aktipis et al. set themselves a difficult task. To support this claim, I first provided a demonstration based on conceptual reasoning. However, this issue can also be evaluated on empirical grounds, as I will explore in chapter 2, where I examine the scientific output this framework has generated within comparative oncology.

Grounding cancer in the framework of multicellularity offers one approach to considering it in evolutionary terms. While adopting such a broad scale is not strictly necessary to study the organismal evolution of cancer, this is precisely the approach taken by the authors, which is why their work captured our attention. Reading the paper by Aktipis et al. revealed the potential for conceptual analysis and philosophical inquiry in this area. The question of where cancer can be found across the tree of life is merely the tip of an iceberg, deeply submerged in evolutionary waters. This dissertation aims to provide an exhaustive state-of-the-art review on the topic of organismal evolution of cancer and its related questions—particularly by exploring the field of comparative oncology.

1.4 Why philosophy can help to shed light on cancer evolution

A question remains: why, and in what ways, can philosophers contribute meaningfully to these scientific problems? The ways in which philosophers can contribute to these points closely align with the effort to develop a new framework for understanding cancer, as discussed thus far. First and foremost, philosophers have a well-known and recognized expertise in defining concepts. Second, philosophy of science has a long tradition of critically assessing the relevance of various accounts, approaches, methods and frameworks in science (Laplane et al., 2019).

Has philosophy already attempted to work on the issue of cancer evolution? The philosophy of biology has long been synonymous with the philosophy of evolutionary biology (Gayon, 2009; Pradeu, 2017). Philosophers, traditionally drawn to theories and concepts, have found evolutionary biology to be an ideal domain for applying their conceptual tools. Moreover, examining biological phenomena on evolutionary scales offers one way to adopt a broad, integrative perspective. Evolutionary biology provides insights into the spatial and temporal relationships among species, traits, populations, and selective mechanisms, offering what could be described as a "big picture view" of the biological world. Such a perspective exemplifies the contribution that philosophy can make to science. Cancer, due to its complexity and impact on human life, represents a major phenomenon of interest within the biomedical sciences. Given its significance and the historical inclination of philosophers to engage with evolutionary accounts, it would be unsurprising if some philosophers of biology have attempted to explore cancer through an evolutionary lens. Yet, despite its importance, cancer remains a relatively underexplored topic in the philosophy of science compared to areas like biological individuality or the units of selection (for an overview, see (Godfrey-Smith, 2014)). In recent years, the amount of philosophical work on evolutionary accounts of cancer has grown significantly. Philosophers specializing in cancer, such as Lucie Laplane, Anya Plutynski and, to a lesser extent, Marta Bertolaso, have contributed extensively to this area. Laplane has explored topics such as clonal evolution (Germain & Laplane, 2017; Laplane, 2018; Laplane & Maley, 2024; Sepich-Poore et al., 2022) while Plutynski, in addition to dedicating a chapter of her book to evolutionary accounts (Plutynski, 2018), has authored several articles framed within an evolutionary perspective (Lean & Plutynski, 2016; Plutynski, 2021, 2023). Additionally, Samir Okasha has examined levels of selection as they apply to cancer, further enriching the philosophical discourse on the topic (Okasha, 2021). Yet, except Samir Okasha's article, all of these productions focused on what I labelled earlier as tumour evolution. But no philosopher has ever chosen to study organismal evolution of cancer in depth. One might speculate that cancer's complexity or specificity makes it less amenable to evolutionary

accounts, which could explain the relative scarcity of philosophical work in this area. However, evolutionary biology has proven to be a productive framework for studying cancer, addressing questions such as: When did cancer first emerge? Has cancer shaped the evolutionary trajectories of certain taxa? Can cancer itself be considered an evolutionary process? or, have some species evolved specific cancer resistance traits? There may not be a specific reason for the lack of philosophers addressing this topic, but that is ultimately beside the point. What truly matters is the identification of gaps in the scientific literature and the aim of this dissertation is to address them. In this sense, I am breaking new ground as the first philosopher to specifically and thoroughly engage with the subject of cancer organismal evolution – a topic which is here contrasted with, and complementary to, the excellent and influential work done by philosophers on tumour evolution.

Before diving into how science attempted to address those topics, I must justify why studying cancer from an evolutionary perspective is worth the investment.

1.5 What can we expect from the study of organismal evolution of cancer?

I have outlined how certain biologists have engaged with questions concerning the evolution of cancer. Drawing on conceptual work examining cancer across the tree of life, I demonstrated how conceptual and theoretical approaches can shed light on this topic. While I have established that philosophers are particularly well-equipped for such conceptual work, the relevance of the study of organismal evolution of cancer remains to be demonstrated. To do so, it is valuable to highlight the areas where the study of cancer's organismal evolution offers the most meaningful insights. In particular, I aim to show that the study of cancer's organismal evolution provides a promising avenue for reconceptualizing the disease—a strategy that should prove useful given the current state of cancer research. In this section, I spell out the different questions to which the study of organismal evolution of cancer can answer.

1.5.1 Why is there cancer?

This question can be understood in two ways. We can either wonder why there is cancer in the first place, that is, the origin of the phenomenon. Or we can wonder why there is still cancer, i.e., why evolution did not get rid of it. If the first formulation is obvious in that it expresses a causal concern about a biological phenomenon, the second one needs to be explained. Why would

evolution ever get rid of cancer? It would imply that cancer is a problem great enough to weigh on the fitness of certain species. However, if cancer is most of the time lethal in mammals, it is less obvious for other taxa (e.g., if plants were indeed getting cancers, it clearly has a less dramatic fitness effect) (J. H. Doonan & Sablowski, 2010; J. Doonan & Hunt, 1996). Overall, does cancer affect fitness enough to constitute a genuine selective pressure? In fact, it could be argued that predation or pathogens are likely to affect fitness much more than cancer, and thus selection would be more driven by those threats than by a set of processes that happen most of the time after the reproductive age. The upcoming chapter will cover more about this topic.

1.5.2 What is cancer?

The question of redefining cancer is obviously one the main features of a reconceptualization as it targets the core component of a conceptual framework: the definition of the object of interest. How is evolution able to help clarify and refine our understanding of cancer? What is commonly referred to as 'cancer' today is a group of very different phenomena which seem to share certain features. Several important cancer research organisations define cancer as a group of several diseases. On the American Cancer Society's website, one can read: "Cancer is more than just one disease. Cancer is a group of diseases characterized by the uncontrolled proliferation of cells." (*What Is Cancer?* s. d.). Looking at cancer through evolution allows us to address the question of what cancer is at a larger scale. It leads us to enquire about the phenomenon beyond where we have been used to find it, that is to say, the human species. One can be compelled to wonder what cancer is because they started to study it through an evolutionary lens and because they need to determine in the first place what phenomenon they should expect to find in diverse species. On the contrary, one could adopt this evolutionary lens because one is looking for a way to reframe cancer's definition and think evolution provides a framework general enough for the task. In any case, cancer being a concept that we usually see through an anthropocentric prism, the further we go from human cases, the more important the question "what does cancer look like in other species?" becomes. Instead of building a definition based on only one species (typically, *homo sapiens*) one can build a definition based on several groups. If we multiply the number of instances of an X, knowing that not all instances display X in the same way, we will have a more exhaustive understanding of X because we will see which characteristics appear the most and how cancer can be multiply realized. However, circularity can appear when we don't have any criterion to sort which characteristic belongs to X or not. I argue that, in the case of cancer in several species, adopting an evolutionary framework can help to avoid this

circularity. In fact, evolution allows us to draw a genealogy of traits or phenomena starting from a very general scale (the tree of life) to a more local scale (a species) and it enables us to draw the evolutionary history of a trait or a species. Therefore, characteristics of cancer that turn out to be ancient and conserved are more likely to be core characteristics of cancer.

Some like Aktipis et al. (2015) adopt what we might call a 'foundationalist' approach, that is defining an object *a priori*. An alternative approach to defining cancer involves allowing definition and investigation to co-evolve—that is, adopting provisional definitions that guide inquiry as understanding develops. These working definitions remain in place until knowledge stabilizes and a clearer grasp of the phenomenon emerges. In any case, some form of pre-definition is necessary for any investigative approach, as it enables to identify relevant phenomena. While such pre-definitions are partly arbitrary, they are essential to investigate questions about where cancer occurs and when it first emerged. Once the initial work is done, we should have a more refined definition that corresponds to what we learned from studying cancer across the tree of life. Chapter 3 will explore further the question of definition by proposing a universal definition of cancer.

1.5.3 Where can we find cancer across the tree of life?

Which species have or don't have cancer? This question is not necessarily an evolutionary one, but it is the evolutionary account which will give any answer the explanatory power that we are entitled to expect whenever we ask the question. Studying cancer across the tree of life implies and effort to gain knowledge about the distribution of cancer across taxa but not only that. What is interesting as well are the proportions in which different species get cancer. In fact, species which have a lot of cancers or species that have very few cancers or not at all are both of interest for different reasons (Vincze et al., 2022). This question is crucial because it serves as a starting point for several other questions regarding organismal evolution of cancer. The more data we have on this topic, the better we will be able to answer those other questions. For instance, determining which species don't have a lot of cancer is of direct interest to understand the mechanisms behind this resistance and how that might inform efforts to prevent cancer. Similarly, it goes hand in hand with the problem of the definition of cancer since the pre-definition should allow us to determine what constitutes a *bona fide* case of cancer, and once we identified the species that have cancer and what it looks like in those species, it helps transform this pre-definition into a more consistent definition.

Another related question is: which species develop which types of cancer? Indeed, the features of cancer can be different depending on the organism in which it arises. For example, while humans frequently develop carcinomas that originate in epithelial tissues, dogs are particularly prone to sarcomas (Schiffman & Breen, 2015). More generally, this begs the question of whether cancer is universal or not.

1.5.4 When did cancer appear in evolution?

As hinted above, evolution is useful to localize a biological phenomenon in time. Cancer in humans is pervasive nowadays and has become the second cause of death worldwide. Recent data show that the general rate of cancer is increasing (Bray et al., 2024) especially early-onset cancers (J. Zhao et al., 2023). But it does not mean, naturally, that cancer is a recent disease in *homo sapiens.* In fact, there are historical traces of cancer in humans dating back to antiquity (Aufderheide, 2003; David & Zimmerman, 2010). And, more generally, there is little doubt that cancer existed long before the appearance of humans on Earth. Only an evolutionary account can tell us when — in the history of evolution — cancer appeared. It can also tell us if it appeared several times. To know when cancer appeared is valuable to monitor the impact that it had on the life-history of many species. For instance, if cancer appeared every time multicellularity appeared (i.e., 26 times (Grosberg & Strathmann, 2007; Niklas & Newman, 2013)) as some authors argue, then it is likely that it has impacted the life-history of all multicellular taxa. If that is true, it could explain certain features or similarities between species.

1.5.5 Cancer from the point of view of life-history trade-off

It can be hypothesised—and is presupposed by many in comparative oncology — that cancer has acted as a significant selective pressure for some multicellular organisms, potentially shaping their life-history trajectories. As Kokko et al. (2017) suggest, this perspective may also shed light on broader evolutionary principles. In particular, it offers insights into why humans and other animals remain vulnerable to cancer despite millions of years of evolution. In this sense, it contributes to answer to the question: why does cancer still exist?

If, as Aktipis and colleagues argue, cancer can affect most of the living world, then we are challenged to view cancer not merely as a disease, but as a potential evolutionary force in its own right. If this is the case, it suggests that many different strategies for resisting or tolerating cancer have evolved throughout the tree of life—offering promising avenues for translational research.

Elephants, for example, appear to have evolved extra copies of the tumour suppressor gene *TP53*, which may contribute to their surprisingly low cancer rates given their large size and long lifespans—a phenomenon sometimes called "Peto's Paradox." On the other hand, small, short-lived animals like mice may invest less in such costly defences. This type of reasoning is at the heart of contemporary comparative oncology.

Additionally, evolution through natural selection is commonly summarized as selecting for traits which enhance the fitness of its bearer. While this is mainly true, some traits can have detrimental effect and not every trait selected for is always beneficial at all stages of life. Organisms face trade-offs between investing energy in growth, reproduction (including parental care), and maintenance. The mechanisms that suppress cancer—such as cell cycle checkpoints, DNA repair, and apoptosis—require metabolic investment and may limit other functions like rapid growth or early reproduction. This sets up potential evolutionary trade-offs: species with high extrinsic mortality may evolve to prioritize fast reproduction over longevity and cancer suppression, whereas species with longer lifespans might evolve more robust cancer defences. For examples of evolutionary forces potentially driving cancer risk in organisms, see (Harris et al., 2017) and for a modelization of trade-offs between investment in cancer defence and investment in reproductive competitiveness, see (Boddy et al., 2015).

Earlier, I discussed why cancer has not been eradicated by evolution. One explanation is that cancer is the harmful by-product of otherwise adaptive traits—an example of evolutionary trade-offs. For example, cancer could be the outcome of *antagonistic pleiotropy* i.e. traits that are advantageous early in life (e.g., rapid cell proliferation or regenerative capacity) may increase cancer risk later on.

1.5.6 What are the mechanisms evolved by some species which protect them from cancer?

When considering the question of where cancer occurs, it becomes apparent that its distribution varies significantly both within and between species. This observed disparity prompts investigation into its underlying causes which can be approached from both mechanistic and evolutionary perspectives. From an evolutionary standpoint, the focus is on identifying the selective pressures that may have led to the presence—or absence—of certain biological mechanisms in particular lineages, thereby influencing a species' susceptibility to cancer. Mechanistically, the search involves examining specific traits or biological mechanisms. For example, within mammals, some species like mice exhibit high cancer rates while others like

elephants show low rates of cancer. This contrast suggests the presence of mechanisms (like DNA damage regulation or control over cell proliferation). While some researchers refer to these as "anti-cancer mechanisms" it is preferable to remain careful with this terminology as it implies these traits evolved directly in response to cancer mortality. In this work, I will use the more cautious expression "mechanisms with cancer-protective effects".

The three last questions help to identify the impact of cancer on the evolutionary trajectory of organisms which might have had to deal with cancer. Answering these questions helps to narrate the evolutionary history of cancer (by highlighting which species had to deal with cancer and what the potential solutions selected as a response were).
Yet it must be mentioned that many hypotheses rely on the assumption that cancer was in fact selected against, which as we will explain is chapter 2 is a widely shared but undemonstrated presupposition.

1.5.7 The therapeutic question

I mentioned earlier that cancer is a highly lethal disease. Naturally the research effort in oncology is targeted towards cancer therapy. Therefore, one can expect from an evolutionary account to be able to make a useful contribution from a therapeutic point of view. The question can be asked in different ways. One can ask if an evolutionary account can help us identify reliable solutions to cancer evolved by certain species (the mechanisms with cancer protective effects mentioned above). Alternatively, one can ask whether an evolutionary account could enlighten us about cancer's unknown features which would change the way we go about cancer prevention or treatment. To illustrate the possibility of this second option, one can think about how exploring the immune system outside of the usual range we were used to provided major insights in certain mechanisms which happened to also be present in humans. For example, identification of Toll receptors in Drosophila marked the beginning of a vast exploration into Toll-like receptors and, more broadly, pattern-recognition receptors in mammals (Pradeu, 2020). This question is of importance in that most non-evolutionary cancer biologists and medical oncologists will evaluate the utility of the study of organismal evolution of cancer through its ability to make useful contribution to cancer therapy. As will be seen in chapter 2, the field of comparative oncology will go from questioning the possibility of making useful contributions into an affirmation. In fact, the argument has been made that since cancer is a challenge faced by multicellular organisms and given that not all multicellular organisms develop cancer—and that significant disparities in cancer

susceptibility exist among those that do—it suggests that evolutionary solutions to cancer have emerged in certain lineages (Caulin & Maley, 2011; Kokko et al., 2017; Nunney et al., 2015) This assumption has yet to be demonstrated, and I will return to it in chapter 2 to show that it is a presumption that many comparatists have yet remains unaddressed for now.

Overall, every question that I mention above can be tied together in order to draw an evolutionary history of cancer. Each question, if solved, unveils a piece of a puzzle which is still incomplete in the most part. As discussed earlier, much energy and time have been invested to cure cancer, less to explore its history. The history of a biological object provides context and this context even if valuable by itself could actually serve a translational purpose. eAs a consequence, I argue that what one can expect from the study of organismal evolution of cancer is an evolutionary history of cancer. Then what one can do with this history depends on the goals of the people studying it. Chapter 2 will explore how the field of comparative oncology is using parts of this story for translational purposes.

Yet, one might ask whether exploring cancer in an evolutionary way is beneficial. Couldn't an evolutionary account introduce even more heterogeneity to an already complex phenomenon? One counterargument is that, on the contrary, an evolutionary account reduces heterogeneity by shifting the focus from outcomes to processes. Instead of focusing on a multiplicity of features which depends on the context in which cancer emerges (the organism, the type of tissue, the organ, …) one could focus on what is common to all instances of cancer across species. By adopting a process-oriented definition of cancer and expanding the range of study to the entire tree of life, the shared biological processes become more apparent, thus offering a unifying framework. In this sense, evolution provides a solution to the heterogeneity of cancer as it brings order to the disorder. This is, among other things, what is at stake in the adoption of an evolutionary account of cancer: to be able to reconceptualize cancer within a contextual and historical framework which should benefit the field. In chapter 3, I will argue that the concept of affordance, rooted in evolutionary thinking, is particularly useful for making sense of this diversity.

In this section, I listed most questions that an evolutionary account of cancer could answer. Within this evolutionary account, there are several approaches that one can take to answer those questions. I argue that the quality of each approach is determined by the number of questions it can successfully take address and the degree of thoroughness with which it can do it.

1.6 What is the best approach to the organismal evolution of cancer?

Understanding cancer through the lens of organismal evolution requires us to engage with a range of approaches, each offering unique perspectives but also facing important limitations. If Aktipis et. al's approach attempted to apply social evolution theories to cancer, I demonstrated in section 1.1 of this chapter that this approach is unreliable, so we will not further consider it here. In this section, I explore the two other approaches to the evolutionary study of cancer at the organismal level: the study of fossil records and comparative biology applied to cancer (see Figure 3.). These approaches differ in the kind of data they use, the questions they are equipped to address, and their conceptual and translational relevance. To justify one's investment in comparative oncology, one must first demonstrate how the other two approaches are far more limited.

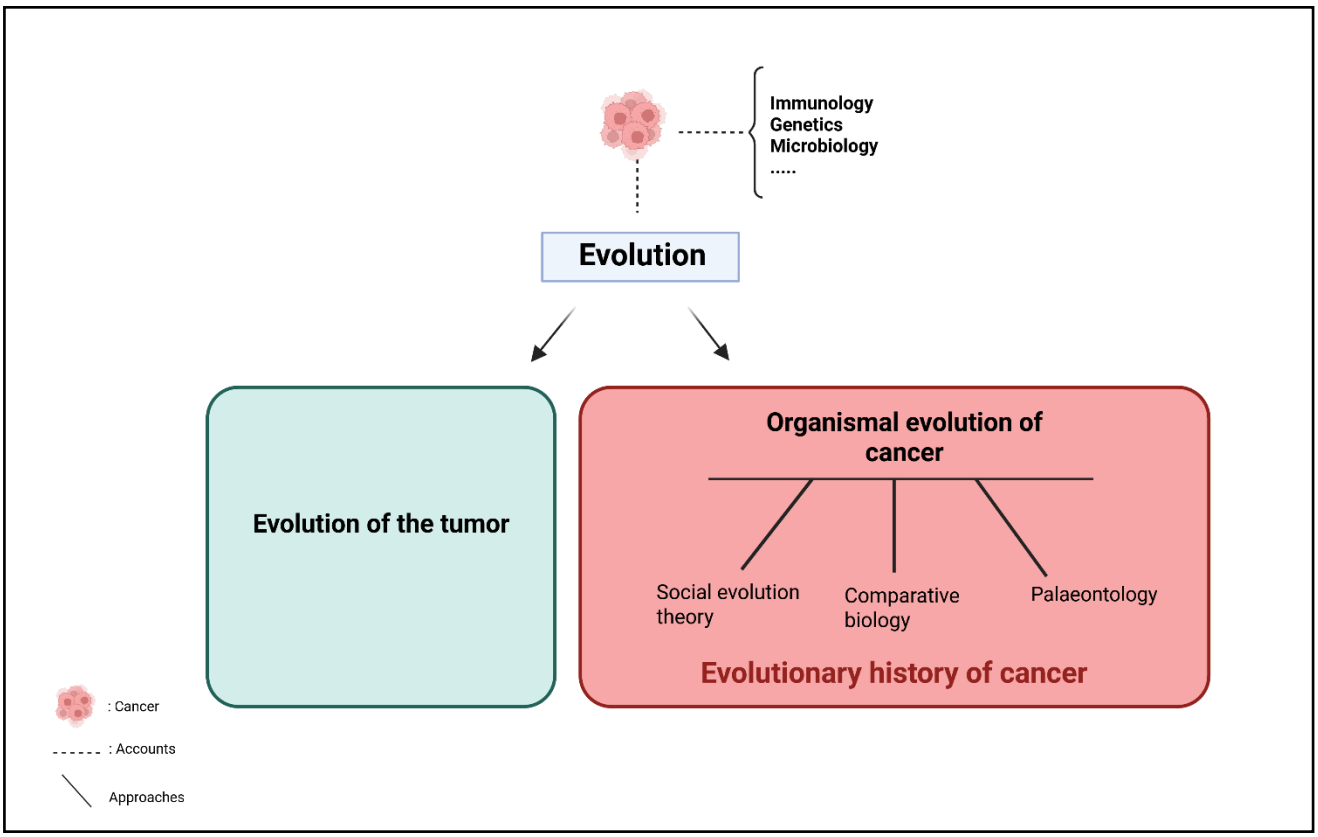


Figure 3. **How to study organismal evolution of cancer?**

There are different accounts of cancer depending on the branch of biology that studies it. My dissertation focuses on evolutionary accounts. There are two kinds of evolutionary accounts: evolution of the tumour which apply evolutionary concepts to the process of cancer; and organismal evolution which tries to understand cancer as a phenomenon with an evolutionary history which happened across the tree of life. In this work I mention three approaches of organismal evolution of cancer. Applying tools of the social evolution theory, adopting comparative methods (comparative biology), and studying fossil records (palaeontology).

1.6.1 The study of fossil records is too limited

Evidence of tumours has been discovered in extinct species such as dinosaurs (Rothschild et al., 1999, 2003) giant sloths (Barbosa et al., 2021) and ancestral turtles (Haridy et al., 2019), where bones dating from 240 million years ago are suspected to be affected by osteosarcoma, with morphology characteristic of metastatic bone tumours (Rothschild et al., 1999). In a study with around 10,000 bone specimens, it was shown that cancer affected dinosaurs from the hadrosauridae family, with evidence of metastases in the Edmontosaurus genus (Rothschild et al., 2003). More closely related to modern humans, evidence of cancer has been repeatedly found in extinct species of homini living 1.8–1.6 million years (Odes et al., 2016).

What insights do fossils provide about cancer? Fossil evidence demonstrates that cancer is not a modern disease but has existed across various species long before the industrial era. However, the information derived from fossils is very limited. Skeletal remains represent only a fraction of the cancers that may have afflicted ancient organisms, and their rarity restricts our understanding of the diversity of cancer types in the past. Among dinosaurs, the majority of identified cancers have been found in Hadrosauridae, or duck-billed dinosaurs. While hypotheses have been proposed to explain this prevalence—such as their conifer-based diet or endothermic metabolism—the limited data render these theories speculative and difficult to assess. Additionally, most cancers do not affect bone, the tissue most commonly preserved during fossilization, or they manifest in the bone's interior, complicating their detection in ancient specimens. Further challenges arise in differentiating bone tumours from other pathological conditions or injuries in fossils, with malignancies in soft tissues being nearly impossible to identify from skeletal remains alone (Capasso, 2005). Moreover, bone tumours often increase skeletal fragility, reducing the likelihood of their preservation in the fossil record (Barbosa et al., 2021).

If we look at the types of questions to which this approach can answer, its limits appear even clearer (see Table 1.1). Studying fossils cannot provide any insights about why there is cancer or any of the life-history questions. The only question to which it can theoretically answer is *when did cancer appear?* But even in that context, the scarcity of data makes it difficult to rely fully on this approach to identify when cancer appeared.

In conclusion, approaching *organismal evolution* of cancer through fossils is not a very relevant and fruitful strategy because of the lack of data and their unreliability. Most of the questions that were listed earlier simply cannot be answered by looking at fossil remains.

1.6.2 A promising candidate: comparative biology of cancer

As explained by Adrian Currie in his book *Comparative Thinking in Biology* (2021), "comparative thinking" is ubiquitous in biology. Since evolution produces variation, the living world is filled with a diversity of different lifestyles and traits. But it is also filled with similarities, including sometimes, in species which are phylogenetically very distant (e.g., birds and bats both evolved wings, which are analogies). Comparative biology takes interest in those similarities or dissimilarities. As mentioned by Currie (Currie, 2021), Paul Griffiths (Griffiths, 2007) rightly described homology as "a phenomenon, that is, an explanation-demanding pattern". But what do comparatists compare and explain?

Currie defines what he identifies as a major "style of thinking" (p.9) in biology: "comparative thinking". This style of thinking manifests itself through "the comparative method" which Currie describes to be a set of techniques which "typically involves combining phylogenetic information with methods of organizing data about variation across taxa (such as principal component analysis)" (*idem*). Currie explains that most, but not all biological comparisons involve ancestry and adaptation. This focus on the evolutionary history of organisms leads to the predominance of concepts such as homology[6] and homoplasy[7]. The traditional comparative method is deeply rooted in this evolutionary tradition; therefore one might expect cancer to be a perfect item for this type of approach, yet I will show that it is not the case.

How does the comparative method apply to cancer? There are two major elements about cancer that differ from the majority of objects usually studied via the comparative method. First, the comparative method is about traits. Currie illustrates his point via comparative functional morphology, using the comparison of skulls and dental features across various mammalian species as an illustrative example. Traits like dental disposition can be easily identified and compared with one another. Cancer is not a trait; it is a disease characterised by a series of features (e.g., the dysregulation of the control on cell proliferation). Thus, it is not possible to directly compare two instances of cellular proliferation without first determining whether they belong to the same category (in that case, the category *cancer*). Second, traits like dental disposition have been

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[6] "Two traits in two lineages are homologous just when that trait is also held by the most recent common ancestor of those lineages." (Currie, 2021) p.15.

[7] Two similar traits in two lineages are homoplastic just when that trait is not present in the lineages' common ancestor." (*Ibid*) p.16.

selected. They are the product of an adaptive history. This is why the comparative methods often study the ancestry of the species which display the trait of interest and also why phylogenetic analysis became a pervasive method of comparison in biology from the 1970's (Cornwallis & Griffin, 2024; Cornwell & Nakagawa, 2017). In terms of explanatory power, comparing an item (in most cases, a trait) which has been selected is more powerful than comparing an item which wasn't selected since identifying why the item was selected gives a strong explanation to its presence (or absence in species which didn't share the same selective pressures). For this reason, I argue that a comparative approach to cancer cannot be directly aligned with conventional comparative methods used in biology. It does not mean that no comparative approach is possible; it means that an efficient comparative approach of cancer will be different from those we usually found in biology since its explanatory power will not rely on the ancestry and adaptive history of cancer.

When considered across the tree of life, cancer becomes a difficult object to explore because of the anthropocentric prism usually adopted by researchers. In fact, in biomedical sciences, the vast majority of cancer research is directed towards clinical outcomes. The results of this anthropocentrism will be explored in chapter 2 and 3, but we can already highlight the problem of definition that it poses. In fact, looking at cancer across the tree of life raises the question of what should be looked for. It could be argued that comparative approaches of cancer should be similar to comparative psychology of consciousness since consciousness is also a complex phenomenon composed of various features. In fact, the study of consciousness has known a similar fate as the study of cancer in that consciousness was also at first defined based on the case of *homo sapiens*. The rationale for this anthropocentrism, however, differs from the cancer case. It does not stem from the fact that consciousness is a health issue requiring resolution, but rather from the assumption that consciousness was a unique characteristic exclusive to the human species (LeDoux & Sorrentino, 2019). However, even if consciousness, just as cancer is a complex phenomenon, it was selected for multiple times in multiple groups (Godfrey-Smith, 2016, 2020). Therefore, if there are some similarities between comparative oncology and comparative psychology of consciousness (mainly because people searching for cancer across the tree of life face similar conceptual struggles as people searching for consciousness across the tree of life), the fact that cancer might not have been selected prevent us from using consciousness as a model to build a comparative study of cancer.

Relatedly, it should be mentioned here that the question "where cancer can be found across the tree of life" doesn't necessarily imply an evolution-based answer. Yet, I argue that a comparative

approach of cancer will be enhanced if it can bring in some evolutionary insights. Comparisons of cancer rates across species (like (Compton et al., 2025; Vincze et al., 2022)) provide useful data which might help isolating key components of cancer causality; but invoking evolutionary tools to make sense of those disparities isn't negligible. The different methods and objects of comparison will manifest different degrees of explanatory power. This leads to our next question: what can be compared in cancer?

- The conditions of cancer: This involves identifying the necessary and sufficient conditions for cancer to arise.
- The causes of cancer: This includes both intrinsic factors (e.g., genetic mutations, cellular dysregulation) and extrinsic factors (e.g., environmental exposures, pathogens). Comparing these causes across species or populations can reveal shared or unique pathways contributing to cancer development.
- Cellular mechanisms: Examining the molecular and cellular processes underlying cancer—such as proliferation, apoptosis evasion, and metastasis—can highlight conserved or divergent mechanisms across different organisms.
- Mechanisms with cancer-protective effects: Many organisms have evolved strategies which happen to suppress or mitigate cancer, such as enhanced DNA repair, cellular senescence, or immune surveillance. It is up to debate whether those mechanisms were specifically selected for defending organisms against cancer or not. Some in comparative oncology argue that comparing these mechanisms can provide insights into solutions to cancer that may have translational potential for human medicine. As will be seen in the next chapter, there are two approaches to cancer comparatism, an evolutionary one and an ecological one. The ecological one will focus on environmental pressures which explains certain features of cancer. In the case of the mechanisms with cancer-protective effects, this distinction can be witnessed in the kind of explanation provided for such traits (e.g., the cellular regulatory mechanism displayed by the naked mole rat is likely due to its subterranean way of life).
- The distribution across groups: As mentioned above, investigating patterns of cancer prevalence and diversity across taxa can uncover evolutionary tendencies, such as why certain groups are more resistant or susceptible to cancer.

Table 1.

List of the questions that the study of organismal evolution of cancer can address and which approach can effectively address them or not.

Question	Can the comparative approach answer to this question?	Can the fossil approach answer this question?	Can the social evolution approach answer this question?
Why is there cancer?	Yes	No	Yes
What is cancer?	Yes	No	Maybe
When did cancer appear?	Yes	Yes	Yes
Which species get cancer?	Yes	Partially but yes	If conjoined with a comparative approach yes
Was cancer a selective pressure?	Yes	No	No
Is cancer the outcome of an evolutionary trade-off	Yes	No	No
What are the "anti-cancer" mechanisms evolved by some species?	Yes	No	No
Which cancer types start at what life stage?	Yes	No	No
What is the impact of cancer in terms of mortality?	Yes	No	No
Which species have which types of cancer?	Yes	Maybe	No

How to cure/prevent cancer?	To be determined	No	No

Chapter 2 will explore how the field of comparative oncology is actively trying to endorse the comparative approach to study cancer. I will draw the historical development of the field, demonstrating that scientific interest in neoplasms occurring in non-mammalian organisms dates back to the early 1900s, when biologists first began documenting and analysing tumour phenomena in diverse species. Relying on this historical context I will propose a critique of the field with a focus on the adequacy of its goals with its methods and objects of enquiry

Chapter 2

Chapter 2 : What is comparative oncology?

Comparative oncology broadly defined studies cancer in non-human organisms. Some might say that it does so across the whole tree of life, but most of the field focuses on vertebrates and more specifically on mammals. Comparative oncology brings together biologists with vastly different methods and perspectives. It includes scientists studying tumour mechanisms in single species (such as the naked mole-rat or the *Drosophila* fly), those developing evolutionary models of cancer risk, and others examining cancer incidence within or across taxa. Moreover, not all researchers demonstrate the same degree of comparativeness in their work. As noted earlier, some focus on a small number of species, while others — such as Aktipis et al. (2015)— compare tumours across different taxa. Consequently, it is not immediately clear why all these researchers should be grouped under a single banner. If one adopts a permissive account, any work studying cancer in non-human organisms could be considered part of comparative oncology. However, this approach is unsatisfactory, as one can study cancer in non-human species without making any comparisons. For instance, during my review of the literature, I encountered numerous references to articles that merely described unusual cases of cell proliferation, occasionally questioning their nature. Therefore, I propose a more restrictive definition of comparative oncology: the study of cancer (or presumed cancer) in non-human organisms, conducted with the intent that insights from one case may shed light on others. This definition encompasses works that compare cases of cancer across different taxa or within specific groups (e.g., mammals in Seluanov et al., 2018). It also includes conceptual or theoretical contributions aimed at examining cancer across taxa, such as (Aktipis et al., 2015). However, it excludes purely descriptive studies of cancers, even when such descriptions involve individuals from different taxonomic groups.

Interestingly, if one asks when the field of comparative oncology started, at least two different answers are usually proposed. On the one hand, the more experimentally oriented comparatists will answer that it started recently in the early 2010's with a series of papers from Vera Gorbunova and Andrei Seluanov's laboratory on cancer resistance in two species of mole-rats (Seluanov et al., 2008, 2009). This spark of interest in non-human species then got ignited by a group of researchers exploring Peto's Paradox and cancer across the tree of life (Aktipis et al., 2015; C. A. Aktipis et al., 2013; Caulin & Maley, 2011). On the other hand, for the more theoretically oriented, it started with Peto's Paradox itself (Peto, 2016)[8] since it opened up a new

[8] I was not able to find the original article from 1977; the 2016 version adds a few comments from Peto.

path of enquiry, consisting in explaining the differences in cancer distribution across species. Obviously, the answers differ in function of the interests of the community. However, it is interesting to note that, to this day, no historical work has been done about comparative oncology, which suggests that this difference in answers might also have to do with a lack of knowledge concerning the historical roots of the field.

This second chapter will explore the whole history of the field from its "prehistory" which starts around 1910 to its current state which is characterized by the adoption of an explicit evolutionary framework. As will be seen, the history of the field is marked by two fundamental ruptures: 1) the identification of patterns in cancer susceptibility thanks to Peto's Paradox and 2) the quick 'evolutionarization' of the field (i.e., its adoption of a clear evolutionary perspective) which started as late as 2010. Both phenomena constitute a rupture with the way comparative oncology was practiced during its 'prehistory,' namely a natural history approach, the latter being a consequence of the former. When I talk about 'natural history' I endorse Bartholomew's view who wrote that "a naturalist, studies the world by observing plants and animals directly" (Bartholomew, 1986). In other words, it is a scientific activity which favours the collection and description of biological items over experimental approaches. We will see that, by becoming more evolutionary, comparative oncology offers a genuine way of studying organismal evolution of cancer. As foreshadowed in chapter 1, the question that should drive this enquiry about comparative oncology is: how is the field answering to questions regarding organismal evolution?

2.1 The many faces of comparative oncology

Comparative oncology is quite heterogeneous in its methods as well as in its goals. There are several reasons which can lead to studying cancer in non-human species. I identified five. As was hinted at above, not all goals are fulfilled to the same degree, and I will demonstrate in this chapter that the intensity to which some goals are achieved varies through time.

1. Production of knowledge on cancer across the tree of life

As every scientific discipline, comparative oncology aims at extending our knowledge on a specific topic: cancer. The difference with traditional oncology is the extension of the framework, which in the case of comparative oncology goes way beyond cancer in humans. Every article we have explored for this thesis participates to some extent in this goal. Reviews like (Aktipis et al., 2015; Albuquerque et al., 2018; Nunney et al., 2015) are good material to get a general picture of

what is known about cancer across the tree of life. From an historical perspective, this goal has been fulfilled since the very beginning of comparative oncology either through the conceptual work of Smith (Smith, 1916a, 1916b, 1916c, 1925) or the experimental work on *Drosophila* from Stark (Stark, 1918, 1919b, 1919a).

2- Exploration of cancer under an evolutionary framework

Considering the purpose of this PhD dissertation, this goal is the most interesting for us. The first piece of evolutionary thinking can be found in Peto's article in which he suggests that "a final understanding of carcinoma induction must tie in the low event rates which humans suffer with evolutionary considerations" (Peto, 2016). If this observation was commented upon in the following decades, it is only in 2008 that the Seluanov and Gorbunova lab (Seluanov et al., 2007, 2008, 2009) started to test Peto's Paradox experimentally by experimenting on cancer resistant rodents. This spark of an evolutionary framework was ignited a few years later thanks to (Caulin & Maley, 2011) who clearly identified a whole avenue of research around which comparative oncology could structure itself: exploring Peto's Paradox's implications. It led to the different types of studies that I listed above. If everything participates to some extent in improving our evolutionary understanding of cancer, there are certain studies that specifically use tools and concepts from evolutionary biology. The work of Hanna Kokko for instance uses mathematical modelling to test hypotheses on the effects of cancer on life-history and the effects of certain traits on the robustness against cancer (Erten et al., 2020; Erten & Kokko, 2020, 2021; Kokko & Hochberg, 2015). This life-history approach also possesses a strong predictive power since, once some hypotheses about the relationship between certain traits and cancer susceptibility are confirmed, they can be tested in other species. For instance, (Erten et al., 2020) demonstrated that some cancer risk/body size ratio did not meet the standards of Peto's Paradox in birds; therefore they found a hypothesis to model. This hypothesis was that cancer robustness in birds was the consequence of an evolutionary lag that slowed down the loss of the cancer resistant legacy that birds inherited from their massive dinosaurs' ancestors. This hypothesis can be and is currently being explored in other species. Thanks to the data gathered on cancer risks in mammals (Vincze et al., 2022) and, beyond, in vertebrates (Compton et al., 2025), comparatists can select resistant species and examine if their ancestors had big body masses and long lifespans.

3- Finding new therapeutical and prevention avenues (translational goal)

This third goal is crucial because it is via this one that comparative oncology expects to set up a dialogue with traditional oncology. Comparative oncology makes a promise that can be

witnessed through many recent papers. The field repeatedly says that it can make valuable contributions to cancer treatment (Abegglen et al., 2015a; Compton et al., 2025; Gorbunova et al., 2014; Kokko et al., 2017; Sepp & Giraudeau, 2023). I found two different arguments used to justify the feasibility of this promise. First, by shedding light on the mechanisms behind cancer resistance, comparative oncology hopes either to identify unsuspected pathways that could be used to cure cancer in humans or to be able to identify mechanisms that could be mimicked or implemented in humans (like using elephant P53 proteins to induce more efficient apoptosis (Abegglen et al., 2023; Preston et al., 2023)). The second argument is well summarized by Hannah Kokko who wrote: "Finally, most cancer treatments focus on the somatic level—killing a particular type of cell or disrupting a particular pathway. However, by identifying the organizing principles of cancer across taxa, in addition to learning something general about biology and cancer evolution, we may also identify new treatments and prevention approaches." (Kokko et al., 2017, p.252). Yet, as I will explain in more detail, even if to this day comparative oncology has highlighted several "anticancer mechanisms", none of them, to the best of my knowledge, were successfully applied to produce efficient therapies.

4- Understanding how cancer impacts wildlife (conservation biology's goal)

If cancer is highly lethal for humans, it is also extremely lethal for other organisms. One goal of comparative oncology is to monitor how cancer affects those organisms and to assess if the high rate in certain species is the consequence of human activities (Giraudeau et al., 2018; Hochberg & Noble, 2017). A paradigmatic example is the work of (Mikaelian et al., 1999) and (Martineau et al., 2002), who suggest that the high number of tumour rates found in beluga whales in the St. Lawrence estuary could be the consequence of pollution of the water via aluminium smouldering. The assessment of the oncogenic impact of human activities on wildlife is important for ethical, public policy, and conservation biology reasons. It is worth noting that, in wildlife, cancer is far less lethal than predation or infectious diseases, which makes this goal of comparatively lower immediate relevance which likely explains the comparatively smaller volume of articles addressing this aspect.

5- Producing knowledge on cancer in pet animals and exotic species in order to better cure them (veterinary goal)

Finally, because cancer is also an issue for pets, part of comparative oncology aims at studying and curing cancer in domestic animals. Some domesticated species are particularly vulnerable to cancer, including dogs. Because of breeding, many races of dogs became vulnerable to specific types of cancers (Schiffman & Breen, 2015). It was also demonstrated that domesticated individuals show higher cancer rates than wild ones from the same species – typically in birds (Effron et al., 1977).

While all five goals constitute comparative oncology, this chapter will mainly focus on the second one, namely the exploration of cancer under an evolutionary framework. I will explain how comparative oncology does (and sometime doesn't) fulfill this evolutionary goal.

2.2 The dawn of comparative oncology: natural history of cancer[9]

In the broad and permissive definition mentioned below, comparative oncology has deep historical roots. When did the idea of exploring instances of cancer in a diversity of animals and, as we will see, plants as well, start? How did the biologists involved in that work conceptualize cancer? Most crucially, what were their goals, and what did they manage to achieve? Characterizing this history is crucial to this research project for a number of reasons, including the need to avoid historical oversimplifications and the opportunity for today's comparative oncology to learn from its own past and perhaps avoid making again errors that have been made in the past. Besides, only a deep scrutiny of the field can allow us to evaluate its ability to properly study organismal evolution of cancer.

If comparative oncology is a small but productive scientific field, it has not always been as productive. In the beginning of the 20th century there was no theoretical grounding behind comparative oncology. Some stumbled upon tumours and chose to describe them, sometimes in non-mammalian organisms. Some attempted to experimentally induce tumours in order to test hypotheses about the origin and nature of cancer.

[9] For conciseness reasons we will assume for now that every mention of tumours present in our review of the literature is a cancer. However, as we will see, the very nature of those tumours is a matter of interest and dispute for many biologists at that time.

2.2.1 Smith's universal definition of cancer: do all cancers come from parasites?

The prehistory of comparative oncology is characterized by a collection of data on tumours in non-human organisms. Before 1900, many biologists assumed that the most fundamental processes of life could be generalized to a wide range of species (Logan, 2002). Most of the comparative work about cancer from the first half of the 20th century seems to endorse this view. The earliest occurrences of comparative work about cancer date back to the beginning of the 20th century. The first author who was studying cancer (or at least what he strongly believed was cancer) in non-human organisms and drawing parallels between tumours of plants and tumours of humans is Erwin Frink Smith. Smith was an American phytopathologist who studied how bacteria induce diseases in plants. It is not entirely clear if Smith believed that the tumour of a Paris daisy he described in his 1907 paper was cancer or not (Smith & Townsend, 1907). He only suggested that this discovery might be of interest to "animal pathologists, especially to those interested in determining the origin of cancerous growths" (p.671). However, from 1916 onwards, Smith was very assertive about his theory. The paper from 1916 is the one in which he provides the most detailed descriptions of his experiments with crown galls. At this point (April 1916), he did not yet argue that what he was describing was cancer as such. However, two months later, he clearly argued for this view and for his own theory of cancer. Smith wanted to demonstrate that the tumours he had found in plants, which were caused by a bacterium he labelled "*Bacterium tumefaciens*", shared most of the characteristics with human tumours. To justify his theory, Smith needed to give a definition of human cancers in the first place, which illustrates one of the main underlying ideas that will be defended in the present PhD dissertation, namely that every serious comparative work requires at least a minimal definitional work. Thus, Smith wrote that what he defines as cancers or malignant tumours are characterized by:

i- Unlimited and uncontrolled cellular proliferation;

ii- Presence of "destructive" secondary tumours;

iii- "Destructive action on surrounding tissues, which are crushed, absorbed or poisoned" (E. F. Smith, 1916a).

The conclusion he drew from his observations of crown galls is that, if the phenomena are so similar, it must be because the cause is the same. Since Smith found out that galls were caused by parasitic bacteria, he concluded that all cancers were certainly induced by parasitic activity. His theory stems not only from his observations but also from an account of the overarching principles

behind living things. This account is best described in his paper from June 1916 published in *Proceeding of the National Academy of Sciences of the United States of America* (Smith, 1916a). According to him, both animals and plants possess an equilibrium manifested by "*growth* and *decay*". He believed that the two processes would be driven by a single substance, the 'protoplasm'. Thus, Smith was endorsing the protoplasmic view of life. Protoplasm is a concept widely used in 19th century biology and which could have several meanings. The way Smith uses it resembles how Thomas Huxley defines it in his 1868 essay: "On the Physical Basis of Life" (Huxley, 1868). Huxley defined protoplasm as "the physical basis of life" which consisted in a "semi-fluid matter" shared by every living system and that were the most basic constituent of living things. Smith, in the same spirit as Huxley, believed that plants and animals were similar in almost every possible way. This is also very much reminiscent of Claude Bernard's thesis in *Lectures on the phenomena of life common to animals and plants* (Bernard, 1878). Smith saw similarities in reproduction, division of labour between the tissues, regeneration, nutrition, and respiration. Assuming this high degree of correspondence, it appears obvious to him that "discoveries relative to the fundamental cell-mechanics of animals apply equally to plants, and vice versa" (Smith, 1916a). Thus, since he believed to have proven that plants can have cancer in the form of crown gall and that this pathology is due to a parasite, then it is likely that all other cancers are also caused by parasites.

"If I can convince you that crown gall is a cancer, you will then be ready to admit that what requires a schizomycete[10] for its production in the plant is not likely to be produced in any very different way in man and animals." (Smith, 1916b)

In an article from 1936 (Levine, 1936), Michael Levine who already commented on Smith's work (Levin & Levine, 1918) explains that not only the parasitic theory was very popular at the time but also that none of the other advocates received as much attention as Smith and his collaborators.

"Perhaps no explanation of the cause of this disease has had greater vogue than the parasitic theory. [...] None of the advocates of the parasitic theory of cancer ever received the attention that was paid the work of Smith." (Levine, 1936)

[10] Schizomycete is an archaic term once used in biology to describe bacteria when they were still thought to belong to the plant kingdom.

Smith's approach offers an excellent and very early illustration of one major option for comparative oncologists, which can be dubbed the 'causal unification' of cancer across taxa, and which consists in asserting that what makes cancer similar in different species is that cancer in such species results from similar causes. This approach is distinct from, and complementary to, what can be called the 'phenomenal unification' of cancer across taxa, which says that what makes cancer similar in different species are the observable similarities between cancerous tumours (as opposed to the underlying causal processes that explain their appearance). Although the two claims can perfectly co-exist, they are conceptually distinct, as it is entirely possible to claim that cancers across species have different causes while sharing major observable features, and/or that cancers across species have similar causes despite the fact that they look different in the end. Another major topic of interest in early comparative oncology is the scope of comparison, that is to say how far across taxa should we be looking at cancer. Smith is the first author, to our knowledge, to elaborate a theory of cancer that advocates for adopting the broadest scope possible.

I must note that this work was not only a precursor to future works in comparative oncology but also a criticism about how cancer research was conducted at the time. Smith criticizes the fact that too many cancer specialists are pathologists and not biologists while the problem of cancer is "peculiarly and preeminently a biological problem" (Smith, 1916b). The issue with the pathologists, according to Smith, is that they are satisfied with purely descriptive characterisations of cancer. For Smith, remaining at the descriptive level was the zenith of pessimism and he preferred to believe that we could someday provide explanatory knowledge about cancer. Despite what the provocative and polemic tone of his papers suggested, Smith wasn't arguing that pathological and morphological studies were to be stopped. Instead, he claimed that the study of cancer had been overtaken by pathologists who only provided morphological and hence descriptive knowledge about tumours. Therefore, there should be more space for explanatory work which is better provided by more biology-oriented spirits. As we will see, this idea that decentring cancer research from medicine and the clinic should be helpful to provide explanatory knowledge about cancer still roams through contemporary comparative oncology. It is expressed when people argue for the benefits of exploring cancer across the tree of life and casting evolutionary lights on this phenomenon (Aktipis et al., 2015; Kokko et al., 2017; Nunney et al., 2015).

2.2.2 *Drosophila melanogaster*: breaking new grounds in comparative oncology

Aside from Smith's work on plants, early comparative oncology also included observations and experiments on insects, especially on *Drosophila melanogaster*. *Drosophila*'s case is quite peculiar since it is the only species of insect that became a model organism for cancer study. If most of the comparative work belonged to the natural history category, that wasn't the case of what was done in *Drosophila*. In fact, the fruit fly bears a central role in genetics and developmental biology; however, its role in oncology is less appreciated. In the early 20th century, while many biologists used this model to investigate the role of chromosomes, the link between those and phenotypes (Bridges, 1916; Morgan, 1910), as well as association between gene mutation rate and X-ray exposure (Muller, 1928), Mary Stark was investigating a different question, namely: "Do chromosomes carry the cause of cancer?" (Villegas, 2019). In 1918, she published the first paper which described tumours in *Drosophila* larvae. Her early studies, inspired by Bridges' identification of the lethal 7 strain in *Drosophila* (Bridges, 1916), marked the beginning of fruit fly cancer research. Stark noted black tumour-like masses in fly larvae, and carried out meticulous analyses of tumour size, number, and spread, pioneering surgical attempts to remove them. Stark even attempted to transplant these masses, laying the groundwork for in vivo tumour experimentation despite limited success due to high surgery-induced lethality. Later, Stark extended her research to include non-lethal, benign tumours, documenting the spread of primary tumours via blood to secondary sites, forming metastases. (Stark, 1919b).

Building on Stark's findings, Ira T. Wilson, with input from early geneticist Fernandus Payne, identified hereditary tumour traits in flies, suggesting that multiple genes contribute to tumour development—an early glimpse into oncogene cooperation (Wilson, 1924). Subsequent studies in the mid-20th century explored the genetic and environmental factors influencing tumour formation, recognizing variables like temperature, diet, and population density as contributors to tumorigenesis (Russell, 1940).

Elizabeth Gateff further propelled the field by identifying the first tumour suppressor gene, l(2)gl, in the 1960s, with lgl mutations leading to malignant, invasive tumours. Collaborating with Howard Schneiderman, Gateff established that lgl behaves as a tumour suppressor (Gateff & Schneiderman, 1969, 1974), and she perfected a method for serial tumour transplantation, which remains influential today. Gateff's advocacy for fly models in cancer research emphasized Drosophila's utility in both genetic and emerging epigenetic studies. Her approach set a precedent for using genetic tools to explore tumour biology and contributed to a broader understanding of

how genes control cellular growth and differentiation. This work has had lasting impacts, informing research into cancer and other diseases linked to genetic mutations. What made Gateff's contribution distinctive was her decision to treat *Drosophila* not only as a convenient genetic system, but as a genuine cancer model at a time when this was far from obvious. She demonstrated that tumours in flies were not merely abnormal growths but exhibited the same invasive behaviour and transplantable properties as cancers in vertebrates, thus legitimizing invertebrate systems as serious model for oncology. Moreover, her insistence on the genetic origins of tumour formation anticipated the later gene-centred view of oncology, decades before tumour suppressors and oncogenes were widely accepted in human cancer research. In this sense, Gateff's originality lay both in her methodological creativity and in her pioneering role in the study of cancer genetics.

The description that I provided above depicts *drosophila* as a successful model for cancer genetic, therefore it might be expected that no debates exist on the nature of those tumours. Yet, the question of whether flies' tumours are *bona fide* cases of cancer is still debated today.

2.3 Looking back on fifty years of exploration: do invertebrates have cancer?

The exploration of cancer across different species carried on for decades. The 1950's do not constitute a rupture with how comparative oncology had been done in the past but a phase of retrospection with two reviews about tumours in invertebrates, one by Berta Scharrer and Margaret Szabó Lochhead in 1950, and the other by Violaine and Jacques Wautier in 1953 and 1955. This type of work is precious because it is rare; even more so since these papers give a central role to the comparative method. Indeed, they compare the tumours of several dozens of species. Moreover, those reviews make references to specific works on the same topic from the first half of the 20th century. Since I could not retrieve those works despite intense effort, the reviews act as a testimony of probably unknown comparative studies of tumours in invertebrates. In this section, I shall dive into those reviews in order to understand what type of questions invertebrates' tumours were raising in the 1950s but also what type of research had been conducted in the first half of the 20th century in comparative oncology.

2.3.1 Scharrer and Lockhead, 1950

The review, which Scharrer and Lockhead (1950) labelled as a "critical survey of the present status of the problem of invertebrate tumours", starts with a general reflection on what cancer is, what invertebrates can tell us about it, and a general line of thought on why invertebrates are poorly studied in comparative oncology. It is a keystone work in our historical exploration of the roots of comparative oncology for several reasons: 1) it focuses on a branch of the tree of life that remains underexplored today, 2) it argues for a comparative approach to cancer, and 3) the authors produced a definition of cancer. This type of conceptual work is important because it is rare in the field of comparative oncology. That being said, we will see in what follows that even this work proves insufficient, as several blind spots remain, especially on the difference between the terms "tumour", "cancer", and "neoplasm".

The introduction to this review was composed of two elements: an argument for why we should study cancer in invertebrates and more generally across the tree of life; and an explanation of why the study of cancer in this branch is arduous. The authors share with Smith several pieces of argument. Firstly, just as the botanist, they argue that since cancer is a disease of abnormal tissue growth it should be apprehended in a biological way. That is to say, it should "include representatives from all groups of living organisms" (p.403). There are two claims in this argument: one is that cancer is a problem of tissue growth, and that tissue growth is a biological phenomenon therefore biological approaches are best suited to deal with it. They write: "the study of tissue growth, normal and abnormal, constitutes the central problem of tumour research which thus becomes essentially a biological problem." (p.403). The second claim is that a biological approach is characterised by an increase in the scale of analysis. This increasing of scale is done by expanding the number of species in which we study cancer. They write: "When approached from this broader point of view, an analysis of tumorous growth should include representatives from all groups of living organisms." (*Ibid*). Contrary to Smith, the authors are not clear whether biological approaches are better in terms of explanatory power or not. However, the authors seem to imply that a wide-range comparison is one of the main tools of biology (as opposed to medical approaches which usually aren't comparative). An element they brought up as an argument in favour of comparatism is the benefits in understanding tumorigenesis thanks to plant studies. They write: "It has been recognized that the study of plant tumours yields significant results." (*Ibid*). It is not clear what results they have in mind since they do not cite any specific paper. I believe that they might have been referring to Smith's work which was known and had been commented upon

some years before this article (De Ropp, 1951; Levine, 1936). The similarities in the justification for comparative study of cancer reinforce our intuition that they did read Smith's work.

The main reason they invoke for the lack of work about cancer in invertebrates is the common idea at the time that this branch of the tree of life simply could not develop tumours. They refer to a comprehensive work on the question from a German researcher named Engel[11],[12] who argued that invertebrates cannot develop tumours. According to the authors, Engel wrote that invertebrates cannot have tumours because of differences in nervous and vascular systems on the one hand and because of the embryonical nature of invertebrates' cells on the other hand. There are traces of the embryonic theory of tumour growth in Smith's work as well. However, as they argued and demonstrated throughout their paper, numerous instances of tumours have been described in many different species of invertebrates from the previous 50 years. Therefore, the question was – and, in fact, still is: are those tumours "true cancers"? Engel's arguments were outdated according to Scharrer and Lochhead and the question of whether invertebrates can have cancer needed to be brought on the table anew.

However, the study of cancer is made difficult by two issues. First, as they rightfully explained, the data on tumours in invertebrate species are controversial. That is to say, it is one thing to identify an unusual and abnormal cellular growth, but it is another to label them as cancer with certainty. The reason for this is that two kinds of knowledge are required in order to correctly identify a cancer in invertebrates. On the one hand, one must know the physiology of the species to identify what counts as abnormal or not; on the other hand, one needs to have a comprehensive knowledge of what cancer is to deliver proper descriptions of the neoplastic phenomena. The problem according to Scharrer and Lochhead is that the cases wrote upon in their review either came from pathologists who don't know invertebrate biology sufficiently well or from zoologists who are "inexperienced in the diagnosis of tumour growth" (p.403). The second difficulty is that the terminology for tumours has been developed for mammalian pathology. The analogy between mammals and invertebrates' physiology wasn't clear enough at the time, according to the authors, to reliably use this taxonomy[13]. In this, they depart from Smith's idea that analogies between all branches of the tree of life can be drawn because of a common principle. Yet I should be careful

[11] ENGEL, C. S. Warum erkranken wirbellose Tiere nicht an Krebs? Ztschr. f. Krebsforsch., 32:531-43, 1930

[12] Unfortunately, we were not able to find this document which might not have been translated in English or French anyway

[13] We argue that it is still an open and unexplored question

in my interpretation. They did not mean that no analogies could be drawn at all but rather that it remained to be demonstrated that there are enough common processes between invertebrates and vertebrates to make such analogies. The consequence of this thesis is that it might be necessary to develop a new taxonomy, specific to invertebrates, to correctly identify and describe cancer in this group.

Are all the phenomena reviewed in this paper genuine cases of 'cancer'? Certainly not, according to the authors. They are categorical on this matter. Some of the diagnoses were doubtful and some researchers having studied them were explicitly hesitant to label the phenomena as "cancer". At the beginning of the paper, they affirm that every growth that is the outcome of an inflammation, cases of hypertrophy of cells due to parasites and insects mycetomes[14] were not true tumours. They write that "in a number of instances a variety of formations in invertebrates which cannot be considered as true tumours have been designated as such." (p.404) and that in most cases those are the "result of what is called an "inflammatory process."" (*Ibid*). Later, they write that "Evidence of the true neoplastic character of such alleged tumours is often scanty; reasonably convincing signs of malignancy are found only in a small number of reported cases"[15]. Does it mean that "true" means cancerous? Is 'neoplasm' synonymous with 'tumour'? Why are tumours that arise from an inflammatory process not considered genuine tumours? Also, if signs of malignancy is one of the criteria for cancer, then are benign tumours found in invertebrates not cancers? Since very few cases explored in this review constitute genuine cases of metastatic phenomena, it confirms that, if one remains consistent, most of this review is in fact ultimately not about 'cancer'.

The authors could have argued that this type of work is of value for cancer even though the cases described here were not cancers. However, they did not, which is a problem in so far as it is not obvious what everything that was learned in this review truly teaches us about cancer. This question is still relevant today as invertebrates are understudied. Some comparative oncologists justify this focus on vertebrates arguing that since tumours in invertebrates are not cancers, they don't see the relevance of studying them if the goal is to understand or even cure cancer. In Chapter 3 I will address this point in more details.

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[14] "Structures caused by intracellular symbiotes", *idem.*

[15] P.404, 407 & 410.

Scharrer and Lochhead's work will be cited a few years later by Harshbarger and Taylor (Harshbarger & Taylor, 1968) Harshbarger being a productive researcher on invertebrate's tumours. Furthermore, it is worth mentioning that it is also present in more contemporary works in comparative oncology. For example, (Albuquerque et al., 2018; Robert, 2010) use it as a reference for instances of tumours in *Drosophila melanogaster* while (Aktipis et al., 2015) refer to it in a larger context, as a review on tumours in invertebrates (which they believe are cancers).

2.3.2 Wautier & Wautier, 1953-1955 (J. Wautier & Wautier, 1955; V. Wautier & Wautier, 1953)

The authors open their discussion with a brief characterisation of cancer: "cancer is characterized by anarchic proliferation of certain cells which slowly invade the organism until it dies." Several conceptions of cancer are mentioned throughout the article. They endorse Gustave Roussy's distinction between benign and malign tumours, the latter being what is truly known as 'cancer' (Roussy, 1939). According to Roussy, cancers are "tissular reactions in which exaggerated proliferation of elements plays a predominant role and leads to persistent tissue overproduction with a tendency to increase". (Roussy, 1939) as quoted by (V. Wautier & Wautier, 1953). According to Roussy, the distinction between benign and malignant tumours is as follows: benign tumours are characterised by "their tendency to develop at its birthplace and its slow growth", while malignancy is defined by "quick, invasive and destructive proliferation". At this occasion, we learn that Roussy, in an article with Wolf, published a list of species in which cancer had been observed (Roussy & Wolf, 1920). This list was mostly about mammals but still contained mentions of cancers in birds, reptiles, amphibians, fishes and *Drosophila*. According to the Wautiers, Roussy and Wolf regarded cancers in other animals as "essentially identical to cancers in humans" (Roussy & Wolf, 1920). This means that such a major figure of human oncology as Gustave Roussy recognized the nature of cancer in non-human organisms, even in non-mammal species.

According to those definitions and considering the data exposed in the first section of their article, only tumours in Sipunculus, *Pygarea*, mayflies and *Drosophila* in the case of the lethal 7 lineages would count as malign. They also seem to imply that lethality is a major criterion to define cancer. Yet it doesn't mean that the cellular proliferation found in those species should be considered as genuine cancers for sure. They are, instead, what looks the most like cancers. The conclusion they draw from the data they gathered in the paper is that, among all the works cited in

which invertebrates were put into contact with carcinogenic substances, a total of five different outcomes could be observed:

1) No reaction was observed.
2) An inflammatory reaction was observed.
3) A stimulation of the normal proliferative process was observed.
4) An abnormal proliferation was observed.
5) No reaction was observed, but offspring were affected.

Among all the abnormal proliferations, none of them were convincing enough to qualify as cancers according to the Wautier couple. Their conclusion is the following: "based on the current state of our knowledge, it does not seem possible for us to affirm that they are truly of a cancerous nature." (J. Wautier & Wautier, 1955) (p.121).

An important question that is sketched by the Wautier couple is the evolutionary explanation behind the scarcity of tumours in invertebrates. They ask: “Should the predisposition to cancer be considered as an evolutionary trait?”. This question that will later become one of the core topics of contemporary comparative oncology remains unanswered in this paper as the Wautier seemed reluctant to take too strong a position on a question that probably could not be answered at the time. Yet, they interestingly tied this question to the question of the loss of regenerative abilities, wondering if the evolutionary explanation should be found in some sort of consequence of high regenerative abilities.[16] However, they remained consistent in their cautious attitude, suggesting that this relation between cancer and regeneration doesn’t always seem to hold. As they pointed out, fishes, amphibians and reptiles display regenerative power of different degrees and yet most attempts to make them develop cancer failed. *Polychatea* which possess high regeneration powers do manifest tumours. In conclusion, whether or not cancer is a consequence of high regenerative abilities was a topic that still needed to be addressed in detail. The link between cancer and regeneration is a major topic of interest today (Oviedo & Beane, 2009).

Overall, what can be learned from those reviews?

1) They constitute an attempt to reconsider the possibility of cancer in invertebrates and to break with the general assumption that this group was unable to develop *bona fide* cancers.

[16] This evolutionary analysis is an exception at that time.

2) We learn that tumours in invertebrates were extensively more studied than it is the case today. Experimental works in this group extend over a wide range of species, especially insects.
3) According to the regular taxonomies of cancer, almost none of the cases of cellular proliferation mentioned in those reviews (which spans over several decades) can be considered as a *bona fide* cancer. Those that could be considered cancers should still be taken with caution.
4) However, the mentioned taxonomies are grounded in mammalian physiology, which creates difficulties in weaving parallels between mammal and invertebrate tumours.
5) As emphasized by Smith two decades before, there is a need for a definitional work about the concept of cancer. However, unlike Smith the two reviews do not argue for a universal definition of cancer but rather for one which is better suited for working on invertebrates' tumours.

According to those articles, several constraints weighted on comparative oncology at the time.

- Conceptual constraints: The conceptual framework was not adjusted to invertebrates.
- Theoretical constraints: There was (and still is) a widespread assumption that invertebrates cannot have cancer. This kind of assumption constrains research by discouraging attention to non-vertebrate systems.
- Epistemic constraints: Scharrer & Lochhead noted that pathologists often lacked sufficient zoological knowledge, while zoologists lacked training in tumour diagnosis. The Wautiers, in turn, catalogued ambiguous proliferations but concluded they could not affirm them as cancers. The division of knowledge made it difficult for anybody to diagnose any phenomenon with accuracy.
- Methodological constraints: Without molecular or genetic tools, researchers could not move beyond morphology to demonstrate that invasive behaviours are driven by somatic mutations or epimutations.

2.4 The first evolutionary rupture: Peto's Paradox

As has been demonstrated in the beginning of this chapter, evolutionary methods, tools or concepts are completely absent from early comparative oncology – apart from the Wautiers, who raised the idea that cancer sensibility could be the evolutionary by-product of high regenerative abilities. In the 1977, a paper published by Richard Peto (Peto, 2016) laid down the first foundation

of an important shift towards an evolutionary account of cancer that would happen thirty years later. Sir Richard Peto is an epidemiologist particularly well-known for his influential work in the field of cancer epidemiology, smoking-related diseases, and large-scale clinical trials. His interest was not in comparative oncology nor in evolutionary biology applied to cancer, yet one specific passage of his 1977 paper became the keystone of contemporary comparative oncology. This paper was designed to be an introduction and a defence of multistage models of carcinogenesis. In a section entitled "General Considerations about Mutagenicity and Mitosis", Peto develops a series of thoughts about the link between cancer rates and exposure to genetic damage endorsing a multistage account of oncogenesis. Several decades later one of those thoughts would generate a major research program in comparative oncology.

2.4.1 Are our stem cells a trillion times more cancerproof than murine stem cells?

Richard Peto believed that the number of mitoses inside an organ cannot be the only factor of the susceptibility of that organ to develop cancer. He implies that exogenous factors must be invoked. In order to demonstrate the soundness of this argument, he uses an evolutionary approach. Epithelial cells are known for their high division rates, which would logically suggest a higher prevalence of cancers in organs with rapid cell turnover, such as the small intestine. Yet, he notes that cancers in this organ are relatively rare and that all animals share the characteristic of displaying a high number of mitoses in the small intestine. Therefore he concludes that animals must have developed mechanisms that ensure the ability for rapid cell turnover without much cancer risks (such as "a precise program […], involving stem cells in crypts with low or moderate turnover rates shedding daughter cells which proliferate rapidly and are then lost before they have time to give rise to carcinomas") (Peto, 2016) (p.627). Therefore, the ability of certain organs to resist cancer can be explained by adaptations induced by a high risk of cancer in our ancestors. As a consequence, he concludes that organs in which cancer can be commonly found are those for which there was no selective pressure in our ancestors, but which are currently exposed to DNA-damaging substances (such as tobacco). As he puts it "we have therefore evolved stem-cell hierarchies and territorial imperatives, based on crypts and villi, that allow this without much risk of cancer, whereas perhaps no such mechanisms have been evolved to protect the bronchi because apes and coelacanths did not smoke cigarettes" (*ibid*).

He nonetheless warns against unwise use of evolutionary explanation arguing that "you can usually demonstrate whatever you want by a plausible evolutionary argument" (Ibid) even

though such type of argument is required to explain the low rate of carcinoma induction in humans (even if they might not be sufficient). The observation he makes is the following: “A man has 1000 times as many cells as a mouse (although the ratio of our epithelial stem-cell numbers is not known), and we usually live at least 30 times as long as mice […] However, it seems that, in the wild, the probabilities of carcinoma induction in mice and in men are not vastly different.” (Ibid). In other words, cells within an organism divide at a relatively consistent rate, leading to the accumulation of mutations over time. Given that mutations are causally implicated in cancer it would be expected that the risk of cancer would correlate positively with both the number and rate of cell divisions. Consequently, larger and longer-lived organisms, which have more cells and experience more cell divisions, should theoretically exhibit a higher incidence of cancer. However, these organisms do not develop cancer at higher rates than smaller, shorter-lived species. From this observation comes a question “Are our stem cells really, then, a billion or a trillion times more "cancerproof" than murine stem cells?” (*Ibid*). This observation was later known as Peto’s Paradox (Nunney, 1999). Peto assumes that in the case of mice and men, the difference in cancer rates is a consequence of men having a higher body mass and longevity than mice. At that time, Peto was not able to provide any explanation of this fact but for him, the solution was to be partly found in evolutionary biology. What remained to be demonstrated was how high body mass and long-lived species achieved cancer resistance.

2.4.2 Peto’s paradox as a tool for cancer prevention?

In a recent work, the philosopher Anya Plutynski identified a debate between two strategies for explaining this “paradox” and for preventing cancer. On the one hand the first strategy favours natural selection as an “endogenous mechanism of control” while on the other hand, the second “blackbox” strategy favours exogenous factors (Plutynski, 2023). Richard Peto already hinted at the later in 1977 and will confirm his position in a 1984 article. As Plutynski demonstrates, “Peto argues that preventing cancer by appeal to the “mechanistic” strategy has been less effective than what he calls the “black box” strategy.” (Plutynski, 2023) (p.1246).

“*Thus far, not only has almost all the important knowledge about how to prevent cancer comes from “black box” epidemiology, but so too has the most direct evidence that most human cancer are avoidable*.” (Peto, 2023)

As will be evidenced later in this chapter, comparative oncology will take the path of the mechanistic strategy, focusing on tumour suppressor mechanisms in long-lived and/or massive species. Yet, even if Peto was convinced of the superiority of the epidemiological black box strategy, he acknowledges in a short paper from 2015 the progresses allowed by the comparative approach which favoured mechanistic explanations. He writes: "the question is starting to be addressed successfully by observation and experiment, informed by increasingly detailed species-specific knowledge of genotypes, DNA repair mechanisms and stem cell biology." (Peto, 2015) (p.2).

Mechanistic demonstrations were performed throughout the two last decades (Abegglen et al., 2015a; Compton et al., 2025; Vincze et al., 2022) and Peto's Paradox was confirmed to be true decades after the initial formulation by Peto. One might wonder why there is such a gap between the formulation of the paradox and the extensive corpus of work about it. It is fair to say that testing Peto's paradox by studying cancer risks across species or finding genetic, molecular or cellular processes that explain cancer resistance, could not have been done in the 70's. In fact, as we will see, comparative genomics will be crucial to identify genetic patterns that lead to cancer resistance. Peto also argues that biological knowledge was not sufficient to correctly address this topic.

"Although the issue has been discussed for decades, it could not previously be investigated properly because far too little was known about normal cell biology, cancer cell genetics, mammalian stem cell kinetics and the many mechanisms for avoiding accumulation of permanently fixed mutations in stem cells." (Peto, 2015) (p.2)

In addition, no database about cancer incidence was available at that time, Peto's observation was based on empirical works on mice models to monitor the impact of aging on cancer risks (Peto et al., 1975).

As highlighted by Plutynski, a crucial question formulated by Peto in 1984 is: which strategy is the best for cancer prevention? Should one focus on exogenous parameters such as the exposome? Or should one focus on cellular mechanisms which prevent cancer? Comparative oncology based a major part of its research endeavour to study adaptive trait which helped organisms to resist to the selective pressure of cancer. This question is important to determine whether comparative oncology is an efficient way to prevent or cure cancer.

2.4.3 The role of Peto's Paradox in contemporary comparative oncology

In 1999 Leonard Nunney coined the expression "Peto's Paradox" (Nunney, 1999). He argues that the solution to Peto's Paradox is evolutionary and he ties it to the topics of multicellularity and kin selection. As Nunney writes: "Peto's paradox is difficult to resolve if the cellular mechanisms preventing cancer are viewed as fixed properties of cells" (p.493). In fact, one must presuppose the existence of different mechanisms which impact mitosis and which arose because cancer constituted a selective pressure. Nunney's argument makes sense because he thinks of cancer as a problem that any multicellular organism must face, conjuring the literature on major evolutionary transitions (Michod, 1999; Szathmáry & Smith, 1995). Before working on cancer, Nunney had worked on group selection, therefore it is not surprising that he conceives cancer as an instance of cheating at the level of cells. In fact, if one conceives the level of cells as a level where kin selection's principles apply and cancer as an instance of cheating, then cancer can be considered to be a selective pressure that will induce evolution of cellular mechanisms that help prevent it. Yet, I do not believe that this framework is necessary to accept the hypothesis that cancer has been a selective pressure.

Nunney's work had a major influence on the way Peto's Paradox was received and treated in comparative oncology. His influence is twofold. First, he has been a vocal advocator of the mechanistic strategy to resolve Peto's paradox. Second, his later collaborations with major actors of the field like Carlo Maley must have helped the multicellularity framework to settle in the field which influenced the research agenda of the field towards (1) an exploration and explanation of Peto's Paradox, (2) the adoption of a framework based on kin selection and cheating. Furthermore, the fact that he coined the term Peto's Paradox which was later endorsed by the entire field and the fact that he was cited in papers like (Seluanov et al., 2009) or (Caulin & Maley, 2011) further indicates that Nunney's work was influential.

Overall, Peto's paradox's interest resides in two things:

a) Its predictive power: The paradox does not predict *per* se, but the observations it makes allow us to predict that long lived and/or large organism will have less cancer incidence than short lived and/or small ones. The prediction also applies to organs since organs which have more cell turnover (e.g. small intestine) should have more cancer.
b) Its explanatory challenge: It raises the question of why cancer rates differ across species.

Part of comparative oncology effort was put into testing the predictions enabled by the paradox. Works like (Compton et al., 2025; Vincze et al., 2022) which collect data about cancer in different species can tell whether or not the paradox holds. Another part of the effort is put into answering b) by exploring cancer prevention mechanisms. However, comparative oncology employs an approach not accepted by all the scientific community. Specifically, comparative oncology assumes that evolutionary and mechanistic explanations are sufficient to account for the observed disparities in cancer rates, both between species and across different organs. Alternatively, to make sense of the paradox, some favour the black box approach mentioned earlier which studies "patterns of correlation between exposure and incidence" (Plutynski, 2023).

The mechanistic strategy is well illustrated by (Caulin & Maley, 2011) who reviewed some mechanisms that could explain the paradox while proposing new ones. At that time very little work had been done but there was still a significant number of proposed mechanisms such as lower somatic mutation rates, redundancy of tumour suppressor genes, different tissue architecture, more efficient immune systems.

A problem with the paradox remains. It relies on the assumption that the clonal model of evolution is true, that is to say that somatic cells evolve by natural selection. This question of whether or not this model is true has been debated by both scientists and philosophers (Laplane & Maley, 2024). Several issues have been pointed out both on theoretical and empirical grounds. Some in comparative oncology like (J. S. Brown et al., 2015) are aware of this issue. Brown et al. use mathematical modelling to evaluate how life-history traits might influence cancer suppression. Their findings refined the understanding of Peto's Paradox. In fact, they found that cancer suppression mechanisms are not directly tied to body size but to broader life-history traits like fecundity, longevity, and reproductive timing. In addition, taxonomic differences and evolutionary trade-offs (e.g., between cancer suppression and other survival traits) are critical in shaping cancer suppression. Finally, humans' increased cancer rates reflect an evolutionary mismatch, emphasizing the role of life-history traits in modern disease patterns. Those results led them to make new predictions and propose new research avenues that would test those predictions. They propose to investigate cancer-suppression adaptations in long-lived species with unique life histories (e.g., large reptiles, certain fish, and naked mole rats) and explore age- and organ-specific trade-offs in cancer suppression. Overall, life-history studies, such as (J. S. Brown et al., 2015) indicate that variables beyond lifespan and body size should also be considered when examining cancer risks across species.

On another note, (Ducasse et al., 2015) offer a strong critic of Peto's Paradox. They share most of the concerns portrayed above but also propose new ones which they believe invalidate Peto's Paradox as it is. They describe what they believe to be flawed assumptions of Peto's paradox. They disagree with the assumptions that: "(i) the number of dividing-cells in an organism is strictly proportional to its size, (ii) each dividing cell has the same risk of mutation, and (iii) only mutations induce transformation to malignancy" (p.3).

Their claims can be summarized as follows:

1) Transformation rates aren't uniform. Different cell types and tissues show varying mutation rates, driven by factors like oxidative stress and DNA repair efficiency;

2) Environment participates in cancer risk. Exposure to pollutants, diet, and infections significantly alters cancer prevalence, as seen in beluga whales in polluted habitats and humans compared to wildlife;

3) The role of the tumour microenvironment must be considered. The local environment of cells, including the immune context and tissue organization, plays a critical role in cancer suppression.

While these criticisms appear reasonable, they stem from the assertion that "the accuracy of Peto's Paradox assumptions are rarely valid and/or quantifiable" (p.1). However, studies such as (Vincze et al., 2022) provide evidence that Peto's Paradox applies to more than just a handful of species. Thus, even if the criticisms are valid and the theoretical framework underlying Peto's Paradox has limitations, the paradox still allows to make accurate empirical predictions.

It is worth examining why Peto's paradox has not provoked as much debate as Tomasetti and Vogelstein's 2015 paper (Tomasetti & Vogelstein, 2015). First, Peto's paradox is fundamentally an observation and it does not attempt to make a causal claim. In contrast, Tomasetti and Vogelstein make a strong causal claim, arguing that a large fraction of cancer risk is explained by the number of stem cell divisions, which inevitably sparked controversy. Second, the way Tomasetti and Vogelstein framed their results was more provocative. By emphasizing the role of random mutations, their argument seemed to diminish the importance of environmental exposures, despite these being a well-established and pressing public health concern. This framing naturally invited pushback from researchers and clinicians who saw it as minimizing preventable risk factors. Finally, context played a role: comparative oncology has remained somewhat orthogonal to the attention of most biologists and clinicians. In contrast, Tomasetti and Vogelstein play a

central role in today's oncology and their work appeared in *Science*, reaching a broad audience. Importantly, one promising goal for comparative oncology would be to enable inter-organ comparisons similar to those of Tomasetti and Vogelstein, but extended across multiple species (i.e., combining inter-organ and inter-species perspectives). Such an approach would make it possible to assess whether observations made in humans also hold true elsewhere and to generate hypotheses about the causes of any interspecies differences.[17]

The significance of Peto's Paradox in comparative oncology lies not only in the experimental studies that test the paradox or explore its evolutionary implications, but also in the theoretical works that synthesize current knowledge, identify key questions, and outline potential avenues for future research. The paradox is still a fruitful source of working hypotheses today.

2.5 The evolutionary stagnation: natural history and the challenge of defining cancer across taxa

After Peto's seminal paper, one may have thought that the field would have taken over the types of questions and hypotheses suggested by Richard Peto. But because none at the time had the same evolutionary conception, the field remained silent on those matters. What I have found is that the years from 70' to 80' were still characterized by a natural history approach, with people collecting data on different occurrences of tumours in many species without an integrative evolutionary perspective in mind. Some researchers like Harshbarger remain in the same path as the two previous reviews (Scharrer & Lochhead, 1950; J. Wautier & Wautier, 1955; V. Wautier & Wautier, 1953) while other papers are pure observations of tumours in a variety of species.

Harshbarger reviewed tumours in insects (Harshbarger & Taylor, 1968) and in what he called lower animals (Harshbarger, 1974, 1977). His 1968 paper synthesizes experimental data on insect neoplasia from 1950-1969, focusing primarily on *Drosophila*. Harshbarger and Taylor categorize insect tumours into four types, noting distinct physiological effects in tumorous strains of *Drosophila*, including reduced fertility and shorter lifespans. Experimental nerve severance in species like *Leucophaea maderae* and *Locusta migratoria* induced atypical epithelial tumours, raising questions about tumorigenesis mechanisms. They endorse Scharrer's past hypothesis that latent viruses could activate following nerve injury, though definitive mechanisms remained

[17] I am grateful to Lucie Laplane for drawing my attention to this point in a personal communication.

unclear. Notably, the review emphasizes the potential for metastasis, challenging the prevailing view that true cancer (as defined by metastasis) did not occur in invertebrates. Conceptually, the authors caution against over-reliance on vertebrate-based cancer definitions and stress the need for refined criteria that account for insect-specific responses to stressors like parasites and radiation. Their conclusion advocates for research into inflammation in insects to clarify the distinction between true neoplasia and inflammatory lesions, contending that studies of insect tumours may enlighten shared oncogenic processes across taxa, despite fundamental physiological differences.

What can be taken away from the works in comparative oncology from the 1970' to 2000'? The reviewed studies collectively emphasize the need to establish cancer definitions that match invertebrate physiology and are adapted to the biological diversity of non-mammalian species. Even if the field was evolving without a specific horizon, the different works mentioned here efficiently highlight a key topic of the field: the tension between the physiological uniqueness of each taxon and the shared oncogenic mechanisms that might be conserved across the tree of life.

2.6 The second evolutionary rupture: the end of natural history in comparative oncology

If the first decade of the 21st century is characterised by even more recollection works[18], the years 2008 and 2011 mark a rupture for comparative oncology. Some experts in comparative oncology identify this moment as the true beginning of the field. In 2008, the Seluanov and Gorbunova lab publishes an article (Seluanov et al., 2008) in which they implicitly address Peto's Paradox, starting from the premise that larger and longer-lived organisms should, theoretically, face higher cancer risks. The authors hypothesize that such organisms must have evolved more complex tumour-suppressor systems, using replicative senescence—driven largely by telomere shortening—as a mechanism to inhibit unchecked cell proliferation. While senescence had been studied in some non-human mammals, research on rodents, particularly those other than mice, had been limited. This study, following a previous investigation by Seluanov et al. (Seluanov et al., 2007) linking large rodents with low telomerase activity and small rodents with high activity, focuses on two key questions: (1) whether large rodents with repressed telomerase activity also show replicative senescence in fibroblasts, and (2) whether small, long-lived rodents have distinct

[18] For instance in belugas see (Martineau et al., 2002; Mikaelian et al., 1999), in birds (Filippich, 2004), in small mammals (Greenacre, 2004), in reptiles (Sykes & Trupkiewicz, 2006).

fibroblast proliferation patterns from short-lived, cancer-prone mice. To answer these questions, the authors examine fibroblast proliferation across fifteen rodent species, which vary widely in size and lifespan. This study is the first to specifically investigate tumour-suppressor mechanisms across diverse species and lays the groundwork for future research on cancer-resistant species, including elephants, whales, and naked mole rats. By emphasizing tumour-suppressor mechanisms beyond the traditional mouse model, this work opened a new experimental path in comparative oncology.

Three years later, in 2011, Caulin and Maley publish the first review paper about Peto's Paradox in comparative oncology. (Caulin & Maley, 2011). This theoretical paper is a paradigmatic example of the direction the field will take for the next decade. The authors use Peto's Paradox as a theoretical base upon which comparative oncology should build its research plan. The main assumption is that if one understands how evolution has suppressed or reduced the occurrence of cancer in some species, one might gain valuable knowledge that might be useful for cancer prevention. Starting from the hypothesis that the differences in cancer distribution across taxa can in part be explained as a function of the size and lifespan of the species, they make hypotheses about the reasons behind the resistance of certain species. Sensitivity to contact inhibition had already been found in the naked mole rat (Seluanov et al., 2009), but the reasons were still uncertain. Other hypotheses like the efficiency of apoptotic processes would be demonstrated in the elephant a few years later (Abegglen et al., 2015b, 2023; Callaway, 2015; Preston et al., 2023). Hence, they argue that comparative oncology should focus on those topics. In fact, the field will dedicate the next decade to test Peto's Paradox (Vincze et al., 2022). The mechanisms behind long-lived mammals' resistance will be further explored (Gorbunova et al., 2012, 2014; Seluanov et al., 2018; Tian et al., 2013). To summarize, Caulin and Maley saw in Peto's Paradox an opportunity for comparative oncology to become relevant at a translational level and proposed a series of questions and hypotheses that the field should try to solve. This emphasis on the translational potential of comparative oncology foreshadowed the direction the field would take in the years that followed.

I argue that the two papers mentioned above definitely ended the prehistory of comparative oncology. First because they were the precursors of what became an important part of the field, *ergo* the quest for tumour suppressor mechanisms. Second, because the 2011 theoretical paper sketched a promise which was later endorsed by most of the field: the promise that comparative oncology can play a valuable role in cancer prevention. Today, this promise is at the core of the research agenda of comparative oncology. Finally, whether it be the search for tumour suppressor

mechanisms in long-lived rodents by Gorbunova and Seluanov, or the theoretical framework developed by Caulin and Maley both participate more than any work from the prehistory in fulfilling the second goal that I listed at the beginning of this chapter: making sense of cancer in an evolutionary way.

As was exposed at the beginning of this chapter, comparative oncology has a variety goals. If in the prehistory of the field, the evolutionary goal was absent, it more recently became, together with the translational one, the main goal of the field. Every goal can be addressed by different axes of research, most of those axes being interdependent. The first axis is recollection of data about cancer distribution or cancer risks. Some of them operate at small scales and focus on specific species, while others focus on larger phylogenetic groups such as mammals or vertebrates (Abegglen et al., 2015b; Compton et al., 2025; Vincze et al., 2022). The success of this axis is made difficult because of the difficulty to create or have access to well-furnished databases. Yet efforts have been made to improve disease monitoring in zoos and aquarium which make this type of research easier to achieve (e.g. Species 360 [19] which is an international non-profit organization that maintains an online database of wild animals under human care). For the moment this axis is mostly mammal-centred even though data about tumours in non-mammalian vertebrates are known (Castro et al., 2016; Effron et al., 1977; Filippich, 2004; Sykes & Trupkiewicz, 2006). This focus on mammals could be explained by the difficulty to harness data on cancer prevalence, which is increased outside of mammals. Yet that is only the case for spontaneous tumours since the 20th century was filled with experimental data about tumours in invertebrates. Therefore, I suspect that this focus on mammals is mainly a theoretical bias of the field. Currently, invertebrates appear as way less interesting for the field than vertebrates at least in terms of tumour suppressor mechanisms or in terms of translational potential.

The second axis of comparative oncology is life-history analysis. Once data on cancer distribution has been gathered, comparatists can choose to study cancer resistant or cancer vulnerable species. Both are equally interesting since, as Compton et al. wrote, “species with exceptional cancer resistance [...] can lead to new discoveries of cancer resistance mechanisms outside the traditional human and murine studies” (p.2) while “the discovery of cancer vulnerable species could lead to new insights into cancer syndromes as well as provide spontaneous ’natural’ animal models of disease that can help us gain a better understanding of various types of cancer and their treatments.” (p. 2-3) (Compton et al., 2025). In both cases, life history approaches seek

[19] Formerly known as International Species Information System or ISIS.

to unravel the evolutionary conditions that can explain those differences in cancer distribution. Body size and lifespan are the most studied physiological variations because of the influence of Peto's Paradox on the field (Abegglen et al., 2015b; Caulin & Maley, 2011; Erten & Kokko, 2020; Kokko & Hochberg, 2015; Nunney et al., 2015; Vincze et al., 2022). Yet, those two variables are not the only life-history traits that can be studied. For example, several studies explore the trade-offs between cancer susceptibility and reproductive success (Boddy et al., 2015; Dujon et al., 2022). Symmetrically, one can study the impact of cancer as a selective pressure as was pleaded by (Kokko & Hochberg, 2015) who wrote: "while cancer biologists should not remain unaware of evolutionary principles evolutionary ecologists should also not remain ignorant that multicellular organisms evolve both life-history traits and cancer protection traits, given cancer risks" (p.8).

The third and last axis explores the resistance identified in the first one by providing empirical scrutiny of the genetic and cellular mechanisms that cause cancer resistance. The most well-documented species are the naked mole-rat, the blind-mole rat and elephants. Naked mole-rats achieve cancer resistance via early contact inhibition (Seluanov et al., 2009) and secretion of high-molecular-mass hyaluronan (Tian et al., 2013). Blind mole-rats' cells undergo concerted necrotic cell death mediated by the production of interferon beta-1b (Gorbunova et al., 2012). Elephants possess 20 copies of TP53 and their cells have an enhanced p53-dependent DNA damage response (Abegglen et al., 2015a). For now, only mammals have been chosen as model for this type of research because comparative oncology focused on testing and explaining Peto's paradox and because it was unequivocally demonstrated to be true only in mammals.

2.7 A retrospective view on a century of comparative oncology

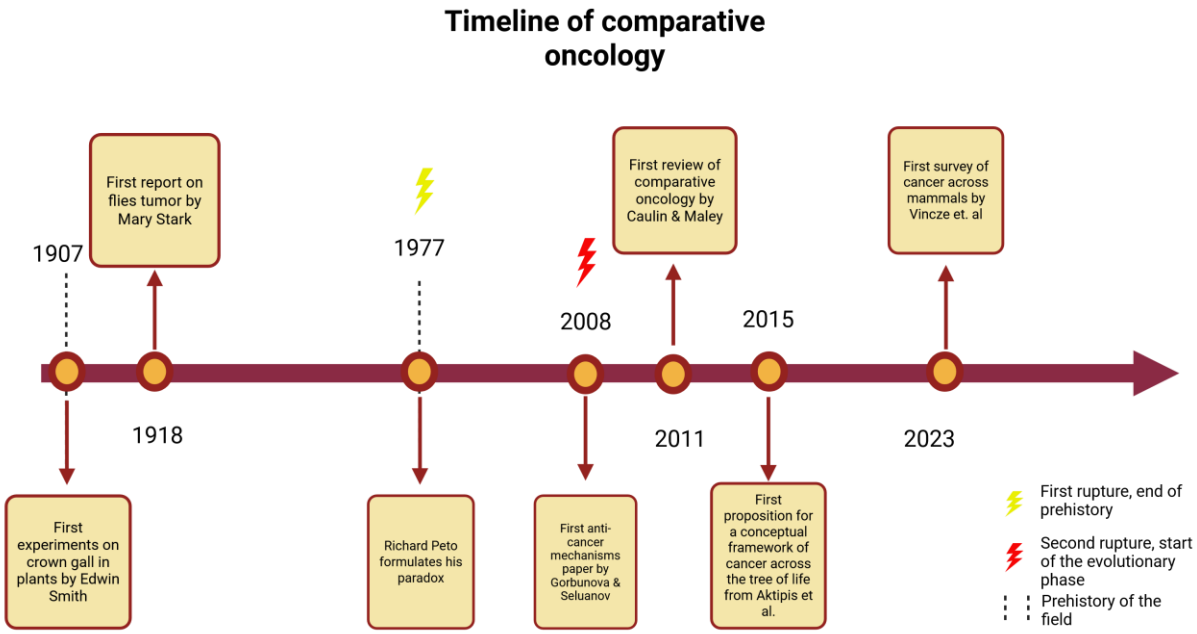


Figure 4. Timeline of comparative oncology

On this timeline, yellow circles mark the major events described in this chapter. The doted lines delineate what is identified as the prehistory of comparative oncology. The two lightning bolts mark the two conceptual ruptures of the field.

In only fifteen years, comparative oncology transformed itself more radically than it had ever done in over a century. The field went from a haphazard search for tumours and their causes, to a more methodical and thorough search for tumour suppressor mechanisms with the explicit goal to participate in developing cancer prevention treatments. What caused this shift are the two ruptures I mentioned, the second one being the consequence of the first one. Because Peto was an epidemiologist, his way of thinking was different from those of biologists usually interested in tumours across taxa. He was interested in patterns of disease distribution which led him to notice the counterintuitive fact that humans have about the same rate of cancer incidence than mice while being bigger and living longer. What is puzzling, however, is the fact that on top of this

observation, Peto already clearly pointed towards an evolutionary explanation[20]; writing: "in the evolutionary diversification of mammals' enormous changes in cellular susceptibility to oncogenesis have developed. These may be nicely illustrated by a comparison of mice and men"; yet evolutionary works about organismal evolution of cancer only appeared from 2011.

At this point it is not clear to us what launched this interest for Peto's Paradox which resulted in a second breath of the field. It can be hypothesised that new methods such as comparative genomics must have led biologists to consider that such research was more likely to be successful. On the one hand, I mentioned that Nunney's work from 1999 had an influence on how the field approached evolutionary questions about cancer. On the other hand, the initial works of both the Seluanov and Gorbunova lab and Caulin and Maley were regarded as good enough to spark this evolutionary and comparative interest in cancer for people who were not working on it previously. Another element which had an influence on the state of the field is the translational promise to which most of the field is subscribing to. I described earlier how Peto's Paradox shaped the field of comparative oncology. I explained that the field is heterogeneous in its methods and goals. In this final section of chapter 2, I will evaluate the strength of the comparative approach to organismal evolution of cancer according to its ability to address the questions listed in chapter 1 and I will explain what conceptual issues stand out of our analysis of the field.

2.7.1 A restricted range of comparison

While the early development of comparative oncology was characterized by a highly comparative approach, involving a wide range of species, contemporary comparative oncology has become more restrictive. In the first half of the 20th century, invertebrates—particularly insects—were a primary focus, as evidenced by the extensive body of experimental work compiled by researchers (Scharrer & Lochhead, 1950; J. Wautier & Wautier, 1955; V. Wautier & Wautier, 1953). In contrast, aside from studies using the *Drosophila* model, tumours in invertebrates are notably scarce in modern comparative oncology. Current single-species studies largely focus on mammals, including elephants, whales, naked mole-rats, and bats. The only two comprehensive reviews of cancer incidence published recently are limited to mammals (Vincze et al., 2022) and vertebrates (Compton et al., 2025). Yet, some argue that the field should embrace a wider range

[20] It should be noted that this observation comes after Peto warned about the reckless use of evolutionary arguments. In fact he wrote: "The trouble with evolutionary arguments is that you can usually demonstrate whatever you want by a plausible evolutionary argument." (Peto, 2016) (p.627).

of comparison (Aktipis et al., 2015; Kokko et al., 2017; Nunney et al., 2015). What explains this loss in the diversity of species taken into consideration? First, it must be noted that there is a general tendency in biology to decrease the number of species used for research (Ankeny & Leonelli, 2020). Second, the field is less conceptually oriented than what it is used to be. Questions of definition were addressed in several comparative works from the first half of the 20th century. Even if Smith's parasitic theory ended up being wrong, he did still produce a theoretical framework to ground his claims. Scharrer and Lockhead faced a difficulty to characterize the cellular proliferation events they were reviewing, which led them to question the relevance of usual definitions of cancer for invertebrates. Currently, with the exception the breakdown of multicellularity framework developed by (Aktipis et al., 2015), no in-depth work of definition exists. As a consequence, the definitions used by comparatists today do not apply outside of vertebrates. Furthermore, some in the field are not convinced by the utility of studying cancer outside of mammals, arguing that data are scarce on invertebrates, which is incorrect if one doesn't focus on spontaneous tumours only. In other words, there is a vicious circle consisting in people not being convinced of the utility of studying cancer outside of mammals, therefore they don't produce any definitional work, and this, ultimately, makes it even harder to start working on invertebrates.

However, to temper our words, it is true that to this day biologists advocating for a high degree of comparison (i.e., including as many branches of the tree of life as possible) failed to provide solid enough arguments to justify their claim. It is not obvious for other comparatists what would be the benefits of studying tumours in, say, algae and fungi. Chapter 3 will be the occasion to develop this point further.

2.7.2 Does comparative oncology constitute a scientific field?

I will argue here that comparative oncology can be labelled a field. I will rely on the work of Elihu Gerson (Gerson, 2013) and Nicholas C. Mullins (Mullins, 1972) two sociologists of science. I will list a series of criteria taken from those authors and examine whether comparative oncology matches them.

- **(1) Shared conceptual framework**. Comparative oncology is anchored in the idea that cancer occurs across a wide range of species making its study across taxa both relevant and informative. The unifying framework is the evolutionary and cross-species view of cancer.

However, the field's actors can disagree on some key ideas such as the definition of cancer and the relevancy of studying cancer beyond vertebrates.

- **(2) Core research agenda**. Peto's Paradox constitutes a legitimate and central object of enquiry. But that is not the only research interest; mechanisms with cancer-protective effects constitute a wide portion of comparative's research. While certain researchers focus on these mechanisms as a way of addressing Peto's paradox, others are motivated by a more general interest in understanding them.[21] Yet beyond these two major avenues, the field's unity remains imperfect, as research often disperses in multiple directions without a clearly defined agenda. However, the field raises questions that diverge significantly from those typically asked in related fields, namely oncology and evolutionary biology. This distinctiveness helps delineate comparative oncology as a field in its own right, with research questions that cannot be fully absorbed by neighbouring disciplines. This also explains why the field has been largely overlooked by both evolutionary biologists and medical oncologists: the questions it addresses lie too far from their respective concerns
- **(3) Community of practitioners.** Specific groups of researchers are well-known for their work in comparative oncology. They interact regularly through conferences, working groups, and seminars, and their collaborations are visible in co-authored articles and some edited volumes.
- **(4) Institutional recognition.** Comparative oncology has some dedicated institutional bases like the Gorbunova and Seluanov laboratory or the Arizona Cancer Evolution Center. However, few departments or funding streams are dedicated solely to comparative oncology.
- **(5) Training and reproduction of members**. There are no widespread or standardized graduate pipeline yet. Entry into the field is often via interdisciplinary detours. There is no specific graduate program in comparative oncology, and no or very few summer schools.
- **(6) Communication channels.** There are no dedicated journals of comparative oncology. Yet, as mentioned, there are dedicated conferences.
- **(7) Methodological identity.** As mentioned above, the field is very diverse There is no specific animal model that is privileged and no distinctive techniques or methods.

[21] Indeed, even if Peto had never formulated his paradox, it is likely that comparatists would still have studied cancer-protective mechanisms, given their translational relevance.

Taken together, these criteria suggest that comparative oncology occupies an intermediate position. On the one hand, it shows many features of an emerging field: a shared conceptual framework centred on evolution and cross-species comparisons, a distinctive research agenda exemplified by Peto's paradox, a recognizable community of practitioners, and some degree of institutional recognition. On the other hand, the absence of standardized training pathways, dedicated journals, or a clear methodological identity indicates that the field is still consolidating rather than fully established. Comparative oncology can therefore be described as a field *in the making*: it has created an intellectual space and gathered a community, but it has yet to achieve the full infrastructure and stability typical of mature scientific fields.

2.7.3 Is comparative oncology now about curing cancer?

During the 20th century, the main justification for studying cancer across the tree of life was that if we gained knowledge about cancer mechanisms in some species, it should enlighten us on the general principles behind this phenomenon. This idea is not gone today but it has been refined into something more medically oriented. Some comparative oncologists believe that one key to solving the problem of cancer is to be found in evolution. Many comparatists believe that multicellularity is a necessary and sufficient condition for cancer to be a threat (Abegglen et al., 2015a; Aktipis et al., 2015; Albuquerque et al., 2018; Erten & Kokko, 2020; Nery et al., 2022; Nunney et al., 2015). Therefore, the fact that some branches do not seem to have to deal with cancer or the fact that some species like big and/or long-lived mammals manage to resist to cancer indicates that evolution found solutions to this problem. Then, the role of comparative oncology would be to identify cancer resistant species, unveil the mechanisms behind their resistance, and potentially bring new therapeutical ideas to general oncology. Throughout several discussions with actors of the field, I realized that even though the field took an evolutionary shift in the way it deals with cancer, it is today mainly driven by a translational purpose. In other words, while the methods and questions are evolutionary, the objectives are not always. This perspective helps to explain why comparative oncology predominantly focuses on studying vertebrates. The underlying assumption is that investigating species phylogenetically closer to humans is more efficient to identify new therapeutic pathways applicable to human medicine. Given this framework, several questions arise. Does comparative oncology possess the necessary tools to make meaningful therapeutic contributions? Is studying cancer in more phylogenetically distant and diverse species truly devoid of translational value? And finally, if comparative oncology were to adopt more evolutionary objectives, what directions should it pursue?

2.8 Is comparative oncology the best way to study organismal evolution of cancer?

In the previous chapter, I outlined the types of questions that an evolutionary account can be reasonably expected to address. I also theorized about the potential of a comparative approach to effectively address these questions (see Table 1, chapter 1). Now that I have examined how this comparative approach is embodied in the field of comparative oncology, I can assess its effectiveness in responding to these questions.

First, it must be noted the differences between the questions it can address and the question it is actually addressing. Currently, comparative oncology does not invest a lot of work into the foundational questions (i.e., What is cancer? When did it appear? Why is there cancer?). The only exception being "which species get cancer?". However, comparative oncology invests a considerable amount of work in more evolutionary questions such as "anti-cancer" mechanisms or the question of whether cancer results from trade-offs. In chapter 4, I will argue that it is curious that the field does not invest in more foundational questions even if it possesses proper tools to do so.

Second, comparative oncology does directly address evolutionary questions. By comparing cancer susceptibility and resistance across species, it reveals how natural selection has shaped diverse mechanisms of tumour suppression, as seen in Peto's paradox and adaptations like elephants' extra p53 copies. It also highlights the evolutionary trade-offs between longevity, reproduction, and disease risk, linking cancer biology to life history strategies. Moreover, the conservation of oncogenes and tumour suppressors across taxa provides insights into the deep evolutionary origins of cancer control. But it would be a mistake to believe that foundational questions do not participate to an evolutionary understanding of cancer.

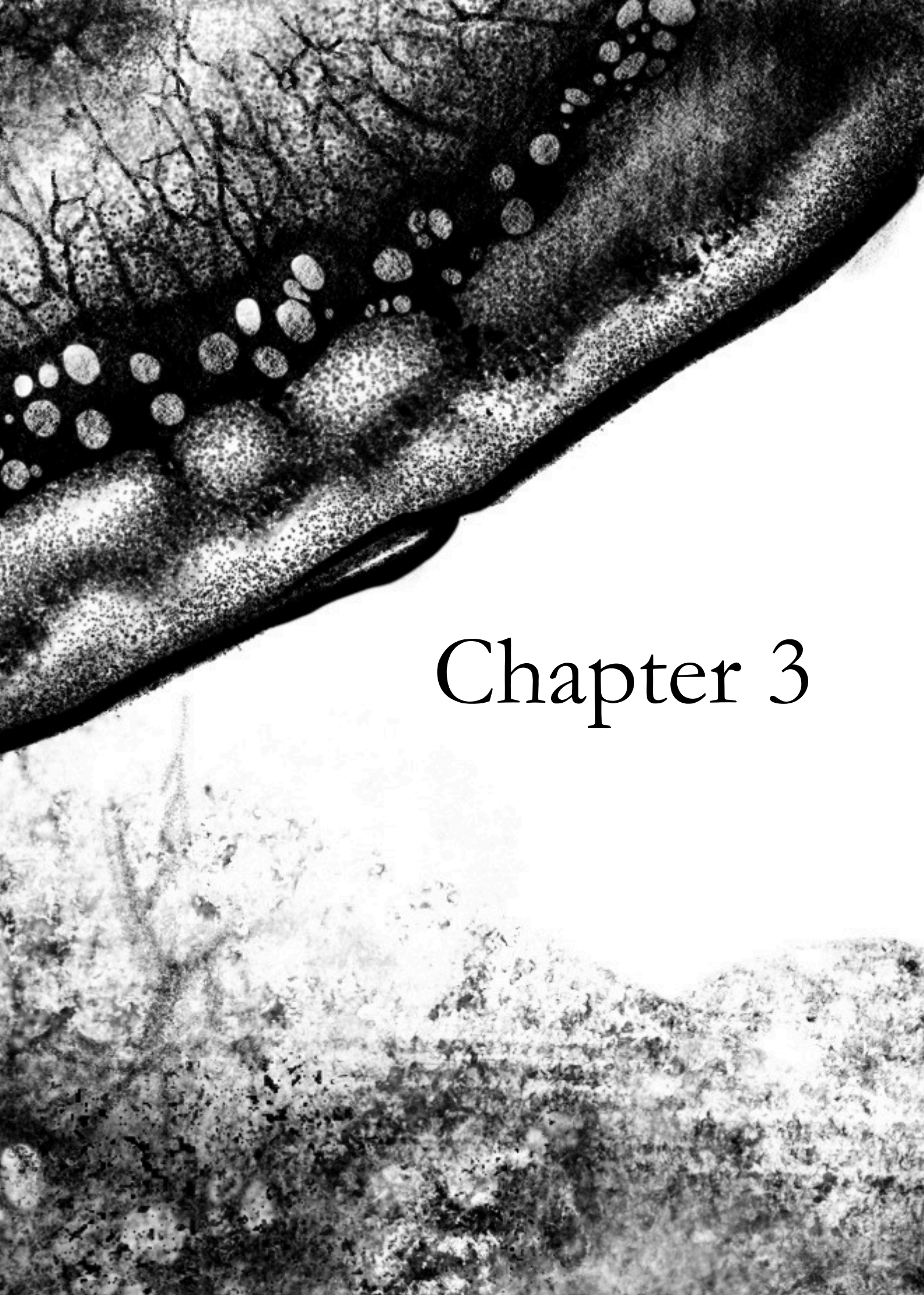
Finally, it appears clearly by looking at Table 2 that comparative oncology is the best approach to organismal evolution of cancer as it can and does address more questions than the other two approaches. However, I argue that it is still imperfect at it does not address the foundational questions which it can and as will be seen, should address.

Table 2

Question	**Can the comparative approach answer to this question?**	**Is comparative oncology trying to answer to that question?**	**Can the palaeontological approach answer this question?**	**Can the social evolution approach answer this question?**
Why is there cancer?	Yes	**No**	No	Yes
What is cancer?	Yes	**No**	No	Maybe
When did cancer appear?	Yes	**No**	Yes	Yes
Which species get cancer?	Yes	Yes	Partially but yes	If conjoined with a comparative approach yes
Was cancer a selective pressure?	Yes	Yes	No	No
Is cancer the outcome of an evolutionary trade-off	Yes	Yes	No	No
What are the "anti-cancer" mechanisms evolved by some species?	Yes	Yes	No	No

Which cancer types start at what life stage?	Yes	**No**	No	No
What is the impact of cancer in terms of mortality?	Yes	Yes	No	No
Which species have which types of cancer?	Yes	Yes	Maybe	No
How to cure/prevent cancer?	To be demonstrated	Yes	No	No

This chapter has traced the historical and conceptual evolution of comparative oncology, emphasizing its transition from an unstructured natural history approach to an evolutionary field putting an emphasis on its translational goal. By exploring cancer across diverse species, the field has grappled with foundational questions, such as how to define cancer in non-vertebrates and whether shared mechanisms underpin oncogenesis across the tree of life. While foundational figures like Smith and Harshbarger made significant contributions, their approaches were limited by the absence of a unified theoretical framework. The emergence of Peto's Paradox marked a turning point, providing the field with a question that demanded evolutionary answers that only a comparative approach could provide. The paradox shed light on the surprising resilience of large and long-lived species against cancer, redirecting to some extent comparative oncology toward the exploration of tumour-suppressor mechanisms. Yet, comparative oncology continues to face significant conceptual and methodological challenges. The limited taxonomic scope of contemporary studies and the persistent reliance on mammalian models reflect a tension between the field's ambitions and its current practices. The absence of a robust definitional framework for cancer across taxa further restricts its potential to encompass the full diversity of life. With chapter 3, I intend to fill this absence by proposing a universal definition of cancer and demonstrate how comparative oncology would greatly benefit from this definition.



Chapter 3

Chapter 3 : A Universal Definition of Cancer and an Explanation for its Diversity across Organs and Species[22]

The many manifestations of cancer have led to multiple definitions, each tailored to specific contexts. While this diversity reflects the complexity of the disease, the lack of a universal definition creates confusion and hinders progress across disciplines. In addition, the usual definitions are not applicable to organisms that do not have basement membranes (plants, fungi, algae, etc.) which precludes comparative approaches to cancer. It raises the question of the possibility of a universal definition of cancer. This chapter argues that a universal definition is possible and offers such a universal definition. The universal definition we propose is the following: *Tumours are uncontrolled cell proliferations due to cell-level evolution, designated as cancers when they are lethal if left untreated.* This definition is justified by the fact that it captures the underlying evolutionary dynamics common to all tumours, offering a unifying framework that applies to any multicellular species and any tissues. This definition also brings order to clinical oncology, notably by introducing the concept of affordances and showing how different risk factors are relevant in different tissues. Finally, this definition provides a necessary tool for studying cancer across the tree of life, a vital step for advancing comparative oncology.

Box of definitions

Tumour: Uncontrolled cell proliferation due to cell-level evolution.[23]

Cancer: Tumour that will be fatal for the organism if left untreated.

Affordances: Properties of the cell–environment system that define the range of potential behaviours available to cancer cells, some of which may be selectively realized depending on their fitness consequences.

There is currently no universally accepted definition of cancer. Instead, the field is marked either by a plurality of definitions that reflect the diverse manifestations of the disease, or by textbook definitions that emphasize specific outcomes and treat them as paradigmatic cases. This

[22] The following chapter results from a collaborative project with Lucie Laplane, Carlo Maley, and Thomas Pradeu, and is adapted from an article currently in preparation. While I was responsible for most of the writing, the core idea underlying our proposed universal definition of cancer originated with Carlo Maley. I am grateful to all my collaborators for their contributions to this work and for granting me permission to include it in this dissertation.

[23] In this chapter, I use the more familiar term “tumour” to refer to what cancer biologists typically call a “neoplasm.” However, I do not restrict the term “tumour” to a mass, as I wish to include liquid neoplasms as well.

conceptual fragmentation has implications across all areas of oncology. In clinical oncology, for instance, it hinders the identification of the core underlying process of the disease. As a result, the field often overlooks the contingent nature of many features it identifies as therapeutic targets. In comparative oncology, the current definitions of cancer are not applicable to organisms that do not have basal membranes (plants, fungi, algae, etc.). Yet, uncontrolled growth and its detrimental consequences affect most multicellular organisms (Aktipis et al., 2015). This limitation has contributed to a strong emphasis on mammals, at the expense of investigating other groups that show signs of proliferative disorders. The aims of this chapter are to 1) demonstrate why a universal definition is necessary, 2) propose such a definition, and 3) show how it can achieve a broad scope of applicability while still being able to account for the specificities of cancer across organs, tissues and species. Section 3.2 proposes a universal definition based on the idea of cell-level evolution and introduces the concept of affordances, which turns out to be crucial to make sense of the specificities of cancer across tissue, organs and species. Section 3.3 explains why the universal definition of cancer proposed here is important for several goals of clinical oncology. Finally, section 3.4 argues that a universal definition is not beneficial for comparative oncology but also for clinical oncology.

3.1 The multiplicity of definitions and why a universal definition is useful

Anyone who takes a look at the current scientific and medical literature on cancer will be struck by the coexistence of many different definitions or characterizations of cancer. Why is that so, and what are the advantages and disadvantages of this multiplicity of definitions? There are different reasons that can explain the multiplicity of definitions of cancer. Anya Plutynski aptly highlighted the challenge of unifying all neoplastic diseases under a single banner, noting that the more we learn about cancer, the more we are led to wonder: “perhaps then cancer is not one but hundreds of diseases.” (Plutynski, 2018).

First, a reason behind the multiplicity of definitions of cancer is the variety of approaches to this disease. In fact, cancer is a highly heterogeneous phenomenon which can be studied under the scope of different biological disciplines (genetics, molecular biology, physiology, immunology, evolutionary biology, and so on). Therefore, it is not surprising that different definitions will be provided depending on the discipline. Yet there is not only a diversity of approaches but also a diversity of characteristics that cancer can display. Cancer behaves differently across organs and displays a lot of heterogeneity within tumours (this is called

‘intratumor heterogeneity’). Depending on the organ and the specificity of the tissue in which it arises, clinicians will not witness the same features (Bianchi et al., 2020). For instance, solid and liquid cancers are different in many ways (Mancini et al., 2021; Montironi et al., 2021). Another example is that while some cancers are heavily hormone-sensitive, others aren’t affected by hormones in any way (Emons, 2022). Therefore, depending on the specificities of each type of cancer, the characterisation will change accordingly. Finally, definitions serve diverse scientific purposes (explaining, predicting, treating, classifying, ...). As a consequence, definitions based on different characteristics will better serve different purposes, which contributes to account for the multiplicity of definitions.

What are the usual definitions encountered in oncology? We must note that a recurring issue for anyone who wants to assess the relevance of existing definitions of cancer is that most of the time, authors are not explicit about their definitions or are not even aware that the way they frame their object of inquiry is contingent upon their scientific purposes. In addition, it is more accurate to speak of ‘characterisations’ of cancer, rather than ‘definitions’ of cancer strictly speaking, for a definition is often understood as setting the necessary and sufficient conditions for something to be what it is while a characterisation describes certain key features or properties of something, often highlighting typical or important aspects, but not necessarily all conditions needed to define it.

Our critique of mainstream definitions of cancer is based upon the definition of the National Cancer Institute as well as common methods to diagnose cancer as we argue they reflect how most clinicians characterize cancer. The National Cancer Institute (*Understanding Cancer - NCI*, 2016) defines cancer as “a disease in which some of the body’s cells grow uncontrollably and spread to other parts of the body”. They add “Cancerous tumours spread into, or invade, nearby tissues and can travel to distant places in the body to form new tumours” in opposition to benign tumours which “do not spread into, or invade, nearby tissues”. Pathologists diagnose cancer using biopsies and cytology samples where they look respectively for invasion through the basement membrane and abnormal nuclei, as well as other features that can be observed through a microscope. However, benign tumours can also display abnormal nuclei, therefore the most important criterion to determine whether a tumour is a cancer is a sign of invasion or metastasis. Also, depending on the organ, pathologists may use criteria that are specific to the particular tissues under consideration.

When put together, the definition and the diagnosing methods denote a focus on invasion indicating the infringement of the basement membrane, which is a typical assumption of clinical oncology. In that regard, they are not universal as they cannot be applied to multicellular organisms and tissues that lack a basement membrane.

The limitation of these definitions lies in their reductiveness, which gives rise to several issues. First, they fail to account for the phenotypic diversity observed both across different types of cancer and within cancers of the same type. For instance, cancers originating in different organs — or even subtypes within a single organ — can display distinct traits such as growth patterns, metastatic potential and patterns, and molecular profiles. These definitions insist on organ specific features, but do not explain why those specificities exist in the first place.

Second, their reductiveness leads to cases of incoherency in the identification of what counts as cancer. According to those definitions, labelling a phenomenon as "cancer" requires the manifestation of traits that are contingent on rare conditions (e.g., the complete cascade of events needed for successful cellular invasion). This problem is not only true for neoplastic phenomena across the tree of life but also for certain manifestations occurring in humans. For example, numerous neoplastic processes initiate in humans during their lifespan, yet most remain clinically inconsequential due to protective mechanisms that stop the progression before the invasive stage. According to current non-universal definitions, those cases should not be labelled as cancers even though they display the same initial mechanisms as acknowledged cancers.

Moreover, conditions such as glioblastoma, leukaemia, lymphoma, non-muscle invasive bladder cancer (Sexton et al., 2010), and carcinoma in situ are classified as cancers despite not involving basal membrane invasion. Although oncologists are generally aware of this inconsistency between the criteria used to define cancer and its classification, this tension further highlights the impossibility for the current framework to make sense of the diversity of cancers.

One could argue that the reductiveness of non-universal definitions does not matter as long as those definitions work well for their purpose(s). This claim is fair in a clinical context because those definitions are indeed efficient to describe diseases whose features depend on the organs in which they arise. However, it is less fair in the context of basic biology. Fundamental biology seeks general principles, not just medical applications. Most of the time in usual scientific practice, biologists use non-universal definitions which are initially framed for the clinic. However, it is not

obvious that those definitions are always the best pick when it comes to studying general and fundamental features of cancer. Our claim is not that biologists should always pick universal definitions of cancer, but rather that non-universal ones should not be the default, implicit, and unjustified choice. In other words, instead of being taken for granted without justification, non-universal definitions of cancer should be used only if they genuinely correspond to the specific research purpose(s) for which they are used in a given scientific context. What I propose in this chapter is that a universal definition offers a solution to this multi-fold issue of narrowness.

Another, and of course not mutually inconsistent, way to argue in favour of universal definitions of cancer is to demonstrate their potential benefits for oncology. Our reasoning proceeds as follows: other biological phenomena—such as immunity or consciousness, for instance—initially relied on narrow and strongly anthropocentric definitions of their object of study but later on they underwent conceptual expansions. The problem lies in that oftentimes scientists initially picked up one occurrence of the phenomenon in their field of study and subsequently hold it as a paradigmatic example of the phenomenon is. In the cases of immunity and consciousness, broadening the conceptual framework led to a more accurate understanding of the phenomena and sometimes revealed crucial conserved features across species that were previously overlooked, precisely because the initial definition was too narrow and too much focused on how these phenomena are realized in humans. For consciousness, several decades of research have now shown that a central nervous system is not required in order to manifest consciousness, contrary to a long-held belief in neuroscience and cognitive science (Godfrey-Smith, 2016, 2020; LeDoux & Sorrentino, 2019). Similarly, immunity was long thought to require lymphocytes and antibodies, but a more encompassing concept of immunity revealed the existence of immunological mechanisms in invertebrates, plants, and even prokaryotes which, crucially, turned out to play a pivotal role in human immunity, too. A paradigmatic example is the discovery of Toll receptors in *Drosophila* which opened a whole continent of discoveries on Toll-like receptors and more generally on pattern-recognition receptors in mammals (Lemaitre, 2004; Pradeu, Thomma, et al., 2024). Crucially, such discoveries, initially made in basic biology, have had important therapeutic impacts.

By analogy with consciousness and immunity, we propose that a broader, less human-centred, definition of cancer is likely to yield comparable advances for our basic understanding of the phenomenon of cancer. If we look at what has already been done outside of humans or mammalian models, it is interesting to take a look at the *Drosophila melanogaster* model of cancer.

The *Drosophila* model has provided insights into cancer biology by uncovering conserved mechanisms relevant for tumorigenesis. Foundational discoveries in oncogenic signalling pathways (e.g., Wnt, Notch, Hedgehog, Hippo) and developmental phenomena like cell competition (Baker, 2020) emerged from studies in flies. More recently, the *Drosophila* model made unique contributions in studying tumour-host interactions, such as identifying conserved regulators of anorexia and cachexia, with insulin-like growth factor-binding protein (IGFBP) homologs implicated in both fly tumours and human/mouse models (Figueroa-Clarevega & Bilder, 2015; Huang et al., 2016; Wagner & Petruzzelli, 2015; Yule et al., 2024). Additionally, a paraneoplastic renal syndrome first characterized in flies later guided discoveries about developmental and functional disorders of the kidney (Xu et al., 2023). To conclude, given the many parallels observed between fly tumours and human tumours, we argue that it would be valuable for any definition of cancer to be applicable to *Drosophila*. In other words, a definition of cancer that would imply that *Drosophila* do not develop cancer would run the risk of making us miss the potentially fruitful lessons that can be drawn from tumour development in this species. Even if the argument made here is particularly convincing in the case of *Drosophila* because of the vast data accumulated over the last two decades, it is reasonable to assume that, should biologists have made similar investigations in other invertebrates, further useful lessons for understanding cancer would have been drawn.

This leads us to our next point: if we argue that a universal definition is required, it is not any universal definition; it needs to be based on an evolutionary understanding of cancer. As we have seen in previous chapters, it would be totally wrong to assume that cancer is a disease that arose recently in *Homo Sapiens*. Evidence of cancer extends to fossil records belonging to many different species and dating back 240 million years, stressing an underlying evolutionary history (Capasso, 2005; Rothschild et al., 1999, 2003). This deep evolutionary history implies that cancer is a recurring outcome across the tree of life. Since evolution likely shaped cancer, any definition needs to account for the various ways cancer manifests in different organisms. Current definitions cannot account for this evolutionary history. Only a universal definition, which virtually applies to most of the tree of life, can account for this evolutionary history of cancer.

As a matter of fact, there is not only an evolutionary history of cancer but also an evolutionary nature of the process. Regarding this point, Brown et al. write in their 2023 paper entitled "Updating the definition of cancer" (J. S. Brown et al., 2023): "These definitions tend to describe what cancer 'looks like' or 'does' but do not describe what cancer 'is' or 'has become'". Brown and colleagues define cancer as "a disease of uncontrolled proliferation by transformed

cells subject to evolution by natural selection." While we agree with the claim that usual definitions focus on the outcomes, we wish to avoid endorsing any essentialist stance about cancer. Therefore, we reformulate their point as follows: current definitions focus on the outcomes of carcinogenesis (particularly invasive behaviour) rather than the processes that led to those outcomes. We wish to follow in Brown et al.'s footsteps in two regards. First, we also want our definition to be based on an evolutionary understanding of cancer. Second, our definition aims at an ontological unification, that is, we believe that envisioning cancer as a causally unified category can be beneficial.

3.2 Our universal definition of cancer, and why it must be complemented by the concept of affordance

3.2.1 Tumours as the product of cell-level evolution

While the multiplicity of definitions serves a scientific purpose—namely, equipping researchers with the most effective conceptual tools—none of these definitions is universally applicable. Instead, we propose to follow Brown and colleagues' proposition, but also to go further. We propose that tumours are *any uncontrolled cell proliferations driven by cell-level evolution*. This evolution includes processes such as natural selection, genetic drift, mutation, gene flow, and takes into account the influence of the microenvironment. For example, genetic drift can play a significant role in neoplasia. Jackpot mutations that occur during development can generate clonal expansions simply due to the organism's growth—a phenomenon that, according to Meza and colleagues (Meza et al., 2005), accounts for ~20% of human tumours. Furthermore, our definition incorporates the influence of microenvironmental control on neoplasia (Bissell et al., 2002; Bissell & Radisky, 2001), where an abnormal environment signals neoplastic cells to proliferate abnormally. This is because the microenvironment acts as a form of inheritance (Laplane & Maley, 2024), shaping the evolutionary trajectory of the cells. Importantly, the rate of cell-level evolution can be measured by assessing cell population sizes, generation/turnover rates, mutation rates, clonal expansion rates, and heritability (Fortunato et al., 2016). Building on this framework, we further propose that cancer specifically refers to a tumour that will be *fatal* for the organism if left untreated. In human medicine, for instance, tumours that have not yet invaded through the basement membrane are rarely fatal (with glioblastoma multiforme being a notable exception). Thus, physicians recognize invasion through the basement membrane as a clinically significant transition, marking the point at which a benign tumour becomes life-threatening and is then called

"malignant"[24]. However, in other organisms, different biological constraints may determine which neoplastic features pose a mortal threat.

Even though we want to take the same path as Brown and colleagues, our definition differs from theirs in that they define cancer as a "disease of uncontrolled proliferation by transformed cells subject to evolution by natural selection". Their concept of "transformed cells" are cells that have acquired the hallmarks of cancer (Hanahan & Weinberg, 2000). Our definition, in contrast, does not specify a set of characteristics, like the hallmarks. Those are the results of the evolutionary process given the affordances of the particular tissue in which cells are evolving. We agree with Brown et al. that the key component of cancer is the cellular evolutionary process. But in some organisms, and perhaps even some human tissues, some of the cancer hallmarks do not evolve. In addition, even though they aim for a definition that describes not what cancer looks like but what cancer is, they ground their definition in the hallmarks of cancer framework. In doing so, they fall into the very pitfall they sought to avoid, since, as we will see below, the hallmarks are consequences of the cancer process rather than the process itself[25].

So, for the purposes of this chapter, we propose to adopt the following universal definition of cancer: *Uncontrolled cell proliferation due to cell-level evolution that will be fatal for the organism if left untreated.*

3.2.2 Affordances: why are tumours so different from one another?

As we argued before, a benefit of any universal definition of tumour and cancer is to be able to pinpoint the core principle of being a tumour or a cancer (lethal tumour), i.e., to find what is common to all instances of tumours or cancers. However, by reaching this higher level of definition, there is a risk of losing the ability to make sense of more specific and local features. This is why we propose to combine our universal definition of tumour and cancer with the use of the concept of *affordance*. This concept helps to explain the local features of tumours and cancers, which are not part of the universal definition of tumour even though they can be crucially important for the understanding of a given tumour. We will first explain why the concept of affordance is

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[24] Blood cancers and brain cancers do not require invasion through a basement membrane but would be considered cancer by our definition.

[25] We can at least rule out immune invasion as not all immune systems can detect cancers (e.g. plants or algae), invasion since not all organisms allow for invasions and neoangiogenesis as not all organisms have a circulatory system.

relevant to better understand the specificities of both tumours and cancers, and, in a second step, how it can be specifically used to shed a new light on cancers (tumours that are lethal if left untreated).

We borrow the concept of affordance from the field of ecological psychology, and in fact our own definition of affordance is much inspired by Thomas Stoffregen[26] (Stoffregen, 2003) who wrote: "Affordances are opportunities for action; they are properties of the animal–environment system that determine what can be done" (p.124). Taking inspiration from Stoffregen's formulation, we develop our own view and propose a taxonomy of affordances relative to tumours and cancers. In fact, it is essential to distinguish between what is possible for a tumour cell and what is likely to occur. We propose to use the concept of affordance to describe the range of actions available to a tumour cell within its environment which provides the contextual opportunities that the cell could, in principle, exploit. However, not all affordances are equally realized. Fitness plays a key role in favouring certain affordances since those that confer a selective advantage—by increasing reproductive success or survival of the cell and hence fitness—are more likely to be consistently observed due to convergent evolution. In this sense, affordances define a space of possibilities, i.e. more specifically the space of potential behaviours, while fitness explains the patterns of actual behaviours. We distinguish three types of affordances in tumour: (i) those that are impossible for a tumour cell to achieve, due to environmental constraints; (ii) those that are possible but unlikely to be realized because they do not confer a fitness benefit; and (iii) those that are both possible and fitness-enhancing, and thus likely to be selected. (i) and (iii), because they describe plausible outcomes, give us strong predictive and explanatory powers since they allow us to explain patterns observed across diverse tumour cases.

Therefore, we define affordances in the context of a tumour as the set of conditions that are salient to a cell's fitness. *Affordances in tumours are properties of the cell–environment system that define the range of potential behaviours available to tumour cells, some of which may be selectively realized depending on their fitness consequences* (see Figure 5).

[26] Thomas Stoffregen is a professor of experimental psychology with a specific interest for the psychology of perception and action.

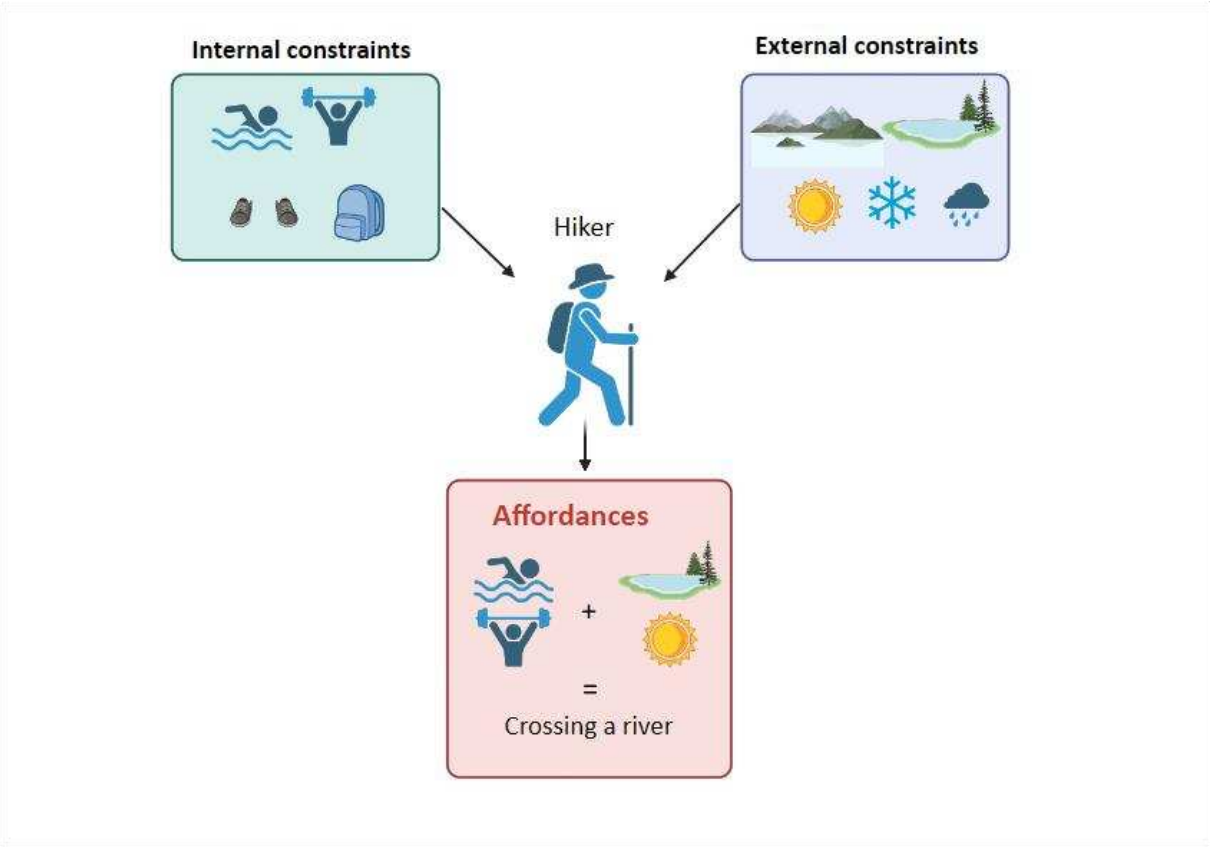

Figure 5. An analogy for affordances

In this figure, the hiker's possible actions are shaped by two types of constraints. Internal constraints include their physical abilities (e.g., swimming skills, cardiovascular fitness) and equipment (e.g., climbing rope, hiking shoes). External constraints include the characteristics of the terrain (e.g., plains, mountains, rivers) and weather conditions (e.g., sunny, rainy, snowy). Together, these constraints determine what the hiker can do. For example, if there is a river, the weather is favourable and the hiker knows how to swim and is in good physical condition, they can cross it. The hiker's ability to cross the river (or not) or to climb a mountain (or not) represents their affordances.

This leads us to distinguish primary properties from secondary properties of a tumour. Indeed, the process of cell-level evolution tells us what the primary properties of tumours are and shows us that all tumours correspond to the same process. Affordances tell us what the secondary properties are based on the analysis of cell-environment systems. It shows us that, because of the different affordances of the tissue or organism in which a tumour arises, outcomes are different. The combination of both allows us not to mistake secondary properties for primary ones. For example, if we are right, dissemination is a secondary property while it is considered a primary one by clinicians.

Although the concept of affordance is useful to shed light on both tumours and cancers, as we just saw, in what follows we focus on cancers (i.e., lethal tumours), because this is where we believe the concept of affordance proves most useful for the field of oncology.

A good example of affordances in cancer can be found in how cancer stem cells express stemness. Lucie Laplane, including in collaboration with haematologist Eric Solary (Laplane & Solary, 2017, 2019), has distinguished four modalities of stemness: (1) stemness is entirely constitutive to the stem cells, (2) it is constitutive to stem cells but its manifestations are triggered by specific and localized microenvironmental stimuli, (3) it is triggered by specific microenvironmental stimuli in any cell that would locate in those specific microenvironment, (4) it is regulated at the system level (see also Laplane, 2016). Epithelial tissues, for example, are spatially highly structured, with little opportunity for cells to move in the absence of cell death. In those tissues, stemness is a matter of location: cells situated at the bottom of the crypts, in very specific spots, receive signals from the local environment that trigger stemness. In contrast, in the hematopoietic system, where cells are highly mobile, stemness is locked so that stimuli from the bone marrow microenvironment can only activate stemness in a highly restricted number of cells, in a way that does not interfere with differentiation. In cancer, one can shift from one category of stemness to another when tissues change. Cancer creates new affordances for stemness, allowing some cells to behave as stem cells independently of microenvironmental stimuli.

Another example is organ tropism in metastasis. Certain cancer cells have a tendency to metastasize to specific organs. For example, breast tumours often metastasize to the bones, lungs, and liver, while lung tumours frequently spread to brain, liver, bone and adrenal glands. Initially proposed in 1889 by Paget through his “seed and soil” hypothesis (Paget, 1889), we now understand the molecular mechanisms behind organ tropism (Langley & Fidler, 2011; Rondeau et al., 2019). We argue that affordances illustrate the seed and soil hypothesis in a less metaphorical way. Metastasis is a multi-step process. At each step, the environment either allows or blocks progression. In parallel, the cell either has, or lacks, traits to capitalize on that opportunity. One of the crucial insights of the “seed and soil” hypothesis is that metastatic spread is not only about the compatibility between cancer cells and a target organ but also about the routes that make dissemination possible in the first place. Cancer cells can only colonize organs they are able to reach, and this access is shaped by the blood and lymphatic vessels that connect the primary tumour to the rest of the body which are external constraints for the cancer cell. The vascular and lymphatic “conduits” thus constitute a decisive constraint: they create opportunities for some metastatic destinations while excluding others. Moreover, tumours actively remodel their vasculature through

angiogenesis and lymphangiogenesis, thereby opening new routes of dissemination. In addition, cells' internal constraints also define their affordances. For example, in the case of brain tropism, only cells expressing traits enabling blood-brain barrier crossing (e.g., expression of certain sialyltransferases[27]) and the ability to hijack astrocyte functions can access and thrive in the brain. Through this example, we aim to highlight the utility of the affordance framework in capturing the dynamics of cancer, which can ultimately contribute to the development of more effective therapies or preventive strategies.

Mainstream oncology, as we illustrated earlier, mainly focuses on the *consequences* of the cancer process, as opposed to the underlying mechanisms leading to these consequences. The popularity of the hallmarks of cancer is an example of this focus on consistent outcomes, such as sustained angiogenesis and tissue invasion & metastasis. But even if cancer is phenomenologically diverse, there are common alterations of the normal cell's life that justify unifying this diversity of cases under a single banner (Plutynski, 2018). The reason those outcomes are consistent is because they are changes to the cell that benefit the cell's fitness (Fortunato et al., 2016; Greaves & Maley, 2012). The popularity and utility of the hallmarks can be understood as identifying useful patterns, often found in mammalian tumours, but which, in our view, are just the result of specific affordances of mammalian tissues.

In order to make sense of the specificities of cancers, we propose to use the concept of affordances. In a nutshell, the concept of affordance allows us to combine the advantages of a universal definition of cancer with the potential to specify the peculiar and unique characteristics of each cancer. The next section will explain how this conceptual proposition can be used in clinical oncology.

3.3 The utility of our definition for clinical oncology

Up to this point, we have not explicitly articulated the benefits of adopting our specific universal definition of tumour. We argue that most importantly, it is beneficial for clinical oncology, because it makes sense of the heterogeneity of cases. Our universal definition provides a shift in perspective which helps human cancer biologists understand what the selective pressures and affordances in the human body are, and how they might be exploited for cancer prevention

[27] Enzymes that modify cell surface sugars, which can help cells interact with brain endothelial cells and cross the blood-brain barrier.

and therapy. Because affordances allow us to distinguish between primary and secondary properties, as explained above, they also tell us what sorts of therapy are available depending on what the target is. If the target is a primary property (related to the process of cell-level evolution) then the therapy should be universal, if the target is a secondary property, then the therapy should be specific. The fact that one can label some features as secondary properties which are therefore not fundamental to tumour does not mean that they do not constitute proper therapeutic targets; instead, this means that the target is a contingency of tumour due to local constraints and that this therapeutic solution will not work across all types. This likely explains why chemotherapies, which should in theory work on all proliferating cancers, in practice are only effective in a particular subset of cancer types.

While it is widely accepted in oncology that cell-level evolution plays a role in cancer, the definition we propose treats cell-level evolution not as a consequence of oncogenesis— like it is usually treated in oncology— but as its very process. The concept of immunoediting, for instance, is based on the premise that evolutionary dynamics at the cellular level drive an arms race between tumour cells and the immune system. Immunotherapies were unsuccessful for the most part until 2011, when Schreiber and colleagues (Schreiber et al., 2011) presented in vivo experimental evidence confirming how the immune system selects for less-immunogenic tumour variants. This demonstration proved the validity of Burnet and Thomas' hypothesis of immune surveillance and demonstrated that the immune system can suppress tumour growth which can also select for tumour variants that are better at evading immune responses (F. M. Burnet, 1971; M. Burnet, 1957; Thomas, 1982). Schreiber and colleagues proposed to move from the initial concept of immune surveillance to the refined concept of immunoediting precisely because they wanted to emphasize the complexity of this evolutionary process, by which tumour cells can be eliminated by the immune system, be in a state of equilibrium with the immune system, and ultimately escape immune recognition via the selection for resistant tumour cells (these stages are called the "3es" for elimination, equilibrium, and escape). It is the refined concept of immunoediting that has led to the development and refinement of cancer immunotherapies. In particular, checkpoint blockade therapies—such as anti-PD-1 and anti-CTLA-4 antibodies—were developed based on the understanding that tumours in the equilibrium phase exploit inhibitory pathways (e.g., PD-1/PD-L1) to evade immune destruction (Desai et al., 2022). By blocking these pathways, these therapies restore T cells' ability to effectively target tumour cells, shifting tumours from the escape phase back toward immune-mediated control. The key point for our purposes is that the validity of immunoediting relies on the recognition of tumour as a case of cell-level evolution. This is

evidence that the framework we propose through our definition can be beneficial for the clinic in that it changes working hypotheses and even the type of experiments which are conducted.

In addition of being a demonstration that our cell-level evolution framework is fruitful, the example of immunoediting can also be seen as an illustration of affordances. The human immune system relies on a set of proteins called the *major histocompatibility complex* (MHC) to help it recognize harmful changes in the body. In humans, MHC proteins are known as *human leukocyte antigens* (HLA). They work by displaying small fragments of proteins — called peptides — from inside cells to immune cells, so the immune system can check whether these peptides look normal or suspicious. Different individuals carry different HLA alleles, and each allele can present a specific set of peptides to the immune system. As a result, an individual's HLA profile influences which mutations are detected and eliminated in a developing tumour. This creates selective pressure for tumour cells to acquire mutations that produce peptides the host's HLA cannot recognize as abnormal. This means that the affordances of tumour cells in a given tumour are constrained by both tumour cells' mutations and the immune system's ability—or inability—to identify those mutations as threats (See Figure 6)

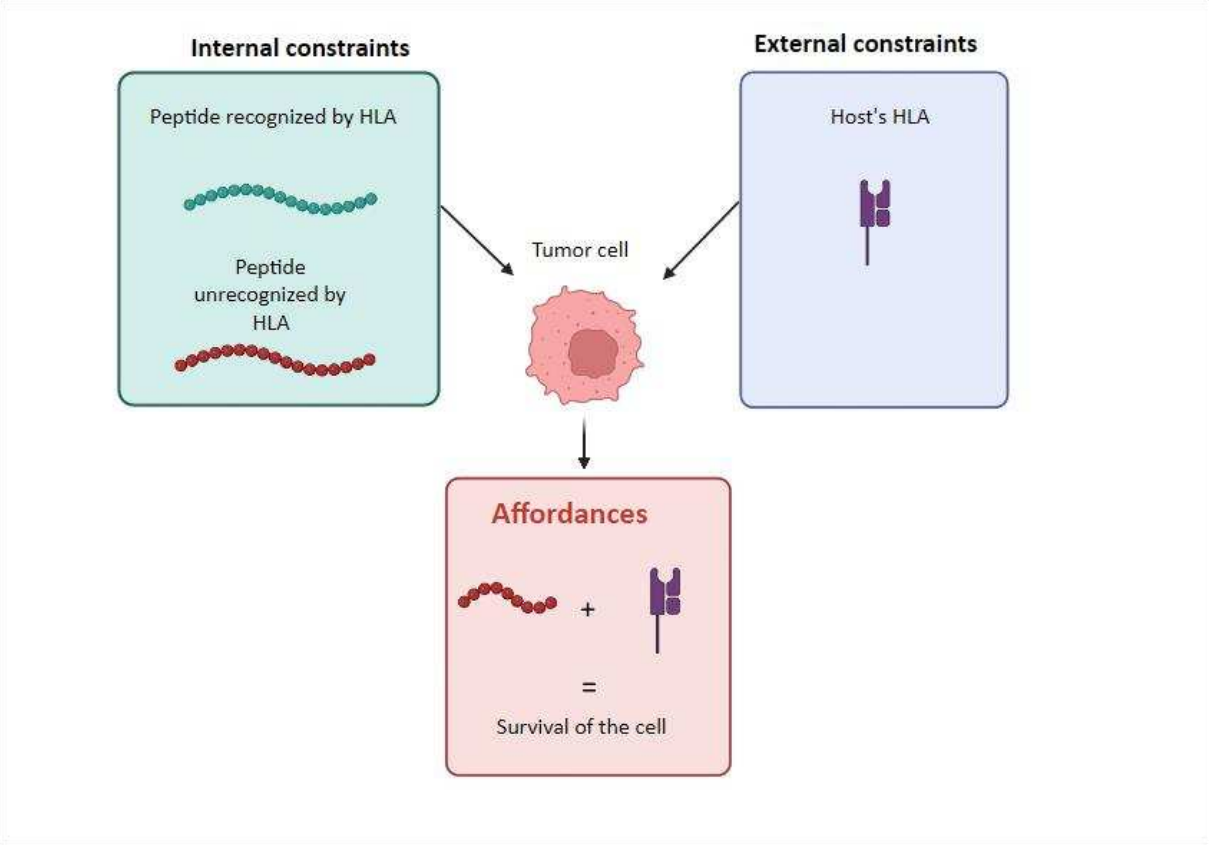

Figure 6. Affordances of a tumour cell in the context of immunoediting

Similarly to the hiker in the previous figure, a tumour cell is subject to internal constraints (e.g., the peptides it produces which depend on specific mutations) and external constraints (e.g., the host's HLA, which can recognize only a limited set of peptides). Together, these constraints define the cell's affordances—the actions it is capable of performing in its environment: in this case, either surviving or being eliminated by the immune system). Notably, the ability of a tumour cell to thrive is, of course, influenced by many additional factors beyond immune surveillance

Another example of the benefits of affordances to clinical oncology is that it can explain the differences between types of tumours within a human since different tissues and cells have different affordances. For example, because there are more constraints on solid tumours, they have to evolve more mutations compared to liquid tumours. Solid tumours must overcome barriers such as limited space for expansion within a tissue, the need to invade surrounding extracellular matrix, restricted access to nutrients and oxygen due to poor vascularization, and mechanical pressures from neighbouring cells and connective tissue. They also need to develop strategies to escape local immune surveillance concentrated in specific tissue environments. In contrast, liquid tumours circulate in the blood or lymph, giving them greater mobility, easier access to nutrients, and fewer physical barriers to proliferation. Because of these additional constraints, solid tumours typically

need to accumulate a larger set of mutations to be able to progress that is to say, they seem to have less affordances than liquid tumours.

Furthermore, this approach also helps to explain differences even between tumours within a particular type of tumour which is important in terms of therapeutic strategy. In addition to the stochasticity of which particular mutations occur, different people's genotypes and exposures provide different affordances for the cells evolving in their bodies. The combination of those random processes and the selective pressures on the evolving cells generate the diversity of tumours that evolve. This is important for personalized medicine since it makes sense of the diversity even within the same tumour type. An example of this are adaptive therapies, which aim not to eradicate all tumour cells but to manage the tumour as an evolving ecosystem. Inspired by ecological and evolutionary principles, adaptive therapy modulates treatment intensity to maintain a stable population of drug-sensitive cells, thereby suppressing the expansion of resistant clones. By treating the tumour as a dynamic evolutionary system, adaptive therapy contrasts with maximum tolerated dose strategies that often accelerate the emergence of resistant populations. This approach, pioneered by Robert Gatenby and colleagues, has shown promise in clinical trials, particularly in metastatic prostate tumour, as a way to prolong disease control while minimizing toxicity (Zhang et al., s. d.). Similar to immunoediting, adaptive therapies rely on the assumption that a tumour embodies cell-level evolution.

In section 3.1, I argued that non-universal definitions are narrow and therefore restrictive because they focus on specific features that do not always happen and in a comparative framework, only apply to a restricted number of species. In this section, I showed that the definition adopted here provides benefits for clinical oncology. In the upcoming section, I will demonstrate that it is also beneficial for comparative oncology since current definitions focus on often rare biological features that apply only to restricted groups of species.

3.4 The utility of our universal definition of cancer for comparative oncology

As defined in chapter 2, comparative oncology is a sub-field of oncology which studies cancer cases (or presumed cancer cases) in non-human organisms, with the intent that insights from one case may shed light on other tumours across species (Vincze et al., 2025). Explicit definitions of cancer are frequently absent in most comparative oncology research. For example, in a representative panel of 28 articles in comparative oncology, 13 provide no definition of cancer. Of

the remaining 15, 8 define it explicitly as a disease of multicellularity or frame their definitions within the context of multicellularity. The most prominent endorsement of this multicellular perspective is found in Aktipis and colleagues' article (2015), which we criticized in chapter 1. This paper has clearly shaped the field definition wise as most studies that define cancer within a multicellular framework cite it. However, this approach serves more as a theoretical framework than a precise definition or a set of criteria for determining what counts as cancer, and some researchers who use it continue to apply traditional criteria such as basal membrane invasion. Beyond this multicellular framework, the field currently lacks tools or definitions that accommodate organisms without basal membranes. In this section, we outline both why the limitations in current cancer definitions pose significant challenges for comparative oncology and also what the benefits of adopting a universal definition would be. Our argument is organized into three points.

3.4.1 Cancer as a universal phenomenon

In Aktipis et al. (2015), cancer was conceptualized as a "disease of multicellularity," occurring when "cheater" cells disrupt the cooperative cellular organization essential for host viability. This perspective, on which we commented earlier on, suggests that cancer represents a universal phenomenon, giving rise to the intuition that a broadly conserved disease exists across multicellular life. Part of comparative oncology subscribes to the idea that cancer — being a disease of multicellularity — virtually concerns all multicellular groups. However, the field is currently unable to test this intuition. Most of the field is currently studying mammals. The comparative method should offer a valuable framework to empirically test this intuition of cancer's universality. However, achieving this requires a comprehensive definition that can apply across different forms of multicellularity, facilitating cross-species comparisons. In addition, if we assume that cancer is to be found outside of mammals, we must make the bet that there is a basic process that is shared by all instances of cancer across species. In other words, we must presuppose an ontological unification. As we explained earlier in this chapter, one of our goals with our definition is to achieve such a unification.

3.4.2 The scope of study is unsatisfactory

As illustrated earlier in this chapter, many existing cancer definitions are anthropocentric, primarily designed around human biology. While it might fit with cancer in mammals, because their physiology is close to the human one, it poses certain problems in the case of non-mammalian organisms. For instance, several reviews about tumours in invertebrates highlighted the impossibility of applying the standard taxonomies of cancer to invertebrates because such taxonomies are framed on the basis of mammals' physiology (Harshbarger & Taylor, 1968; Scharrer & Lochhead, 1950; V. Wautier & Wautier, 1953). As a consequence, it becomes impossible to correctly identify instances of cancer in invertebrates, let alone in plants. In other words, these definitions generally work well for mammals, and possibly most vertebrates, but they may not apply to simpler organisms, such as placozoans (Fortunato et al., 2021), which lack many of the cellular structures found in higher animals. To enable meaningful comparisons, comparative oncology requires a definition that extends beyond vertebrates, supporting a broader and more inclusive approach.

Expanding the conceptual boundaries of cancer to include non-vertebrate organisms broadens the scientific toolkit, opening up new models for research and revealing previously unrecognized mechanisms of tumour development and suppression. Just as studies on elephants and naked mole rats have uncovered cancer resistance mechanisms, a wider phylogenetic lens could yield insights from organisms such as plants, sponges, and molluscs—species that have evolved distinct solutions to uncontrolled cell-level evolution. Moreover, a universal definition enables the development of environmental comparative oncology; by admitting the possibility of non-vertebrate organisms to develop cancer, the definition would allow the field to account for how pollution, ecological stress, and infectious agents influence cancer-like processes across entire ecosystems. For instance, it is well established that molluscs are widely exposed to microplastics (S. Zhao et al., 2024) and other environmental contaminants (Ochoa-Esteso et al., 2024; Pipe & Coles, 1995). However, the exact kind of pathologies this can induce is still understudied.

Finally, comparative oncology can be of utility for the clinic as reconstituting the evolutionary history of cancer can reveal the factors that make organisms more or less susceptible to develop cancer. However, this approach is obviously only possible if comparative oncology adopts a definition of cancer that can be applied broadly across the tree of life.

3.4.3 A universal definition as a methodological asset

A universal definition of cancer would serve as a valuable methodological tool. For instance, it could help botanists and researchers working with non-animal organisms recognize cancer-like processes in their fields. As mentioned earlier, there have been benefits in redefining what behaviour or consciousness are in non-human organisms. In the same way that it was a mistake to assume that all cognition presupposes a nervous system, we propose that it is a mistake to assume that all neoplasia presupposes invasion through a basement membrane. In the same way that non-neuronal cognition has now informed us about cognition, including human cognition, in decisive ways, we propose that non-basement membrane-dependent cancer has the potential to bring us decisive information on how cancer originated, how it is multiply realizable, and how we might eventually intervene on mechanisms that lead to cancer. This could lead to novel research questions, such as whether orthologs of human cancer genes function similarly in non-animal species.

Such a definition would also aid comparative oncologists in identifying homologies and functional parallels across species. The concept of affordances should enable us to make such parallels. By examining the similarities and differences in how various organisms enable or inhibit tumorigenesis, researchers could predict the types of cancer that might evolve in different biological environments.

Finally, a standardized definition would benefit comparatists by helping them understand the different affordances which lead to different manifestations of the same phenomenon. Understanding how cancer manifests across species could help clarify why certain tumour characteristics evolve as they do within the unique physiological context of the human body.

To summarize, this chapter has argued for the necessity and value of a universal definition of cancer grounded in the concept of cell-level evolution and enriched by the analytical framework of affordances. Such a definition captures the common causal process underlying all cancers while allowing us to make sense of their immense diversity across tissues, organs, and species. We have shown that this approach benefits clinical oncology by clarifying the distinction between primary and secondary properties of cancer, thus guiding more targeted and context-appropriate therapeutic strategies. We have also demonstrated its importance for comparative oncology, where it enables consistent identification of cancer across the tree of life, supports cross-species comparisons, and opens new avenues for understanding the evolutionary origins and constraints of tumorigenesis.

By integrating evolutionary thinking with an affordance-based perspective, our framework offers a conceptual and methodological tool that can benefit research programs with diverging goals, expand the scope of comparative studies, and ultimately enrich both our theoretical understanding and practical management of cancer.

Chapter 4

Chapter 4 : What potential future for comparative oncology?

I have suggested that cancer research currently faces a period of stagnation. While the precise reasons for this remain unclear, I propose that a fundamental issue lies in a misguided conceptualization of cancer. Adopting an alternative framework—particularly one grounded in evolutionary theory—could help overcome this stagnation. In that regards, evolutionary biology offers a useful framework for understanding cancer, as it provides both historical context and a comparative perspective across species. As was demonstrated in the previous chapters, comparative oncology represents the most promising approach for applying this evolutionary view. Yet, I also argued that comparative oncology has not yet been able to make significant translational breakthroughs. This raises the possibility that other frameworks may be better suited to address the practical challenges of cancer prevention and treatment. This does not diminish the value of an evolutionary account of cancer, however, since its strength may lie elsewhere: in its capacity to better characterize and define a phenomenon and to situate it within the broader context of life's history. Therefore, I am not suggesting that comparative oncology should abandon translational research altogether. Rather, I propose that, since this is not where its primary strengths lie[28], the field should direct more attention toward areas where its evolutionary expertise is best applied and most fruitful.

In our recently published article in *Nature Reviews Cancer* (Vincze et al., 2025) (reproduced in the Appendix of this PhD dissertation), we outlined potential future directions for the field of comparative oncology. The primary objective of that paper was to serve as an accessible and compelling introduction for both evolutionary biologists and cancer researchers: to encourage the former to consider cancer as a legitimate object of evolutionary inquiry, and to invite the latter to adopt an evolutionary perspective in their understanding of cancer. Given its persuasive intent and its multidisciplinary audience, we intentionally avoided overly speculative propositions. In contrast, this dissertation embraces speculation—not only because the format allows it, but because engaging in such conceptual exploration is a central task of philosophy. One of philosophy's contributions lies in offering a systematic mapping of issues. This kind of mapping entails a careful exploration of the full range of possibilities—not to propose irrelevant or arbitrary ideas, but to delineate an optimal space of inquiry. Such a space should be sufficiently general to

[28] Achieving therapeutic breakthroughs in cancer research is notoriously difficult, even within clinical oncology. Given that comparative oncology has only been pursuing this goal for about two decades, it is reasonable to expect that any substantial translational advances will take years, if not decades, to be developed.

capture all relevant dimensions of the issue, while remaining specific enough to avoid diverting attention to unproductive avenues.

Following philosophy of science's approach, the upcoming chapter will make proposals for future avenues in comparative oncology, in hope that it will help the elaboration of the field's research strategy.

4.1 Reconceptualizing cancer through evolution

The first and simplest proposition I want to make for comparative oncology is to invest more time into reconceptualizing cancer. As argued in chapter 1, organismal evolution is a framework of choice to reconceptualize cancer. However, in chapter 2, I showed how little comparative oncology, mostly because of its focus on the translational question, is actually trying to work towards such reconceptualization[29]. I would like to argue that, because of this focus, the field has neglected what I call foundational questions such as "what is cancer?" or "when did cancer appear?". Yet foundational questions are those which participate the most in reconceptualizing cancer since they provide context about its origins as well as cues about the nature of the process. Comparative oncology has undoubtedly been fruitful in terms of production of empirical knowledge, but the field has tended to significantly overlook its conceptual potential. In principle, evolution provides some of the best tools to propose new paradigms for a biological object. Thus, we should expect from comparative oncology to induce such a shift for cancer research.

To successfully exploit its potential, it is crucial for comparative oncology to propose new definitions of cancer which align better with the study of cancer across taxa, but which also have an impact on general oncology. A reconceptualization is efficient only if it enables the generation of new working hypotheses for the researchers studying the object that is reconceptualized. Such was our contribution in chapter 3. We proposed that a tumour (cancer being a tumour which becomes lethal) is an uncontrolled cell proliferation driven by cell-level evolution. This definition attempts to achieve the same goal as Aktipis et al., which is to provide a conceptual unification of

[29] To their credit, Aktipis and colleagues' conceptual proposition is an attempt to reconceptualize cancer and this attempt was successful in that it was cited many times within the small field of comparative oncology. Yet this does not invalidate my claim, as I have demonstrated that their conceptual framework contains several flaws. Moreover, I argue that a single attempt at reconceptualization does not fully harness the conceptual potential of an evolutionary account.

cancer. Conceptual unification in the case of cancer has two benefits. The first—though, in my view, the less important—is that it appeals to those who believe in the existence of a natural kind called "cancer." Our definition is broad enough to encompass all instances of cancer across the tree of life, yet not so broad that it becomes a catch-all concept, as in the proposal by Aktipis et al. However, I do not consider the notion of cancer as a natural kind to be particularly relevant for the present work. Debates about whether scientific categories correspond to real divisions in nature have little practical impact on researchers: they do not guide experimentation, nor do they help settle controversies over how a term should be used. As outlined in the introduction, the guiding principle of philosophy in science is the relevance of philosophical contributions to scientific activity; its goal is to contribute to the development of solutions to scientific problems (Pradeu, Lemoine, et al., 2024). In this light, questions about natural kinds are primarily ontological and, in my view, offer little value for cancer research. What matters more is whether a new conceptualization of cancer can influence how scientists understand and investigate the phenomenon.

The second benefit is that it facilitates the building of bridges between different domains of oncology. A shared conceptual framework functions as a common language, allowing researchers in disparate subfields to compare results more directly, to transfer insights or methodological innovations across contexts, and to integrate findings into a more coherent picture. In this sense, conceptual unification is less about ontological precision than about facilitating collaboration and translation between domains.

Finally, comparative oncology should not be reduced to its translational dimension. On the contrary, the field's most important contribution to cancer research right now could be conceptual unification. The field has a foundational role: it can generate knowledge about the fundamental features of cancer. Through its evolutionary perspective, it should work towards new definitions or characterisations of cancer based on what has been unearthed from the study of cancer across species.

In this dissertation, I have attempted to propose a way to reconceptualize cancer through evolution. My proposal aims to redefine cancer so that all instances of the phenomenon can be unified under a single concept. I am confident in the ability of both biologists and philosophers to develop alternative approaches to reconceptualizing cancer in evolutionary terms. Nevertheless, I hope that the justification developed in chapter 3 for my proposal will capture the interest of the comparative oncology community and that, in the absence of alternative proposals, some researchers may adopt it or build upon it.

4.2 From general to comparative oncology: questioning the paradigm

The conceptual framework is not the only element on which comparative oncology can induce a shift. Theoretical frameworks can also be flawed if they are based on inaccurate assumptions or if the overarching theory behind the field is incomplete or biased. In fact, a theoretical framework relies among other things on foundational principles, that is to say, beliefs, ideas, or premises that are accepted as true or valid, often serving as a foundational basis for the reasoning in the field they come from. I argue that theory-wise, comparative oncology possesses the tools to challenge assumptions in general oncology. For instance, comparative oncology has already disproved the assumption that cancer is primarily a human disease. Yet comparative oncology itself is not free from certain inadequate assumptions. Therefore, in addition to assumptions in general oncology that comparative oncology can help to challenge, I will propose assumptions within comparative oncology that the field itself needs to address.

4.2.1 Assumptions in general oncology

i) A more detailed molecular investigation of cancer will lead to a better understanding of the disease

This assumption underpins much of contemporary cancer research, which ranges from classical genetic experiments to large-scale 'omics' analyses and single-cell studies. While those approaches are important if properly used, the current phase of stagnation of oncology is a clue about the limits of relying on this level of study only. It gives us a knowledge of specificities of cancer (sometimes specificities of a certain type of cancer), but prevent the development of a broader, more integrated kind of knowledge. And if cancer is an instance of cell-level evolution, as we claim, then knowing the molecular and mechanistic details of the process is not telling us much about the basic nature of the process. This suggests that understanding cancer also requires attention to higher-order biological organization: tissue architecture, immune systems, evolutionary pressures, and species-specific life history traits. Comparative oncology can help to realize that more data is not always better data—and that broader conceptual models may yield greater explanatory power than deeper molecular zoom.

ii) Cancer is fundamentally a genetic disease

The idea that cancer is caused primarily by DNA mutations has dominated biomedical thinking for decades. While many recognize that cancer involves more than genetic changes, it is still widely assumed that its fundamental cause is genetic. If genetic changes are undoubtedly involved in tumorigenesis, comparative oncology shows that mutation alone does not fully explain cancer risk. As mentioned in chapter 2, some species maintain low cancer rates despite high mutational burdens. Bowhead whales (*Balaena mysticetus*) accumulate enormous numbers of somatic mutations over their long lifespans yet exhibit remarkably low cancer incidence which may be underpinned by enhanced DNA repair mechanisms (e.g., in ERCC1 and PCNA genes) (Firsanov et al., 2024). Comparative oncology has also revealed differences in cancer risk between major ecological groups—for example, between carnivores and herbivores—suggesting that ecological and metabolic factors, rather than genetic mutation rates alone, shape cancer susceptibility (Vincze et al., 2022). More generally, several mechanisms with cancer-protective effects highlighted by comparative oncology do not act at the genetic level (see Figure 1 in the Appendix) which reveal that cancer susceptibility depends not only on genetic mutation rates but also, critically, on factors such as DNA repair efficiency, immune surveillance, extracellular matrix properties, and secreted anti-cancer factors—all of which vary greatly across species. Comparative oncology thus reframes cancer as a system-level phenomenon, not merely the outcome of genetic mutations.

Together, those three assumptions constitute what one could call the current paradigm of oncology. If comparative oncology can successfully challenge them, it would demonstrate that the current paradigm of cancer research is flawed – or at least problematic. By doing so, comparative oncology would challenge the overarching assumption that the current oncology paradigm is correct providing a good orientation for cancer research. This would confirm the potential of evolutionary approaches to identify where oncology is mistaken at a theoretical level. As a philosopher, I cannot help but emphasize the importance of articulating theoretical and conceptual work with experimental research.

iii) Cancer is an enemy that must be eradicated.

Framing cancer as an enemy from within or an aberration to be “eliminated” reflects a military metaphor popularized by Richard Nixon in 1971 and which is still deeply ingrained in biomedical discourse. Most treatments aim at eradicating completely any trace of cancer. However, comparative oncology resituates cancer within a more general history, a life-history which involves selection and adaptation. Many animals, including humans, carry dormant or

harmless tumours throughout their lives without clinical consequences. Some species, like bats and whales, appear to deal with cancer thanks to mechanisms that manage rather than destroy abnormal cells (See Article in Appendix). Comparative oncology can shift the framework from eradication to management, proposing that controlling cancer, as some species appear to do, might be a more realistic and sustainable strategy than seeking complete elimination. The example of adaptive therapies which we mentioned in chapter 3 is a good instance of such shift.

4.2.2 Assumptions in comparative oncology

i) Cancer is fundamentally a genetic disease

In the last sub-section, I wrote that comparative oncology can, and to some extent does, challenge the assumption that cancer is fundamentally a genetic disease. However, it does not mean that the field itself is immune from such assumptions. In fact, comparative oncology remains largely dependent on the somatic mutation theory. Peto's paradox which has been a major topic of inquiry in the field is a paradox only if cancer incidence is relative to cellular division as more cellular division induces more mutations in key genes controlling cell proliferation, apoptosis, and DNA repair. This perspective is not without merit. Even a limited survey of somatic mutation rates across 9 species found a strong correlation between somatic mutation rates and cancer prevalence (Compton et al., 2025). The field inherits the gene-centric framing of general oncology by focusing on mutational burden, cancer-related gene families, and DNA repair pathways as primarily explanatory.

Many cross-species studies make use of comparative genomics to identify the reasons behind cancer resistance in certain species. For example, Caulin et al (2015) surveyed mammalian genomes and looked for positive correlation of tumour-suppressor genes with increasing body mass and longevity. Tollis and colleagues (2017) reviewed comparative genomics of cancer and explain that more efficient DNA repair, and a higher sensitivity to DNA damage, are recurring mechanisms across cancer resistant species. Overall, genes and mutations remain central in comparative oncology, and while the field does not reduce cancer solely to these factors, the influence of general oncology's gene-centric framework is still evident. I suggest that the field should stay mindful of this tendency to overemphasize gene-related questions and recognize that the comparative perspective offers precisely the kind of space where greater freedom and flexibility in inquiry are possible.

ii) Cancer manifests uniformly across all species

A frequent implicit premise in comparative oncology is that cancer represents a single, unified biological process in all multicellular organisms—defined by uncontrolled cell proliferation, tissue invasion, and metastasis. My attempt to produce a universal definition of cancer goes in this direction. However, I also demonstrated that because of the differences in affordances, cancer can vary in phenotypes across species. *Ergo* it is true that cancer is fundamentally the same thing in every species, but it is not true that cancer looks the same in every species. For example, coral tumours do not necessarily disrupt organismal function the way mammalian cancers do, and in some cases, they may not progress beyond benign tumours. This assumption—that cancer should manifest identically across taxa—is one reason why some remain sceptical about its occurrence in invertebrates. When cancer in these organisms does not resemble its mammalian form, it is often dismissed as not being "true" cancer. In subsection 4.3, I give arguments for extending the scope of comparative oncology to invertebrates.

iii) Resistance mechanisms in one species can be replicated directly into another

A recurring theme in comparative oncology is the search for mechanisms with cancer protective effects such as the extra *TP53* copies in elephants or the high-molecular-weight hyaluronan acid in naked mole rats. These adaptations are often discussed as though they could be engineered in humans to provide similar protection. However, such mechanisms exist within co-evolved physiological systems and replicating them as such may not provide the same benefits—or could even have deleterious side effects. For example, elephants' augmented *TP53* works within their specific cell turnover rates, immune system parameters, and developmental programs; in humans, increasing *TP53* dosage could have detrimental instead of beneficial effects: for instance, it could induce the appearance of early onset aging phenotypes (Tyner et al., 2002) or slower wound healing (Gannon et al., 2011). Likewise, naked mole rats' extracellular matrix properties are concomitant with their low-oxygen subterranean environment and unusually slow metabolism (Fang et al., 2014). In order to facilitate this work, I suggest that the concept of affordances, introduced in chapter 3, can be particularly useful. At first glance, the most reasonable strategy might seem to replicate cancer-protective mechanisms from species genetically closest to humans—namely, other mammals. However, recent surveys of cancer risk across mammals reveal striking disparities and suggest that lifestyle parameters like diet might play a critical role in shaping these disparities (Vincze et al., 2022). Such parameters act as constraints on biological

systems, creating different affordances for potential cancer cells. In other words, even closely related species may present very different possibilities for what cancer cells can or cannot do, depending on the ecological and evolutionary contexts in which such species evolved[30]. For example, the subterranean lifestyle of the naked mole-rat likely favoured the evolution of mechanisms that incidentally provide cancer resistance. Yet these affordances of the naked mole-rat cells are highly specific to its ecological niche and differ greatly from those found in humans. This makes it less plausible that naked mole rat mechanisms could be replicated in humans compared to mechanisms arising in species that share the relevant affordances with us. In other words, it would certainly be useful to give a more central role to the similarity of affordances when assessing the possibility that a cancer resistance mechanism identified in one species may also exist in another species.

4.3 Looking beyond mammals: expansion of the range of analysis

I explained why comparative oncology should attempt to reconceptualize cancer notably via a new definition; for example, one that would be better suited for comparative studies. However, I demonstrated in chapter 2 that comparative oncology is still mainly focused on vertebrates even though many biologists have been vocal since the middle of the 20th century about the possibility for invertebrates to develop cancer. This is obviously a consequence of the lack of conceptual work. Because the field relies on previous definitions of cancer which were framed for mammals, it is actually unable to increase its range of analysis outside of vertebrates. Hence, I argue that the field should explore cancer outside of vertebrates and dedicate more attention to invertebrates. There are several potential benefits to study cancer in invertebrates.

1) Several invertebrate models are already well established, and a wide range of experimental tools are available to study them. Invertebrate models such as *Drosophila melanogaster*, *Caenorhabditis elegans*, and various planarians have played a significant role in experimental biology, owing to their short generation times, genetic accessibility, and the availability of advanced molecular tools (J. Bolker, 2012; J. A. Bolker, 2009). These systems enable large-scale genetic screening, lineage tracing, and high-throughput drug testing, which are often more challenging to implement in vertebrate models. Research using *Drosophila* has shown its usefulness in exploring tumour growth, metastasis, and oncogenic signalling pathways,

[30] Ecological affordances could also explain intra-species variations as well due to the same ecological variations, e.g., population differences in diet.

suggesting that invertebrates may offer valuable systems for investigating conserved mechanisms relevant to cancer (Villegas, 2019). Including these organisms in the scope of comparative oncology could provide additional, cost-effective experimental platforms for studying cancer development and resistance in controlled settings—offering a useful complement to insights gained from vertebrate models.

2) Invertebrates could possess mechanisms with cancer-protective effects of a different kind than vertebrates. For example, organisms such as hydra and planarians display impressive regenerative abilities, which involve tightly regulated stem cell proliferation and apoptosis—processes which we know plays a role in tumorigenesis (Bode, 2003; Cebrià et al., 2011). In addition, specific features which are unique among basal invertebrates, namely the presence of decentralized nervous systems, the absence of complex vascular systems, and clonal reproduction may shape different constraints on somatic mutation accumulation and tissue maintenance. If cancer suppression mechanisms have evolved under distinct selective pressures in these taxa, studying them may offer insights into features that are less apparent—or absent—in vertebrate models.

3) If some invertebrate groups rarely or never develop cancer (e.g., cnidarians), then stronger evidence is required to establish this claim. The current absence of confirmed cancer cases in certain lineages may suggest—but does not yet demonstrate—that these organisms are resistant to cancer. For example, Fortunato et al. (2025) provide evidence that the lack of reports of cancer in sponges may indeed indicate a cancer suppression phenotype as they found that sponges are incredibly resistant to radiation. Without systematic empirical investigation, such an inference uncertain. As argued for by comparatists in the 50's, the criteria used to identify cancer in vertebrates—such as tissue invasion, metastasis, or disruption of organ function—may not apply well to this group, suggesting the need for revised definitions and alternative, more fitted diagnosing tools (Scharrer & Lochhead, 1950; V. Wautier & Wautier, 1953). Moreover, a negative result about the presence of cancer in invertebrate is as much telling as a positive result. It would raise important questions about the affordances of invertebrates which would lead to the absence of any form of cancer. In our research group in Bordeaux, especially through the work of Lea Mascherrek, we are trying to reconceptualize the criteria of cancer so as to make them usable across all metazoans, including invertebrates.

4) Just like immunology, cancer research is an area in which expanding the scope of investigation to include invertebrates could lead to the discovery of new phenomena. To illustrate this, I will examine the particularly relevant case of cell competition. Cell competition was initially

identified in Drosophila in 1975 when Morata and Ripoll (Morata & Ripoll, 1975) discovered that *Minute* Mutant drosophila had slower cell division rates and slower development due to defects in ribosomal protein genes. When mutant cells were placed next to wild-type cells in mosaic tissues, the wild-type cells consistently outcompeted and eliminated their slower-growing neighbours via apoptosis, even though the mutant cells were viable on their own. This led to the key insight that cells within a tissue can be compared in terms of their "fitness", and that less fit cells are actively eliminated, not simply outgrown (Marongiu et al., 2021). A major turning point occurred in 2004 when Moreno and Basler (Moreno & Basler, 2004) demonstrated that *Drosophila* cells overexpressing *dMyc* could eliminate surrounding wild-type cells, branding them as "super-competitors. By 2008, the idea that cell competition might underlie early tumorigenesis was formally proposed, along with potential therapeutic strategies. In fact, even though several obstacles could be foreseen, Moreno hypothesized that cell competition markers could help identify pre-tumoral lesions, allowing intervention very early in the oncogenesis process. From 2008 to 2016, studies expanded this concept beyond flies to mice, zebrafish and finally human cells cultures. Importantly, cell competition was shown to have a dual effect—eliminating harmful or unfit cells in some contexts, while promoting tumour growth in others, particularly via oncogenic pathways like *MYC* and Hippo (Di Gregorio et al., 2016). Those discoveries led to the study of cell competition in human cancers, where similar principles were found to apply. In particular, oncogenic drivers such as MYC and deregulation of the Hippo pathway were shown to endow human epithelial cells with competitive advantages, enabling them to displace their normal neighbours (Parker et al., 2021). Evidence from human organoid models and epithelial tissues such as the skin, oesophagus, and colon has confirmed that competition contribute to clonal selection in premalignant tissues, helping explain how normal tissues gradually evolve toward cancer (Parker et al., 2021; van Neerven & Vermeulen, 2023).

Thus, this example constitutes a good and inspiring example for anyone interested in exploring cancer and cancer-related processes in invertebrates, with a potential application to vertebrates as well. In other words, my objection to many compartments of comparative oncology as it has been developed in the last two decades is that it has remained mammal centric, whereas its full potential resides in adopting a very broad approach to comparisons across taxa. Comparative oncologists who argue that the reason for being focused on mammals is that similarities between humans and mammals are higher than those between humans and invertebrates reiterate a mistake that has been made many times in comparative biology, i.e., the assumption that more similarities mean a higher probability of finding something important and useful for understanding cancer in humans. In fact, we know today loads of things about cancers

in mice, but the capacity to translate this knowledge into human cancers has been incredibly limited and, overall, largely unsuccessful. Why not consider the example of cell competition above as another (and, of course, not mutually exclusive) fruitful avenue, where a phenomenon discovered in invertebrates turns out to be extremely useful for understanding human cancers – despite the myriads of biological differences between *Drosophila* and humans?

5) Finally, research on insects such as *Drosophila melanogaster* has been extraordinarily valuable for advancing human medical oncology. Flies provided the genetic blueprints for several of the most critical cancer pathways—including Hedgehog, Notch, and Hippo—and clarified how Ras/EGFR, PI3K/TOR, and JNK signalling drive tumour growth, invasion, and therapy response. These insights are not just theoretical: they directly underpin approved cancer drugs and inform multiple ongoing clinical trials. Importantly, about 75% of known human disease genes have clear *Drosophila* counterparts, making it possible to model conserved aspects of cancer biology in a whole organism both rapidly and at low cost. This unique combination of genetic tractability and experimental efficiency allows researchers to screen drugs and therapeutic combinations in vivo before moving to more complex and expensive mouse or patient studies, accelerating the pipeline from discovery to treatment.

4.4 Expanding the horizon of comparative oncology: towards a transdisciplinary framework

I have just outlined why comparative oncology should extend its scope beyond vertebrates to include at least some invertebrate species. Yet even with this expansion, the field's overall scope would remain limited—particularly if it does not broaden its methodological toolkit. As was explained above, the field is still partly shaped by a gene-centred framework. Therefore, to enrich its repertoire and move beyond the dominant paradigm of oncology, comparative oncology could become more transdisciplinary by incorporating insights and methods from other disciplines. Complementary to this, cancer research could benefit from integrating comparative oncology's methods and findings into other areas that study the disease.

4.4.1 Incorporating other disciplines into comparative oncology

Immunology:

Instead of focusing solely on immunological mechanisms that could be adapted for human medicine, I suggest that the field could investigate how immune systems in diverse taxa coevolved with the risk of neoplasia in their unique ecological and evolutionary contexts. This mapping should include cellular composition, immune checkpoint repertoires, and functional responses to DNA damage and tumour antigens. It would feed into a broader theory of how immune systems maintain a balance between organismal integrity and cellular proliferation.

In order to help comparative oncology to advance on translational grounds, the collection and comparison of immune system data across different animals could unveil new cancer-resistance strategies that evolved in the wild, like extrusion of damaged cells that placozoans appear to use (Fortunato et al., 2021). These insights might lead to new ways to prevent or treat cancer in humans.

Biology of metabolism:

Metabolism also plays a key role in oncogenesis as tumour initiation and progression rely heavily on metabolic reprogramming (Martínez-Reyes & Chandel, 2021). By situating metabolism in the framework of energy balance and life-history trade-offs, researchers can develop models predicting how resource allocation influences the balance between somatic maintenance and tumour suppression. This would allow the field to explain variations in cancer risk without relying on a gene-centred framework as Kapsetaki et al. (2025) did by looking at the potential link between high blood glucose levels in birds and their low cancer prevalence. In the same line as my proposition above, building a mapping of metabolic profiles across species and linking these profiles to life-history traits and cancer incidence could reveal whether particular metabolic regimes, like those seen in long-lived low-cancer species, provide resistance against cancer.

Microbiome studies:

Integrating microbiota science into comparative oncology reframes tumours as ecosystems that include not only cancer and stromal cells but also microbial actors. It offers a way to address cancer's ecological and evolutionary complexity. It opens novel avenues for cross-species therapeutic strategies such as dietary modulation, probiotics, or faecal microbiota transplantation.

Different species host distinct microbial communities shaped by diet and the environment, providing experiments to test how microbiota influences cancer risk and resistance across the tree of life. This perspective may even participate to explain Peto's paradox by examining whether

unique microbial adaptations contribute to cancer resistance in large, long-lived animals. In translational contexts, companion animals like dogs, which share human environments and microbiomes, can help to understand how microbes shape tumour progression.

Developmental biology:

Some people have proposed to reconceptualize cancer as a developmental disorder (Da Costa, 2001; Rubin, 1985) – extending in fact an idea that has been suggested as early as the 18th century (Rather, 1978). In fact, regeneration and tumour growth both involve activation of developmental programs. By comparing developmental signalling pathways (e.g., Wnt, Notch, Hedgehog) in species with high and low regeneration capacities, researchers could uncover affordances of development that either constrain or promote oncogenesis. Such work could clarify whether there has been a trade-off between high regenerative abilities and cancer risk. By comparing how different species use, suppress, or repurpose developmental programs, researchers may reframe cancer as a potential byproduct or misexpression of conserved developmental mechanisms. In a narrower sense, tumours often recapitulate embryonic processes such as rapid proliferation or loss of differentiation suggesting that cancer may involve the abnormal expression of conserved developmental mechanisms. Comparative oncology could thus examine how different species stabilize or silence these pathways once ontogeny is complete, and whether variations in developmental timing or plasticity alter cancer susceptibility. In fact, differences in heterochrony (the timing of developmental processes) across species could shape when and how tissues are vulnerable to oncogenesis. So, for instance, long-lived species with slow development may have more robust control on cell cycle exit; short-lived species with rapid growth may be more vulnerable to oncogenic mutations during development.

Another way to link development and cancer is through the balance between developmental plasticity and canalization. Plasticity means that development can produce flexible outcomes, while canalization refers to stability and robustness (Hallgrímsson et al., 2002). Cancer can be seen as what happens when this stability breaks down, letting cells grow or change identity without control. Comparative oncology could explore this by looking at species with different balances. Mammals, with more canalized development may maintain tighter control over cell fate decisions and tissue architecture, potentially lowering cancer risk relative to species like planarians (Sánchez Alvarado, 2000) or hydra (Bosch, 2009) where developmental pathways remain flexible or open-ended into adulthood. Studying these contrasts could show whether cancer risk depends in part on how tightly development is kept under control across different lineages.

By comparing how species use, suppress, or repurpose developmental programs, both in regeneration and in embryogenesis, researchers may reframe cancer as a potential byproduct of developmental strategies that are deeply embedded in the evolutionary history of multicellular organisms.

Ecology:

In the past five years, comparative oncology developed a more systematic knowledge of cancer risks across species (Compton et al., 2025; Vincze et al., 2022). However, such knowledge remains often limited to domesticated or captive animals. In addition, researchers have argued that tumour progression and metastasis share similarities with how individuals interact with ecological systems (Polyak et al., 2009). For example, cancer cells colonizing new niches and constructing their microhabitat is analogous to niche construction in ecology (Reynolds et al., 2020). Therefore, I argue that comparative oncology could be enhanced with methods from systems ecology. For example, by treating tumours as ecosystems, researchers can analyse how host-specific variables—such as immune systems, tissue architectures, and microbiota—shape tumour–host interactions in different animals (Korolev et al., 2014). This perspective highlights the importance of tumour microenvironments: as Amend and Pienta (2015) argue in their "cancer swamp" analogy, cancers generate toxic ecological niches that select for the most aggressive clones. Comparative oncology can examine whether some species have evolved mechanisms to prevent such swamp-like conditions from forming, offering insight into strategies of microenvironmental regulation. Finally, ecological theory enriches the evolutionary dimension of comparative oncology by suggesting therapeutic parallels. Korolev et al. (2014) propose that tumours might be "steered" into less aggressive states by manipulating ecological interactions; comparative oncology can test whether certain species already achieve such steering, for instance by immune modulation or stromal competition. Thus, ecological approaches expand comparative oncology's capacity to identify not only how cancer varies across the tree of life, but also which ecological strategies could inspire human therapies.

4.4.2 Incorporating comparative oncology into other disciplines

Application of comparative oncology to onco-immunology:

Comparative oncology has already identified immune-related cancer resistance traits in certain species, such as the enhanced apoptosis triggered by elephants' multiple TP53 copies or

the possible role of high-molecular-weight hyaluronan in naked mole rats' suppression of cell proliferation. These findings could inform onco-immunology by highlighting naturally evolved immune responses that prevent tumours from escaping control. Rather than developing new immune strategies from scratch, immunologists could use these cross-species observations as models for how immune surveillance and tissue maintenance can co-evolve with long lifespan.

Application of comparative oncology to neuro-oncology:

Veterinary and wildlife oncology have recorded the occurrence of central nervous system (CNS) tumours across a variety of species, from gliomas in dogs to meningiomas in cats and brain tumours in wild mammals (Baba & Câtoi, 2007; Schiffman & Breen, 2015). This comparative oncology evidence could help neuro-oncology assess whether certain neural cancer types are widespread across mammals or instead cluster in specific ecological or life-history contexts. These cross-species records could also help identify traits — such as neural regenerative capacity or blood–brain barrier structure — that correlate with lower CNS tumour incidence.

4.5 Why comparative oncology needs to explore more small-scale mechanisms

As seen in chapter 2 and illustrated in our *Nature Reviews Cancer* article (see Article in Appendix), comparative oncology is an interdisciplinary field which makes use of a variety of methods. Some methods operate at small scales, which is the case of the research conducted by the Gorbunova and Seluanov's lab, for example. Comparative oncology could invest more work into the very small-scale level as it underpins precise mechanisms of oncogenesis. For instance, even though I claimed that the hallmarks of cancer are contingencies of cancer in organisms with the required affordances, they remain essential for understanding how malignant transformation unfolds in different biological contexts. Studying these processes comparatively in many species at the cellular and molecular level—such as alterations in cell cycle control, DNA damage responses, or intercellular communication—can reveal which hallmarks are conserved across species.

To conclude, the future of comparative oncology lies in embracing its dual role as both a conceptual and an empirical enterprise. Its greatest strength is not necessarily to make translational progress but to reconceptualize cancer as a disease with an evolutionary history and to challenge recuring and unaddressed assumptions in today's oncology. However, this research agenda

requires extending the taxonomic scope of study beyond vertebrates, incorporating knowledge and methods from other disciplines without losing sight of the evolutionary perspective. By doing so, comparative oncology can move beyond a mammal centric and gene-centric focus, contributing to a more complete scientific understanding of cancer

Chapter 5 Conclusion

This dissertation set out to investigate what the study of the organismal evolution of cancer can contribute to cancer research. In doing so, it aimed not only to survey the emerging field of comparative oncology but also to provide a conceptual and philosophical framework capable of clarifying what is at stake when cancer is studied across the tree of life. The central claim defended here is that an evolutionary perspective at the organismal level does not only add a new layer of empirical data but also forces us to reconceptualize the very object of inquiry, i.e., cancer itself.

The four chapters of this dissertation developed this claim in complementary ways. The first chapter showed that current evolutionary approaches, and in particular the influential framework that conceives cancer as a breakdown of multicellularity conceptualized as "cheating," remains insufficiently precise and overly permissive. Authors involved in this approach have succeeded in drawing attention to cancer' broad distribution across the tree of life, but the conceptual framework they proposed is unsatisfactory. The second chapter traced the history of comparative oncology, from its "pre-history" characterized by a natural history approach to its present-day "evolutionarization," and highlighted the heterogeneity of its goals. It showed how the field, while still consolidating itself, uniquely provides the only empirical and theoretical space within which organismal evolution of cancer can be fruitfully explored.

The third chapter, which constitutes the core contribution of this dissertation, argued for a universal definition of cancer grounded in evolutionary reasoning. By combining the comparative perspective with a process-oriented understanding, it proposed a way of overcoming the anthropocentric and fragmented definitions that have long dominated oncology. This definitional work illustrates how philosophical analysis can serve science by providing conceptual clarity, unifying heterogeneous phenomena, and enabling more robust comparisons (especially across different scientific fields and approaches). Building on this, the fourth chapter proposed a series of avenues that comparative oncology could pursue in the near future. Drawing on the critiques and analyses developed in earlier chapters, I argued for directions that could strengthen the field—both by embracing its conceptual potential and by adopting new empirical practices. In this way, the dissertation moved from critique to constructive proposal.

Taken together, these chapters demonstrate that the organismal evolution of cancer provides a powerful framework with the potential to reshape cancer research conceptually. By situating cancer within the broader context of life's evolutionary history, we gain access to questions that are otherwise neglected: When did cancer first emerge? To what extent has it shaped the life-

history strategies of different taxa? What conserved features of cancer are shared across species, and how should these inform our definitions? Such questions not only enrich the evolutionary biology but also contribute to clarifying the aims and scope of comparative oncology.

Of course, this work has its limitations. It does not propose any major theoretical shift like philosophy can sometimes achieve, nor does it resolve any empirical challenge that comparative oncology faces. Its contribution lies primarily in the academic and conceptual sphere: clarifying definitions, producing an exhaustive and comprehensive mapping of comparative oncology, and identifying where philosophical work can most productively penetrate research on organismal evolution of cancer. Future work could build on this foundation by refining the universal definition proposed here, and by making use of this attempt at conceptual unification to bring greater coherence and consensus across cancer research.

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Perspective

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Advancing cancer research via comparative oncology

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Abstract

In the ongoing battle against cancer, the natural world provides promising inspiration for designing novel therapeutic strategies. The field of comparative oncology offers a valuable source of such inspiration. By combining evolutionary biology, ecology, veterinary medicine and clinical oncology, comparative oncology aims to better understand cancer, especially by highlighting taxa that are strongly resistant or susceptible to cancer and to identify the molecular and cellular mechanisms underlying the remarkable cancer resistance of some taxa. Such studies hold profound implications for human cancer research and treatment, and increase the probability of detecting therapeutic avenues that are non-toxic to healthy cells and tissues. This Perspective underscores the importance of comparative oncology, emphasizes its relevance, and showcases recent breakthroughs in identifying natural cancer resistance mechanisms and opportunities for clinical translation. We advocate for a better integration of cancer research on non-conventional model species into oncology and we urge enhanced cooperation between clinicians and comparative oncologists to advance cancer prevention or treatment strategies.

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Introduction

The landscape of cancer research is evolving rapidly, driven by the mounting prevalence of the disease and by new hopes of finding effective treatments for patients. The field of comparative oncology broadens the conventional view of cancer by shedding light on manifestations of this disease across the tree of life, including mechanisms of its initiation and progression and different resistance mechanisms on a broad scale. Comparative oncology has two main driving goals. The first is to offer a deeper and more integrative understanding of what cancer is. The second is to find novel treatments for patients, by drawing inspiration from natural cancer-resistance mechanisms that have evolved in diverse species. Comparative oncology has deep historical roots. Comparisons of tumorigenic processes and discussions over possible common aetiologies in humans, animals and even plants have been taking place since the late nineteenth and early twentieth centuries (for example, refs. 1,2). Nonetheless, this approach has been largely overlooked, hindering the possibility of drawing fruitful lessons for the clinic. Thus, it is now critical to showcase the value of comparative oncology in order to promote crosstalk between cancer researchers and oncologists.

In this Perspective, we reflect on 30 years of modern comparative oncology and highlight five key insights that can be gleaned from studying cancer across a variety of species. We underscore the invaluable contribution of both truly comparative studies (that is, offering comparisons over a wide range of taxa) and species-specific investigations (that is, focusing on individual non-standard model organisms with unique cancer risk or resistance mechanisms). These approaches not only advance our understanding of cancer from molecular, physiological and evolutionary perspectives, but may also serve as a source of inspiration for innovative cancer therapies in both human and veterinary oncology.

Cancer is an ancient disease

Cancer represents the second leading cause of mortality in humans, with incidences projected to increase in the coming decades[3]. However, contrary to common perception, cancer is not a modern disease. The ancient history of cancer is emphasized by fossilized and mummified records of neoplasia. Evidence of tumours has been found in extinct species such as dinosaurs[4,5], giant sloths[6] and ancestral turtles[7], where bones dating from 150–240 million years ago are suspected to have been affected by cancer, sometimes with morphology characteristic of human metastatic tumours[5]. In a study comprising around 10,000 bone specimens, it was shown that cancer affected dinosaurs from the Hadrosauridae family, with evidence of metastases in the *Edmontosaurus* genus[4].

More closely related to modern humans, evidence of cancer has been repeatedly found in extinct species of *Homini* living 1.8–1.6 million years ago[8]. The oldest written mention of cancer in our own species lies in the Edwin Smith Papyrus, a medical text from ancient Egypt probably written in 3,000 Before Common Era[9]. It describes eight cases of breast tumours that were removed by cauterization. More recently, a rectal cancer has been diagnosed in an Egyptian mummy of the Ptolemaic period (200–400 common era (CE))[10,11].

The scarcity of examples of cancer in ancient times should be understood in context. Most tumours do not affect the bones (which is the only tissue that is generally well preserved during fossilization) or affect the bone interiors, making it more difficult to detect them in ancient specimens. Challenges lie in distinguishing actual bone tumours from other bone-related diseases or injuries in fossils, with soft tissue malignancies being even more difficult to determine based on ancient skeletal remains[12]. Furthermore, bone tumours increase the fragility of skeletal elements, leading to a decreased probability of preservation[6]. Despite these challenges, the available data strongly indicate that cancer has existed for millions of years in a wide range of taxa.

Cancer is widespread across animals

Whereas clinical oncology logically focuses on humans and their close mammalian relatives, the premise of comparative oncology is that much may be learned by extending the scope of cancer research across a broader phylogenetic spectrum. Cancer is known to occur in a very wide range of non-human animals. In non-mammalian vertebrates, unambiguous cancers are found in birds, amphibians, reptiles and fish[13]. Cancer-like phenomena are also seen in invertebrates, as described below[1,14].

Cancers in non-domestic animals are increasingly documented through surveys based on data from zoos[15] and recently from wild populations[16]. Crucially, zoo-based studies have shown substantial variance in cancer risk in different species[15,17]. For instance, certain carnivorous marsupials have an exceptionally high risk of cancer-associated mortality (as much as 57% in the Kowari), whereas even-toed ungulates (such as deer or giraffes) generally have very low risks[15]. Carnivores overall are more prone to cancer-associated mortality than animals with other lifestyles, which raises interesting perspectives on the impact of diet on cancer. Among many other insights, these surveys highlight animals that show unusually high or low cancer rates and can prompt targeted studies to uncover the underlying mechanisms of cancer susceptibility or resistance (see the related discussion below). Moreover, comparative studies underscore how animals across the phylogenetic tree exhibit very different patterns of neoplasia (Box 1), not just in cancer risk, but also the proportion of benign and malignant tumours[18], as well as the risk of mortality associated with these[15]. For instance, birds, reptiles and amphibians appear better protected against all tumour types, including malignant ones, than mammals[18,19].

What is less clear is whether we consider cancer to exist beyond vertebrates. To address this, we must start with definitions of cancer that encompass the diversity of metazoan body plans, and which can be satisfied by evidence from basic pathology. One simple definition of cancer is an inappropriately proliferating group of cells that grows invasively out of its site of origin. By this criterion, cancer can indeed be found in invertebrates. The most species-rich group of animals on Earth are the insects, which include the fruitfly (*Drosophila melanogaster*) where both naturally occurring and experimentally induced tumours match the aforementioned definition[20–22]. Tumours have been documented in other arthropods, as well as Hydra[23,24], mollusks[25] and corals[13], but some of these do not meet all the criteria for the cancer definition provided above (for example, calicoblastic neoplasms of corals or tumours in general in Hydra lack invasive behaviour). Evidence for tumours is currently lacking in phyla such as echinoderms (for example, starfishes), nematodes and sponges, although more systematic studies may reveal their presence. Thus, more work is needed to determine the true scope of cancer across the animal kingdom.

Fruitflies, the best-studied host of cancer among invertebrates, highlight issues that arise when considering species less closely related to humans[21]. Fly tumours share many characteristics with human tumours, including uncontrolled division of immortalized cells, defects in differentiation, destruction of the basement membrane and

migration into neighbouring tissues. Fly tumours also induce remarkably analogous tumour–host interactions, including the co-option of an oxygen supply, recognition by immune cells, and the ability to induce systemic (paraneoplastic) cachexia and coagulopathy (impaired blood clotting)[20]. However, unlike human tumours, fly tumours can result from transformation through only one or two genetic insults and are genetically relatively stable; moreover, as insects do not have a vascular circulatory system, whether they metastasize is ambiguous[26,27]. Thus, it is important to keep in mind the particular anatomy and lifestyle of the organism under scrutiny when looking for cancer features.

Nevertheless, the basic cellular processes that are corrupted by transformation in human tumours are fundamental to all animal cells. Moreover, the tumour-suppressor genes (for example, *TP53*) and proto-oncogenes (for example, Ras genes) that are frequently mutated in human cancers are highly conserved throughout the metazoan lineage, where they play very similar roles. Combined with the widespread tumours documented throughout animal phylogeny, these data support the idea that a comparative approach can shed light on both the ancient origins of tumour formation and the host response to a tumour.

Evolution of cancer-resistance mechanisms

All things being equal, we might expect that the likelihood of accumulating mutations would increase with the number of cells and cell divisions within an organism. Therefore, cancer risk should increase with both body size and lifespan. Nonetheless, the existence of animal species such as whales and elephants that show very low cancer incidence despite lifespans of several centuries or body sizes of up to 100 tons represents a puzzle and highlights that some species have evolved remarkable cancer-resistance mechanisms. Following the foundational work by Richard Peto[28], this phenomenon has been coined 'Peto's paradox'[29]. Some smaller animals including naked mole rats or multiple species of bats have also evolved remarkably low cancer rates[30,31]. Although data from the wild are extremely scarce, the exceptional cancer resistance of certain species, both large and small, is widely acknowledged. Molecular or cellular mechanisms contributing to resistance in these animals represent a rich resource for knowledge and inspiration for novel cancer prevention or treatment strategies.

The number of natural cancer-resistance mechanisms to be found is probably large, as indicated by the diversity of mechanisms identified so far (Fig. 1). Some mechanisms are general to a wide range of organisms, including through the elimination of expanded clones by normal tissues via cell competition, apoptosis or anoikis (apoptosis owing to detachment from the extracellular matrix), and stabilizing selection within young healthy tissues to limit selection for oncogenic mutations or epigenetic changes, as well as others[32,33]. Other mechanisms are more specific to a single species or a handful of closely related taxa, and the primary mechanism can vary from more efficient prevention of cancer (for example, by early contact inhibition (where cell growth and/or migration is slowed or halted when cells come into contact with other cells) in naked mole rats[31], extrusion of tumours from healthy tissue[34], repression of somatic telomerase activity[35] and replicative senescence in large bodied taxa[36,37]) to improved detection and elimination of damaged cells (for example, p53-mediated apoptosis in elephants[38], attenuated DNA methyltransferase 1 (DNMT1) activity and subsequent derepression of transposable elements in response to hyperplasia in the blind mole rats[39], and efficient DNA repair in damaged cells for example, in bowhead whales[40]). All these mechanisms can be framed within what should be an obvious first principle — that animals have evolved mechanisms that limit the fitness impacts of rogue somatic evolution. Thus, natural selection at the organismal level has driven natural selection at the somatic level.

Even intensively studied and more traditional cancer-resistance mechanisms can benefit from the comparative approach. For instance, the immune system and its associated 'cancer immunosurveillance' activity[41] are a current frontier of therapy[42,43]. Foundational work in the 1990s in fruitflies led to insights into the origins of immune responses in animals[44,45], and immune elimination of tumours was recently described in *D. melanogaster*, via both phagocytosis by macrophage-like cells[46] and a humoral-like response based on antimicrobial peptides[47]. In parallel, it has become increasingly clear that immune-based anticancer mechanisms in mammals involve innate as well as adaptive immunity, providing an important complement to the exclusive focus on adaptive immunity of the original immunosurveillance hypothesis[48]. As research turns towards harnessing innate immune responses to cancer in humans[49], studying immunological elimination of non-mammalian cancer opens up the possibility of testing hypotheses of conserved mechanisms and exploring informative connections, such as the parallels between immunological responses to cancer and to persistent wounds[20,50].

A combination of complementary methods

From its deep historical roots, the field of comparative oncology has expanded substantially in the past two decades, in at least two ways. First, it has become truly comparative, in the sense that studies today incorporate data on many taxa (Table 1). Second, it has capitalized on cutting-edge approaches and technologies, especially at the genomic level. Consequently, we now exist in an era when comparative oncology can truly flourish.

The past decade has seen an exponential growth of publicly available databases that allow for cross-species explorations of cancer risk

Box 1 | Transmissible cancers

Whereas most cancers arise within and die with their host, a handful of transmissible cancers have evolved the ability to spread between individuals and continue proliferation and tumour-formation in new hosts. These cancers challenge the conventional definitions of cancer and provide unique insights into immune evasion strategies and clonal evolution of cancer cells. Though absent in humans and generally rare among vertebrates, three lineages of transmissible cancers are currently known in mammals: two in Tasmanian devils (*Sarcophilus harrisii*) known as devil facial tumour disease (DFTD1 and DFTD2) and one in domestic dogs (*Canis lupus familiaris*) known as canine transmissible venereal tumour (CTVT). Both DFTD and CTVT cells evade host immunity and allograft rejection through downregulation of major histocompatibility complex (MHC) molecules. For example, DFTD cells reversibly suppress MHCI expression via epigenetic modifications of histones, effectively becoming 'invisible' to T cells[83]. Similarly, CTVT cells transiently lose MHCII expression during transmission to avoid immune rejection[84]. These adaptations parallel findings in human cancers, where MHC loss is a hallmark of immune escape in metastatic and therapy-resistant tumours[85]. Studying transmissible cancers provides a unique perspective to investigate how malignant cells overcome immune barriers, a process critical to understanding metastasis and immunotherapy resistance in human cancers.

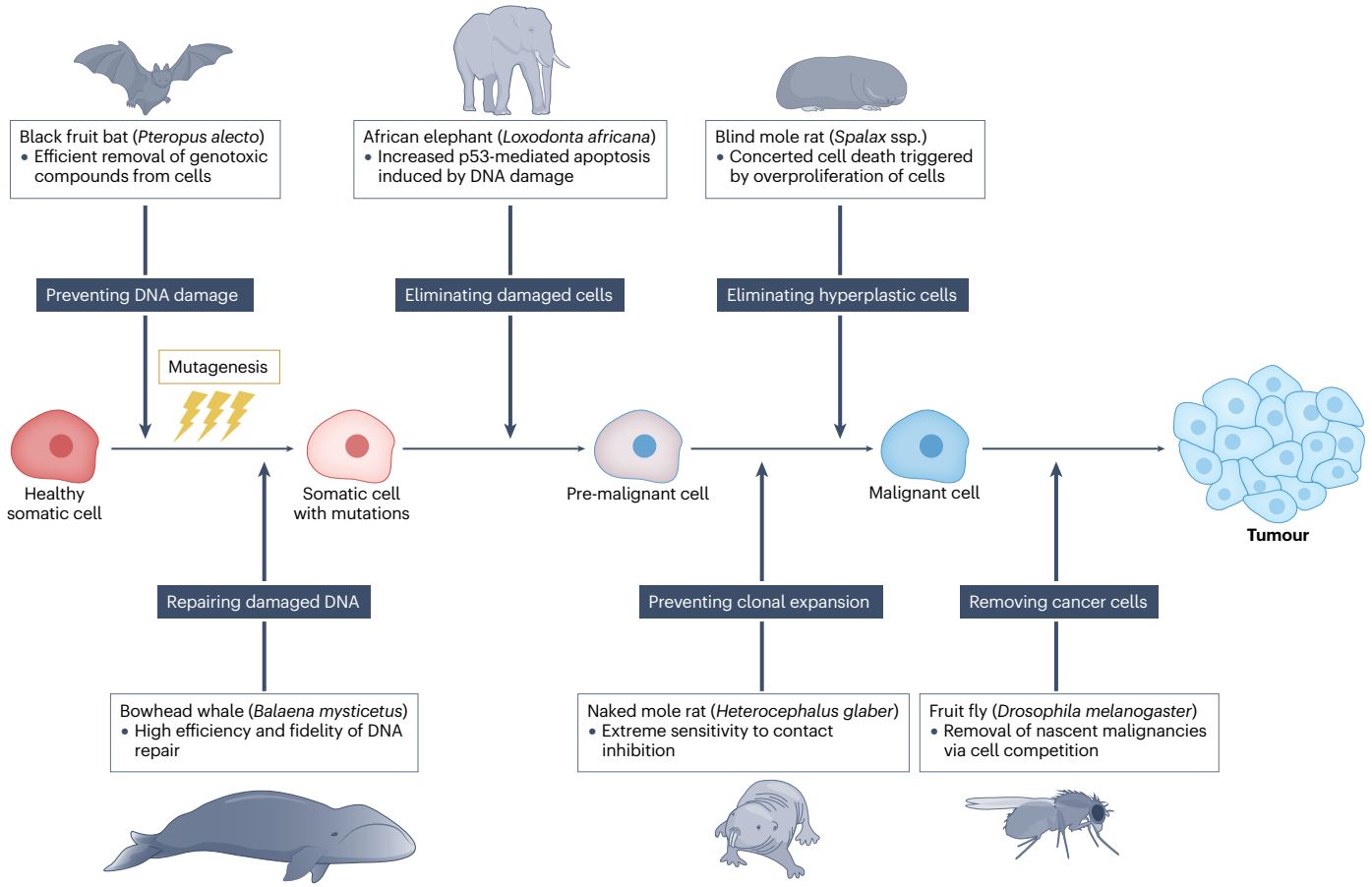

Fig. 1 | Diverse cancer-resistance mechanisms of organisms across the tree of life. Animals exhibit diverse proposed cancer-resistance mechanisms, including DNA protection (for example, efficient genotoxic drug efflux mediated by the ATP-binding cassette transporters in bats protects the DNA[81]), efficient DNA repair (for example, high levels of cold-inducible RNA-binding protein (CIRBP) and replication protein A2 (RPA2) are present in bowhead whale fibroblasts and increase the efficiency and fidelity of DNA repair[40]), effective elimination of damaged cells (for example, high rates of p53-mediated apoptosis in response to DNA damage in elephants[38]), preventing clonal expansion of pre-malignant cells (for example, naked mole rat fibroblasts secrete extremely high-molecular-weight hyaluronan, acting as a signal to CD44 receptors and triggering early contact inhibition[60]), elimination of overproliferating pre-malignant cells (for example, blind mole rat cells express very low levels of DNA methyltransferase 1 that fails to maintain DNA methylation on retrotransposable elements during rapid cell proliferation; derepression of retrotransposable elements triggers the formation of cytoplasmic RNA–DNA hybrids, which activate the cyclic GMP–AMP synthase (cGAS)–stimulator of interferon genes (STING) pathway and induce concerted cell death[39]) and removal of cancer cells via cell competition (for example, in fruitflies, genetically fitter cells eliminate neighbouring abnormal cells, such as those with mutations or altered polarity, through apoptosis triggered by signalling imbalances, innate immune responses or mechanical stress[82]).

and mechanisms of cancer evasion. Data on cancer prevalence in zoo animals allow for the exploration of life-history, ecological and phylogenetic risk factors of cancer using computational analysis and modelling. High-quality reference genomes of several hundred species are now available, which serve as a basis for powerful comparative genomic studies[38], for instance, to study the evolution of tumour-suppressor mechanisms, or cancer predisposition in taxa at the level of genes and genetic backgrounds. Comparative transcriptomics[51] (for example, to identify key regulatory pathways or biomarkers of cancer and potentially pinpoint therapeutic targets), proteomics (for example, to identify shared or unique protein signatures of cancer development, progression or to predict treatment response) and metabolomics (for example, to identify metabolic alterations or metabolic vulnerabilities in cancer) are also more accessible for multiple species. Moreover, advances in high-precision genome sequencing of individuals have improved estimates of mutation rates across different species and tissues[52], helping to understand mutational events driving cancer across the tree of life. The advent of ultra-accurate duplex sequencing methods promises to enable the quantification of somatic mutational processes from any cell type in any species[52,53], removing technical barriers that have hitherto prevented their study. Similarly, epigenetic biomarkers of ageing based on DNA methylation data can be studied across species[54], potentially helping to understand how epigenetic alterations shape the ageing phenotype and the risk of cancer. Finally, artificial intelligence and machine learning is being applied to determine whether conservation of tumour cell morphology across taxa[55] can illuminate shared biology, guide drug development, and improve both the diagnosis and therapeutic targeting in humans as well as

in model and non-model species. All these approaches benefit from improved molecular phylogenies (using DNA sequences to understand the evolutionary relationships between taxa and to reconstruct their evolutionary history) that now provide a robust framework for comparative studies. Collectively, such advances have strongly elevated the opportunities and impact of comparative oncology, enabling researchers to feasibly study how cancer resistance unfolded over evolutionary time.

Using comparative oncology as a starting point, we can now rationally identify novel model organisms for cancer research and perform species-specific studies with the aim of understanding which molecular or cellular mechanisms shape their cancer-resistant phenotypes. The study of such taxa is still in its infancy, with only a handful of non-model organisms having been inspected so far (for example, naked mole rats, elephants, bowhead whales and bats)[30,31,40]. Along with the plethora of methods listed above, advances in gene editing, including CRISPR–Cas9, can be considered to study and validate candidate cancer-resistance mechanisms across species. Although making genetically modified elephants or whales presents both technical and ethical challenges, these organisms can be used to discover novel mechanisms of cancer resistance, which can then be modelled in mice and ultimately translated to patients.

Translatability of cross-species studies

Understanding diverse tumour-suppressive mechanisms and variance in cancer predisposition across animals holds more than just intellectual interest: it can lead to the development of new cancer therapies and cancer prevention interventions for patients. Mechanisms and genetic adaptations identified in naturally cancer-resistant animals can be mimicked in more cancer-prone species using pharmacological or genetic manipulations. Furthermore, investigating how evolution has fine-tuned tumour-suppressive mechanisms in concert with the natural lifespan and reproductive pattern of various animals could reveal how these mechanisms are shaped by natural selection. Such an understanding could provide the opportunity to design interventions mimicking the effects of natural selection. Below we present several examples where comparative biology has resulted in clinically relevant insights.

Autophagy

Studies in model organisms have revealed that genetically enhancing autophagy can extend lifespans and reduce or delay cancer incidence[56,57]. This suggests that variability in autophagy could contribute to variations in lifespans and cancer patterns across organisms. Can we, then, alter the autophagy pathway in individuals to mimic the evolutionary changes brought about by natural selection? Indeed, evidence already exists to support this concept. Lifestyles that enhance autophagy, such as dietary restriction (or altered nutrient intake without restriction) and exercise, have been shown to reduce cancer incidence[58]. Nonetheless, the beneficial effects of autophagy-enhancing strategies may apply primarily to cancer prevention or early intervention, as autophagy can also support cancer cell survival in established tumours[59].

High-molecular-mass hyaluronan

Naked mole rats are exceptionally long lived for their size and are very resistant to cancer with only a handful of cancer cases reported after decades of observation[60]. High-molecular-mass hyaluronan plays an important role in the cancer resistance of naked mole rats by preventing hyperplasia[60]. Remarkably, overexpression in mice of the naked mole rat gene encoding the enzyme involved in hyaluronan synthesis reduced cancer incidence and extended mouse lifespan[61]. This provides proof-of-principle that anticancer mechanisms that evolved in cancer-resistant species can be 'exported' into other species even if this will need to be shown with more phylogenetically distant species.

Downregulation of DNMT1 to activate transposable elements

A second mole rat species, the blind mole rat, which is phylogenetically distant from the naked mole rat even though it too displays long lifespan and cancer resistance, has evolved low DNMT1 activity, which leads to the rapid activation of transposable elements in hyperplastic cells triggering the activation of the interferon response and elimination of affected cells[39]. Remarkably, DNMT1 inhibitors are already actively used in cancer treatment to induce the death of cancer cells through activation of the interferon response[62].

Enhanced p53

Another example of the translatability of comparative oncology to human medicine relates to the extensive work performed on elephants over the past decade. In 2015, two groups of researchers independently concluded that the high cancer resistance of elephants must be related to their genetic architecture[38,63]. These researchers showed that the elephant genome harbours 20 copies of the key tumour-suppressor

Table 1 | Species-specific cancer rates in non-domestic vertebrates

Dataset	Parameter	Taxonomic coverage	Number of species	Number of inspected individuals	Minimum number of individuals per species
Boddy et al.[17]	Neoplasia prevalence; malignancy prevalence	Mammals	41	869	4
Vincze et al.[15]	Cancer mortality risk	Mammals	191	11,840	20
Vincze et al.[15]	Age-standardized incidence of cancer mortality	Mammals	172	11,192	20
Compton et al.[18]	Neoplasia prevalence; malignancy prevalence	Mammals	102	5,843	20
Compton et al.[18]	Neoplasia prevalence; malignancy prevalence	Amphibians	41	3,061	20
Compton et al.[18]	Neoplasia prevalence; malignancy prevalence	Reptiles	71	2,693	20
Compton et al.[18]	Neoplasia prevalence; malignancy prevalence	Birds	113	5,939	20

A list of publicly available primary databases reporting species-specific cancer rates in non-domestic vertebrates (ordered chronologically). Note that all these cancer risk datasets are based on populations under human care (for example, zoos) and provide disease risk at the time of death (obtained through necropsy). Although these databases[15,17,18] were compiled independently, some degree of overlap among the recorded individuals is expected, although the extent remains unknown. The minimum number of individuals per species reflects the minimum number of necropsy reports required to estimate cancer rate.

gene, *TP53*, of which most other animal taxa possess a single copy. This remarkable gene expansion retains one 'ancestral copy' similar to the *TP53* conserved across the animal phylogeny and at least 19 additional truncated copies of *TP53* reinserted into the elephant genome over evolutionary time as retrogenes[64]. *TP53* codes for the p53 DNA-binding transcription factor and plays a prominent role in DNA repair and, when repair is not possible, in triggering apoptosis. Selection for multiple copies of this gene could contribute to the increased protection against cancer in elephants, which are large and long lived, through a more efficient p53-mediated DNA damage response. Although some argue that the *TP53* retrogenes appear to be inactive and not under evident selection[65], recent efforts demonstrate that at least one of these retrogenes can trigger non-transcriptional apoptosis through the mitochondria when expressed in human cancer cells[66], supporting a functional role and potential selection for tumour-suppressor gene amplification as a cancer-resistance mechanism. This discovery triggered research into a clinical application to prevent or treat human cancer using the elephant *TP53*. Initial results have confirmed that transgenic mice expressing elephant *TP53* survive longer following carcinogenic exposure[67].

Enhanced DNA repair

Organisms can achieve cancer resistance not only through enhanced action of tumour-suppressive mechanisms that eliminate the damaged cells, but also by improved genome maintenance. This strategy has recently been described in the bowhead whale, which is very large and perhaps the longest-lived mammal. Compared with other mammals, the bowhead whale shows significantly lower rates of mutations achieved through more efficient and accurate DNA double-strand break repair[40]. This is probably aided by two proteins present in high levels in bowhead whale cells (cold-inducible RNA-binding protein (CIRBP) and replication protein A2 (RPA2)), which increase the efficiency and fidelity of DNA repair even when overexpressed in human cells.

Inherent challenges of translation

The identification of natural tumour-suppressor mechanisms, such as those listed above, highlight the potential of learning from different species to impact the outcomes of patients with cancer. Moreover, given the diversity of life on Earth, it is highly plausible that numerous other natural anticancer mechanisms will be discovered in the near future[68]. Translation of such mechanisms to human medicine should be feasible, especially considering the conserved biological processes common to all animals. Although promising opportunities may exist, we must acknowledge the inherent challenges. For instance, natural selection has had a considerable amount of time to balance the trade-offs associated with alteration to pathways contributing to tumour suppression across species (or in particular species). Such costs of pathway manipulations remain poorly understood and we need to recognize the pleiotropic nature of many genes, pathways and processes. A prominent example is *TP53*, the upregulation of which can lead to premature ageing[69].

Perspectives and future directions

As outlined above, comparative oncology has matured into a well-established research field, with specific questions, rich methods, innovative technologies and robust results. As a cross-disciplinary group of researchers involved in this domain, we outline below key directions that we envision for the advancement of comparative oncology in the coming years.

Interdisciplinary integration

A crucial imperative for comparative oncology is to improve the integration between medical oncology, veterinary medicine, evolutionary biology and ecology. Indeed, a major limitation in the application of knowledge gained from comparative oncology has been that medical doctors often lack a background in evolutionary thinking, whereas laboratory scientists often have limited knowledge of therapies or lack experience of working with non-model animals. Therefore, the translation of findings in comparative oncology can only be achieved by tight cooperation between these groups. In practical terms, we advocate for three changes. International cancer meetings should devote regular sessions to comparative and evolutionary approaches to cancer. Equally, more conferences of an interdisciplinary nature, bringing all of this expertise together should be organized to trigger cooperation among these teams. Major cancer journals should be open to data-driven and conceptual submissions in comparative oncology (instead of limiting such work to evolutionary biology journals, which are generally not read by clinical oncologists). Finally, educational curricula (beginning as early as undergraduate education and continuing through graduate and professional training) should be restructured to reflect the importance of evolutionary approaches to understanding cancer. We must both provide better background in evolutionary biology to medical trainees and convince students of evolutionary biology and ecology to embrace cancer as their main topic of interest. Ultimately, these changes to disseminate knowledge about comparative oncology will drive the connection to clinical practice.

Comparison of the hallmarks of cancer across taxa

Until now, comparative oncology has focused, understandably, on assessing cancer risk and/or cancer mortality across taxa. Yet a plethora of new discoveries awaits those who will explore the phylogenetic diversity in mechanisms for each of the hallmarks of cancer (for example, resistance to cell death, genome instability, deregulation of cell metabolism and replicative immortality[70]). We know very little about the cell biology of neoplasms in non-human animals. Crucially, this line of thought does not assume that these traits co-occur with cancer in all taxa where they are found; for instance, knowing in which taxa and through which mechanisms deregulation of cell metabolism occurs will be informative for understanding the connection between cancer and deregulation of cell metabolism in humans, even if it turns out that some taxa undergo deregulation of cell metabolism without having cancer.

Comparison of cancer-associated traits across taxa

Following observations on the cross-species diversity in cancer risk, researchers were hoping to reveal the key life-history predictors of tumour prevalence across taxa. Although many attempts have been made (by examining the effect of body size, lifespan, reproductive rates and others), and although statistical significance was observed in some cases, the proportion of variance in cross-species cancer prevalence explained by these predictors remains negligible. This highlights that cancer risk is probably influenced by a complex interplay of multiple factors beyond the commonly examined life-history traits, potentially including genetic, immunological and ecological variables that have yet to be fully understood. An important future direction in comparative oncology will be to establish cancer-associated traits across species (that is, traits that co-occur with cancer although they have yet to be defined as hallmarks of cancer). For example, microbiome composition has been linked to both cancer risk and responses to cancer therapies in humans and model organisms[71]. However, can one find

interesting correlations between microbiome composition and/or microbiome disruption and cancer risk and/or mortality across the tree of life? This question should be explored systematically in future research. Interestingly, polymorphic microbiomes have been added as an 'enabling characteristic' in Hanahan's most recent update on the hallmarks of cancer[72]. Other major examples of cancer-associated traits that could be explored across taxa include biological ageing, regenerative capabilities and the regulation of inflammation.

Improving the search for anticancer mechanisms

More rigorous conceptualizations of anticancer mechanisms will be needed to explore their impact on cancer risk and mortality across taxa. One potential classification is based on the stage in cancer progression at which anticancer mechanisms act: some may function to prevent the initiation of neoplasms, others to prevent the transformation of a benign neoplasm into a cancer, or to prevent a cancer from metastasizing or even prevent a cancer from causing death. The analysis of each transition depends on distinct data (for example, prevalence of neoplasia versus cancer mortality) and the mechanisms for preventing each of these transitions are likely to be different. Furthermore, even for already well-recognized anticancer mechanisms, much more specific work is needed, particularly comparative studies involving multiple taxa. For example, immunological surveillance is often described as one of the major anticancer mechanisms, but the most detailed insights into immune components relevant to cancer (for example, T cell subsets, cytokines and checkpoint molecules) derive from human or murine studies. Although in humans the role of specific immune components is relatively well understood, far less is known about how these components function across species and under which evolutionary or ecological conditions they are effective. Future work will certainly generate more precise insights through the development of 'comparative oncoimmunology'. For instance, do phagocytic cells impact cancer initiation and progression in all metazoans (as they do in many mammals and in some insects)? Other promising examples for study include natural killer cells (which are widespread across metazoans[73]) and immune components that, like cyclic GMP–AMP synthase (cGAS) for instance, display strong homologies across a wide variety of taxa[74] and are known to play a role in cancer resistance in mammals including humans[75].

Understanding taxonomic variation in metastatic disease

How much does the incidence of metastasis vary across species and what explains that variation? Natural selection may have prevented cancer mortality by preventing the evolution of metastasis. So far, we have very little data on the risk of metastatic disease. We also lack hypotheses about which selective pressures might lead to susceptibility to metastatic disease. Progress will probably require access to full necropsy reports from zoos and wildlife services, which is currently difficult but not impossible. If there are species that are prone to malignancy but do not develop metastasis, they may have evolved mechanisms to prevent metastasis. We might also test whether the organotropism of metastasis seen in humans is similar or different in non-human animals.

Systematic exploration of the impact of environmental exposures on carcinogenesis

Cancer prevention, although perhaps the ultimate goal, has received considerably less attention than its potential treatment. Environmental stressors and contaminants, including industrial, pharmaceutical or agricultural chemicals, air pollution, radiation or light pollution, markedly contribute to carcinogenesis in humans and animals alike[76–78]. Despite growing global concern, substantial amounts of pollutants continue to be released into the environment yearly, and many currently banned pollutants will persist in our environment for decades or even centuries[79]. A key obstacle is that too little is known about the impact of most environmental factors, for many reasons including long latency from exposure to cancer development, complex interactions between genetic predisposition and environmental contaminants, multiple and often simultaneous exposures to various contaminants, ethical constraints of performing exposure experiments on human subjects, and the difficulty of identifying and controlling all potential confounding effects. By developing systematic comparisons of correlations between cancer incidence and exposure to environmental factors in wild taxa and animals living in anthropogenic environments, such as domestic animals, those living in urbanized areas, zoos or degraded habitats, one might highlight key carcinogenic factors that will elicit the formulation of precise and specific hypotheses that can be tested in the laboratory. Animals that are particularly susceptible to cancer, as well as animals with shorter lifespans, might act as sentinels for the detection of carcinogens that are not yet apparent in humans. Animal populations that experience unusually high cancer rates could also provide insight into environmental conditions (probably resulting from more recent human activities) that have contributed to these greater cancer rates.

Developing studies on cancer in wildlife

Cancer surveillance programmes for wildlife are scarce and must be encouraged. By comparing cancer in animals in the wild versus those in captivity, we might be able to better distinguish the role of environmental factors from intrinsic cancer susceptibility or resistance in shaping cancer prevalence, which may help to inform human cancer epidemiology. However, collecting data on cancer risk in diverse wild animals, especially those living in their natural habitats, is challenging, including due to the diverse nature of cancer and that small but lethal tumours can often go undetected during necropsies. Moreover, recovering carcasses of wild animals alone is very challenging, especially before decomposition or consumption by scavengers, which would affect their suitability for necropsy[78]. Additionally, disentangling cancer-related mortalities from other sources of more obvious deaths, such as predation or other illnesses triggered by cancer-related health issues, poses further issues. Collaborative endeavours involving oncologists, veterinarians and researchers from diverse fields are essential to overcome these challenges.

Studying critical transitions in the history of life with respect to cancer incidence

Clearly the transition from unicellular life to multicellular life was a necessary but perhaps not sufficient condition for the development of cancer[80]. It is likely that the presence of cell turnover in adult tissues is a key cancer vulnerability. For many organisms, such as *Caenorhabditis elegans* for instance, cell proliferation ends with development and the adult body is almost entirely composed of quiescent fully differentiated cells, except for the germline, and such organisms do not develop cancer. What is the relationship across taxa (and across organs) between the presence of adult somatic stem cells and the probability of developing cancer? Is indeterminate growth a risk factor for cancer? What process(es) in mammals make them more susceptible to cancer than other vertebrates?

Comparative genomics and functional validation of putative resistance mechanisms

One of the primary goals of comparative oncology is to discover the mechanisms of cancer resistance that have evolved in non-human species and then translate those discoveries to improved cancer prevention and management in humans. Genomic studies that compare cancer resistant with cancer susceptible species, particularly when closely related, can identify genomic regions that appear to have changed by natural selection independently in multiple cancer-resistant species. Those genomic changes are putative molecular mechanisms of cancer resistance that we can then engineer into cell lines[66] and model organisms to test for cancer-resistance phenotypes.

Fostering methodological advances in non-model organisms to advance human cancer research

Applying and refining state-of-the-art molecular techniques (for example, single-cell DNA and RNA sequencing, metabolomics, proteomics, epigenetic profiling and CRISPR–Cas9) in cancer-resistant, non-model taxa can provide deeper insights into how specific molecular pathways are regulated in cancer-resistant species. Although such approaches have already provided some fascinating results in a handful of cancer-resistant taxa, applying them at a wide taxonomic scale could help identify conserved pathways of tumour prevention, suppression or elimination across species. Such knowledge might help to uncover universal targets for cancer therapy or prevention.

Conclusions

The advancements and insights gained so far from comparative oncology call for a better integration of this approach into oncological research. With increased attention, continuous collaboration between disciplines, and a stronger commitment from conventional cancer biologists and clinical researchers to explore cancer from a new perspective, comparative oncology can truly reshape the landscape of cancer research and treatment.

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All authors researched data for the article. O.V., D.B., A.C., J.D., V.G., C.C.M., J.D.S., A.S., M.G. and T.P. contributed substantially to discussion of the content. All authors wrote the article. O.V., D.B., A.C., J.D., V.G., C.C.M., J.D.S., A.S., M.G. and T.P. reviewed and/or edited the manuscript before submission.

Competing interests

J.D.S. reports being a cofounder and shareholder of Peel Therapeutics outside the submitted work. All other authors declare no competing interests.

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